SparkFun RTK Product Manual

Simple and Cost Effective High-Precision Navigation

This PDF is automatically generated. See https://docs.sparkfun.com/SparkFun_RTK_Firmware/ for the latest version.

SparkFun Electronics - 2023

Table Of Contents

1. Introduction

The SparkFun RTK products are exceptional GNSS receivers out-of-box and can be used with little or no configuration. This RTK Product Manual provides detailed descriptions of all the available features of the RTK products.

The line of RTK products offered by SparkFun all run identical firmware. The RTK firmware and this guide cover the following products:

SparkFun RTK Facet L-Band (GPS-20000)
Hookup Guide

SparkFun RTK Facet (GPS-19029)
Hookup Guide

SparkFun RTK Reference Station (GPS-22429)
Hookup Guide

SparkFun RTK Express Plus (GPS-18590)
Hookup Guide

SparkFun RTK Express (GPS-18442)
Hookup Guide

SparkFun RTK Surveyor (GPS-18443)
Hookup Guide

Depending on the hardware platform different features may or may not be supported. We will denote each product in each section so that you know what is supported.

There are multiple ways to configure an RTK product:

- Bluetooth - Good for in-field changes
- WiFi - Good for in-field changes
- Serial Terminal - Requires a computer but allows for all configuration settings
- Settings File - Used for configuring multiple RTK devices identically

The Bluetooth or Serial Terminal methods are recommended for most advanced configurations. Most, but not all settings are also available over WiFi but can be tricky to input via mobile phone.

If you have an issue, feature request, bug report, or a general question about the RTK firmware specifically we encourage you to post your comments on the firmware's repository. If you feel like bragging or showing off what you did with your RTK product, we'd be thrilled to hear about it on the issues list as well!

Things like how to attach an antenna or other hardware-specific topics are best read on the Hookup Guides for the individual products.

2. Quick Start

Surveyor: / Express: / Express Plus: / Facet: / Facet L-Band: / Reference Station:

This quick start guide will get you started in 10 minutes or less. For the full product manual, please proceed to the **Introduction**.

Are you using Android or iOS?

2.1 Android

1. Download SW Maps. This may not be the GIS software you intend to do your data collection, but SW Maps is free and makes sure everything is working correctly out of the box.

Download SW Maps for Android

2. Mount the hardware:

- For RTK Surveyor/Express/Express Plus: Attach the included antenna to a monopole using the included thread adapter. Clamp the RTK device to the monopole. Use the included cable to connect the antenna to the RTK Surveyor/Express/Express Plus (Figure 1).
- For RTK Facet/Facet L-Band: Attach the Facet to a monopole using the included thread adapter (Figure 1).

Figure 1

3. Turn on your RTK device by pressing the POWER button until the display shows ‘SparkFun RTK’ then you can release it (Figure 2).

Figure 2

4. Please note the four-digit code in the top left corner of the display (**B022** in the picture below). This is the MAC address you will want to pair with (Figure 3).

Figure 3

5. From your cell phone, open Bluetooth settings and pair it with a new device. You will see a list of available Bluetooth devices. Select the ‘Facet Rover-3AF1’ where 'Facet' is the type of device you have (Surveyor, Express, Facet, etc) and 3AF1 is the MAC address you see on the device’s display (Figure 4).

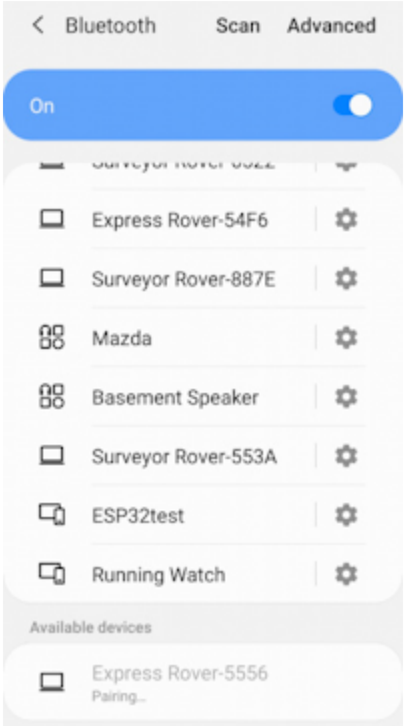


Figure 4

6. Once paired, open SW Maps. Select 'New Project' and give your project a name like 'RTK Project'.
7. Press the SW Maps icon in the top left corner of the home screen and select Bluetooth GNSS. You should see the 'Facet Rover-3AF1' in the list. Select it then press the 'Connect' button in the bottom left corner (Figure 5). SW Maps will show a warning that the instrument height is 0m. That's ok.

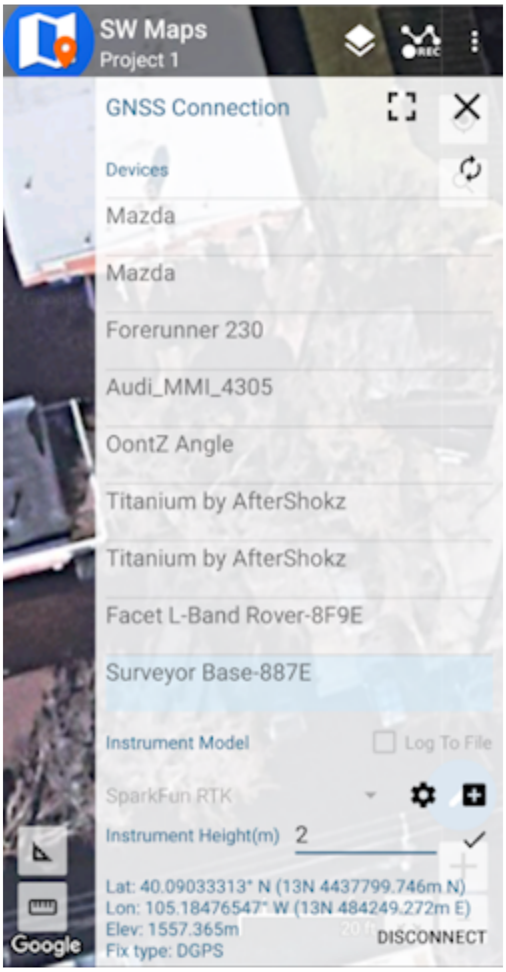


Figure 5

8. Once connected, have a look at the display on the RTK device. You should see the MAC address disappear and be replaced by the Bluetooth icon (Figure 6). You're connected!

Figure 6

9. Now put the device outside with a clear view of the sky. GNSS doesn't work indoors or near windows. Within about 30 seconds you should see 10 or more satellites in view (SIV) (Figure 7). More SIV is better. We regularly see 30 or more SIV. The horizontal positional accuracy (HPA) will start at around 10 meters and begin to decrease. The lower the HPA the more accurate your position. If you wait a few moments, this will drop dramatically to around 0.3 meters (300mm = 1ft) or better.

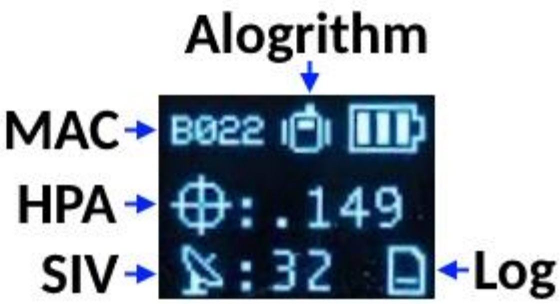


Figure 7

You can now use your RTK device to measure points with very good (sub-meter) accuracy. If you need extreme accuracy (down to 10mm) continue reading the RTK Crash Course.

2.2 iOS

The software options for Apple iOS are much more limited because Apple products do not support Bluetooth SPP. That's ok! The SparkFun RTK products support Bluetooth Low Energy (BLE) which *does* work with iOS.

1. Download SW Maps for iOS. This may not be the GIS software you intend to do your data collection, but SW Maps is free and makes sure everything is working correctly out of the box.

   Download SW Maps for iOS

2. Mount the hardware:

- For RTK Surveyor/Express/Express Plus: Attach the included antenna to a monopole using the included thread adapter. Clamp the RTK device to the monopole. Use the included cable to connect the antenna to the RTK Surveyor/Express/Express Plus (Figure 1).
- For RTK Facet/Facet L-Band: Attach the Facet to a monopole using the included thread adapter (Figure 1).

Figure 1

3. Turn on your RTK device by pressing the POWER button until the display shows 'SparkFun RTK' then you can release it (Figure 2).

Figure 2

4. Put the RTK device into configuration mode by tapping the POWER or SETUP button multiple times until the Config menu is highlighted (Figure 3).

Figure 3

5. From your phone, connect to the WiFi network *RTK Config*.
6. Open a browser (Chrome is preferred) and type **rtk.local** into the address bar. Note: Devices with older firmware may still need to enter **192.168.4.1**.
7. Under the *System Configuration* menu, change the **Bluetooth Protocol** to **BLE** (Figure 4). Then click **Save Configuration** and then **Exit and Reset**. The unit will now reboot.

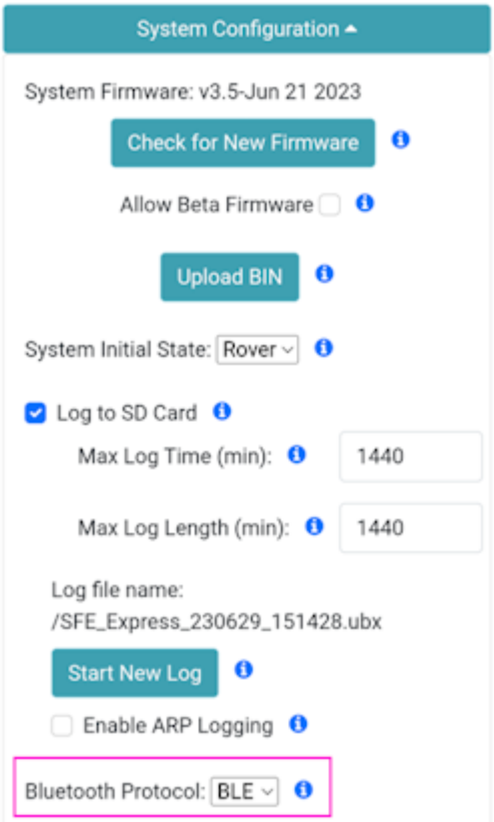


Figure 4

8. You should now be disconnected from the *RTK Config* WiFi network. Make sure Bluetooth is enabled on your iOS device Settings (Figure 5). The RTK device will not appear in the OTHER DEVICES list. That is OK.

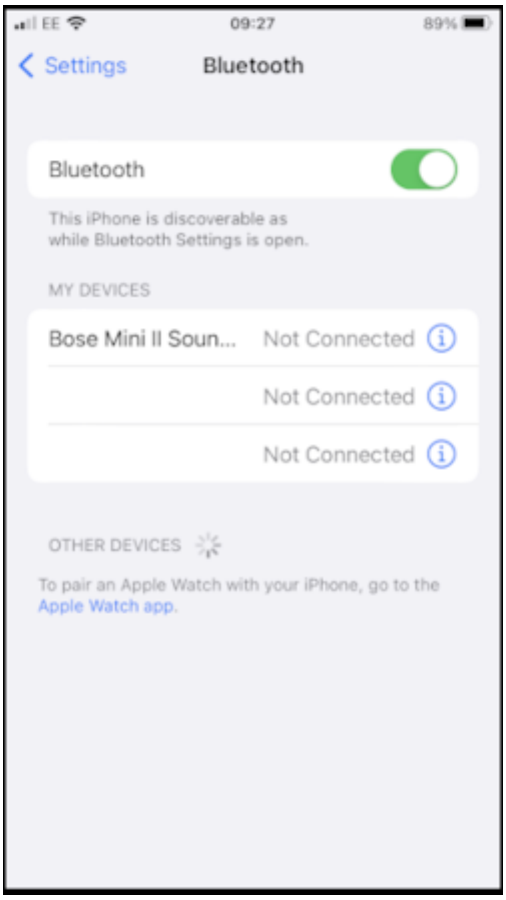


Figure 5

9. Open SW Maps. Select ‘New Project’ and give your project a name like ‘RTK Project’.
10. Press the SW Maps icon in the top left corner of the home screen and select Bluetooth GNSS. You will need to agree to allow a Bluetooth connection. Set the *Instrument Model* to **Generic NMEA (Bluetooth LE)**. Press 'Scan' and your RTK device should appear. Select it then press the ‘Connect’ button in the bottom left corner.
11. Once connected, have a look at the display on the RTK device. You should see the MAC address disappear and be replaced by the Bluetooth icon (Figure 6). You’re connected!

Figure 6

12. Now put the device outside with a clear view of the sky. GNSS doesn’t work indoors or near windows. Within about 30 seconds you should see 10 or more satellites in view (SIV) (Figure 7). More SIV is better. We regularly see 30 or more SIV. The horizontal positional accuracy (HPA) will start at around 10 meters and begin to decrease. The lower the HPA the more accurate your position. If you wait a few moments, this will drop dramatically to around 0.3 meters (300mm = 1ft) or better.

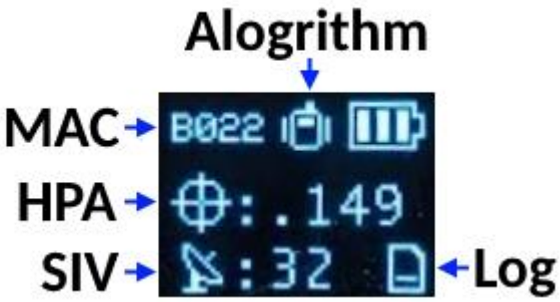


Figure 7

You can now use your RTK device to measure points with very good (sub-meter) accuracy. If you need extreme accuracy (down to 10mm) continue reading the RTK Crash Course.

2.3 RTK Crash Course

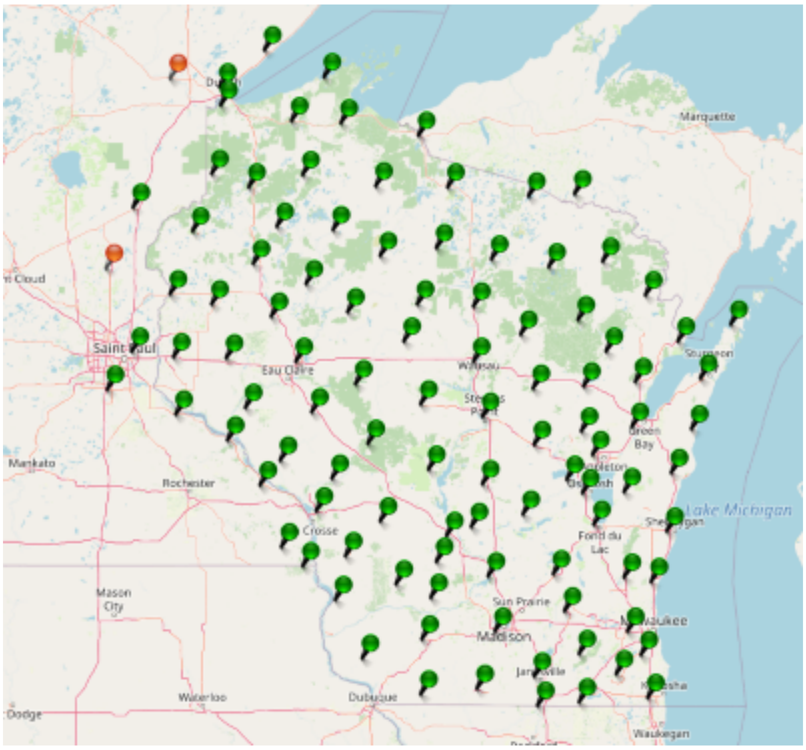
To get millimeter accuracy we need to provide the RTK unit with correction values. Corrections, often called RTCM, help the RTK unit refine its position calculations. RTCM (Radio Technical Commission for Maritime Services) can be obtained from a variety of sources but they fall into three buckets: Commercial, Public, and Civilian Reference Stations.

Commercial Reference Networks

These companies set up a large number of reference stations that cover entire regions and countries, but charge a monthly fee. They are often easy to use but can be expensive.

- PointOneNav ($50/month) - US, EU, Australia, South Korea
- Skylark ($29 to $69/month) - US, EU, Japan, Australia
- SensorCloud RTK ($100/month) partial US, EU
- Premium Positioning (~$315/month) partial EU
- KeyNetGPS ($375/month) North Eastern US
- Hexagon/Leica ($500/month) - partial US, EU

Public Reference Stations



State Wide Network of Continuously Operating Reference Stations (CORS)

Be sure to check if your state or country provides corrections for free. Many do! Currently, there are 21 states in the USA that provide this for free as a department of transportation service. Search ‘Wisconsin CORS’ as an example. Similarly, in France, check out CentipedeRTK. There are several public networks across the globe, be sure to google around!

Civilian Reference Stations

The base station at SparkFun

You can set up your own correction source. This is done with a 2nd GNSS receiver that is stationary, often called a Base Station. There is just the one-time upfront cost of the Base Station hardware. See the Creating a Permanent Base document for more information.

2.4 NTRIP Example

Once you have decided on a correction source we need to feed that data into your SparkFun RTK device. In this example, we will use PointOneNav and SW Maps.

1. Create an account on PointOneNav. **Note:** This service costs $50 per month at the time of writing.
2. Open SW Maps and connect to the RTK device over Bluetooth.
3. Once connected, open the SW Maps menu again (top left corner) and you will see a new option; click on 'NTRIP Client'.
4. Enter the credentials provided by PointOneNav and click Connect (Figure 1). Verify that *Send NMEA GGA* is checked.

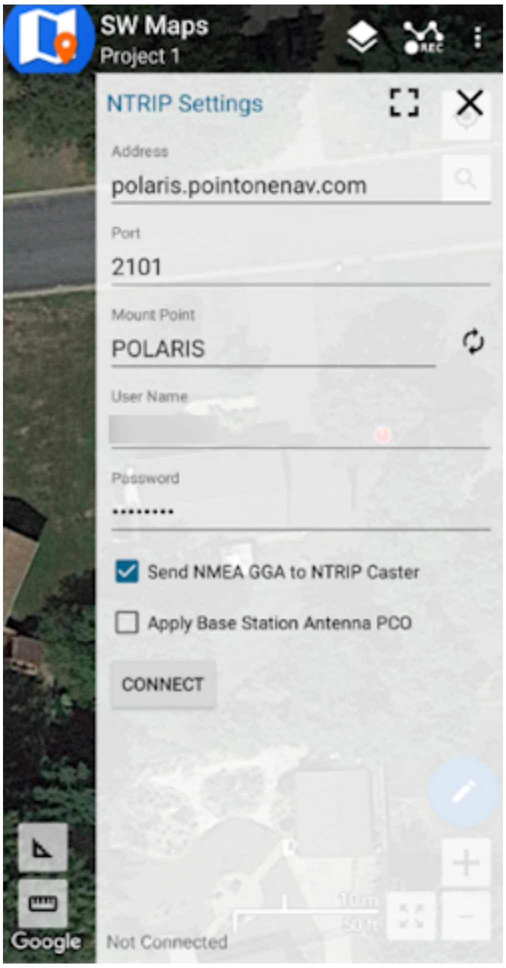


Figure 1

5. Corrections will be downloaded every second from PointOneNav using your phone's cellular connection and then sent down to the RTK device over Bluetooth. You don't need a very fast internet connection or a lot of data; it's only about 530 bytes per second.

Assuming you are outside, as soon as corrections are sent to the device, the Crosshair icon will become double and begin flashing. Once RTK Fix is achieved (usually under 30 seconds) the double crosshairs will become solid and the HPA will be below 20mm (Figure 2). You can now take positional readings with millimeter accuracy!

Figure 2

In SW Maps, the position bubble will turn from Blue (regular GNSS fix), then to Orange (RTK Float), then to Green (RTK Fix) (Figure 3).

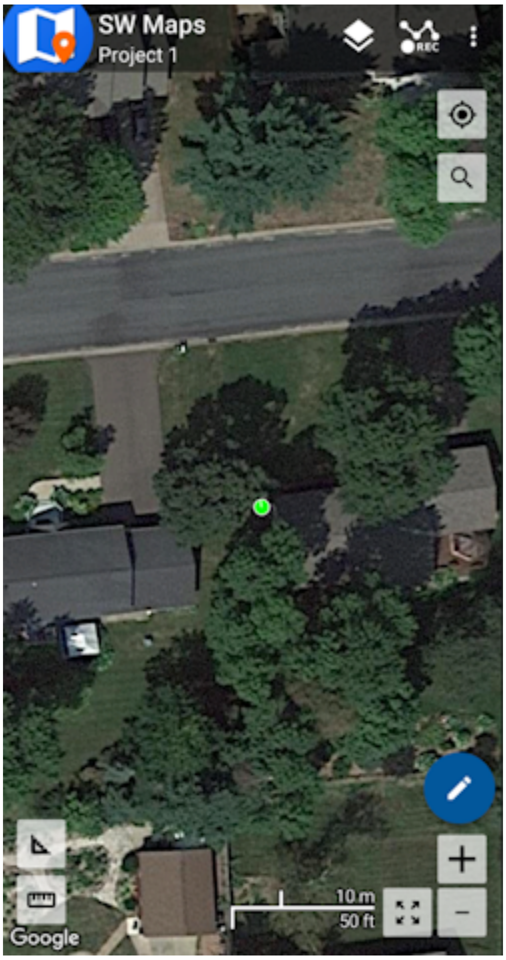


Figure 3

RTK Fix will be maintained as long as there is a clear view of the sky and corrections are delivered to the device every few seconds.

2.5 Common Gotchas

- High-precision GNSS only works with dual frequency L1/L2 antennas. This means that GPS antenna you got in the early 2000s with your TomTom is not going to work. Please use the L1/L2 antennas provided by SparkFun.
- High-precision GNSS works best with a clear view of the sky; it does not work indoors or near a window. GNSS performance is generally *not* affected by clouds or storms. Trees and buildings *can* degrade performance but usually only in very thick canopies or very near tall building walls. GNSS reception is very possible in dense urban centers with skyscrapers but high-precision RTK may be impossible.
- The location reported by the RTK device is the location of the antenna element; it's *not* the location of the pointy end of the stick. Lat and Long are fairly easy to obtain but if you're capturing altitude be sure to do additional reading on ARPs (antenna reference points) and how to account for the antenna height in your data collection software.
- An internet connection is required for most types of RTK. RTCM corrections can be transmitted over other types of connections (such as serial telemetry radios). See Correction Transport for more details.

2.6 RTK Facet L-Band Keys

The RTK Facet L-Band is unique in that it must obtain keys to decrypt the signal from a geosynchronous satellite. Here are the steps to do so:

1. Turn on your RTK Facet L-Band by pressing the POWER button until the display shows 'SparkFun RTK' then you can release it (Figure 1).

   Figure 1

2. Put the RTK device into configuration mode by tapping the POWER button multiple times until the Config menu is highlighted (Figure 2).

   Figure 2

3. From your phone or laptop, connect to the WiFi network *RTK Config*.
4. Open a browser (Chrome is preferred) and type **rtk.local** into the address bar. Note: Devices with older firmware may still need to enter **192.168.4.1**.
5. Under the *WiFi Configuration* menu, enter the SSID and password for your local WiFi network (Figure 3). You can enter up to four. This can be a home, office, cellular hotspot, or any other WiFi network. The unit will attempt to connect to the internet periodically to obtain new keys, including this first day. Then click **Save Configuration** and then **Exit and Reset**. The unit will now reboot.

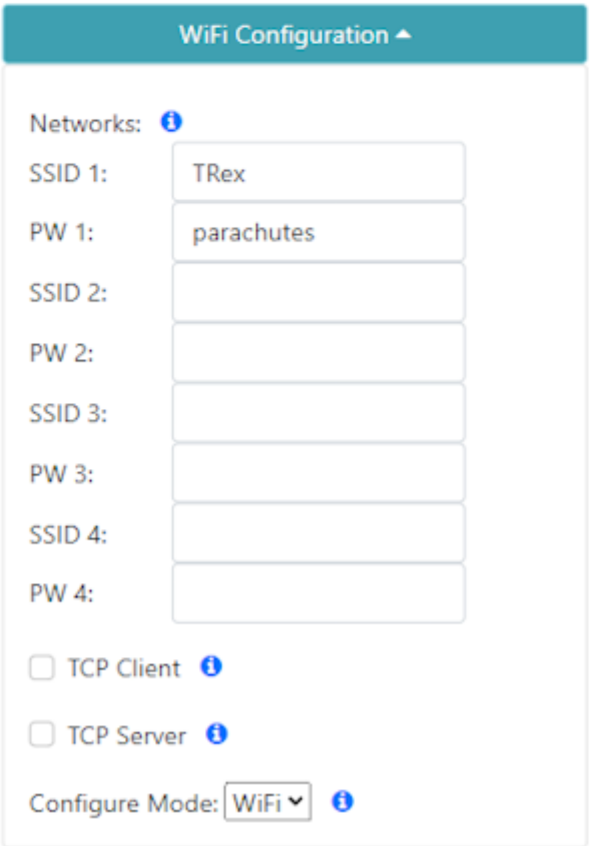


Figure 3

6. After reboot, the device will connect to WiFi and obtain keys. You should see a series of displays indicating the automatic process (Figure 4).

Figure 4

Keys are valid for a minimum of 29 days and a maximum of 60. The device will automatically attempt to connect to WiFi to obtain new keys. If WiFi is not available during that period the keys will expire. The device will continue to operate with expired keys, with ~0.3m accuracy but not be able to obtain RTK Fix mode.

7. Now put the device outside with a clear view of the sky. GNSS doesn't work indoors or near windows. Within about 30 seconds you should see 10 or more satellites in view (SIV). More SIV is better. We regularly see 30 or more SIV. The horizontal positional accuracy (HPA) will start at around 10 meters and begin to decrease. The lower the HPA the more accurate your position.

Figure 5

Upon successful reception and decryption of L-Band corrections, the satellite dish icon will increase to a three-pronged icon (Figure 5). As the unit's accuracy increases a normal cross-hair will turn to a double blinking cross-hair indicating a floating RTK solution, and a solid double cross-hair will indicate a fixed RTK solution. The HPA will be below 0.030 (30mm) or better once RTK Fix is achieved.

You can now use your RTK device to measure points with millimeter accuracy. Please see Android or iOS for guidance on getting the RTK device connected to GIS software over Bluetooth.

2.7 RTK Reference Station

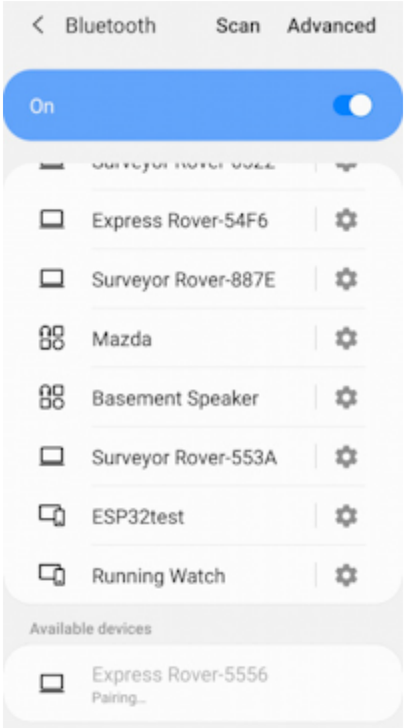
While most of this Quick Start guide can be used with the RTK Reference Station, the Reference Station Hookup Guide is the best place to get started.

3. Connecting Bluetooth

Surveyor: / Express: / Express Plus: / Facet: / Facet L-Band: / Reference Station:

SparkFun RTK products transmit full NMEA sentences over Bluetooth serial port profile (SPP) at 4Hz and 115200bps. This means that nearly any GIS application that can receive NMEA data over a serial port (almost all do) can be used with the RTK Express. As long as your device can open a serial port over Bluetooth (also known as SPP) your device can retrieve industry-standard NMEA positional data. The following steps show how to connect an external tablet, or cell phone to the RTK device so that any serial port-based GIS application can be used.

3.1 Android



Pairing with the 'Express Rover-5556' over Bluetooth

Open Android's system settings and find the 'Bluetooth' or 'Connected devices' options. Scan for devices and pair with the device in the list that matches the Bluetooth MAC address on your RTK device.

When powered on, the RTK product will broadcast itself as either '[Platform] Rover-5556' or '[Platform] Base-5556' depending on which state it is in. [Platform] is Facet, Express, Surveyor, etc. Discover and pair with this device from your phone or tablet. Once paired, open SW Maps.

Bluetooth MAC address B022 is shown in the upper left corner

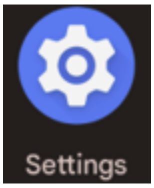
Note: *B022* is the last four digits of your unit's MAC address and will be unique to the device in front of you. This is helpful in case there are multiple RTK devices within Bluetooth range.

3.1.1 Enable Mock Location

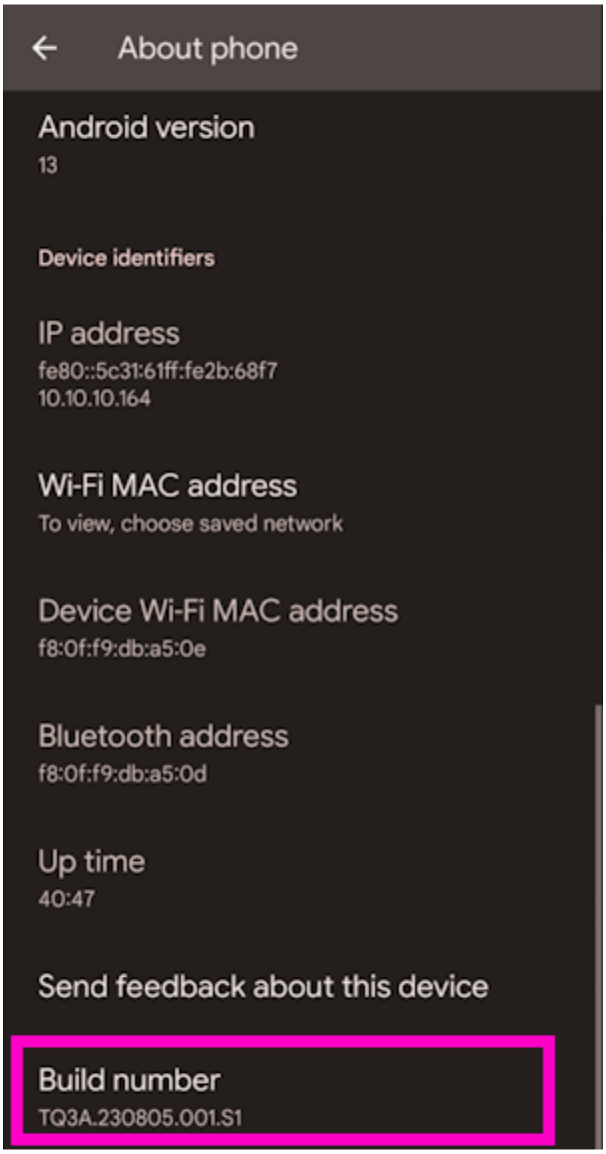
Most GIS applications will gracefully handle the Bluetooth connection to the RTK device and provide an NTRIP Client for getting the RTCM corrections so this section can be skipped. If, in the rare case, a GIS app does not allow NTRIP corrections, Mock Locations can be enabled under Android. Then a data provider like Lefebure or GNSS Master can be used to act as a middle-man.

Before proceeding, it is recommended to have the mock location provider app already installed. So if you haven't already, consider installing Lefebure, GNSS Master, etc.

To enable **Mock Locations**, *Developer Mode* in Android must be enabled. It is best to google the most recent procedure for this but the following procedure should work:

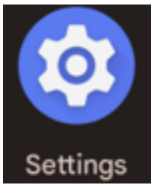
1) Open Android settings 

2) Open *About phone*

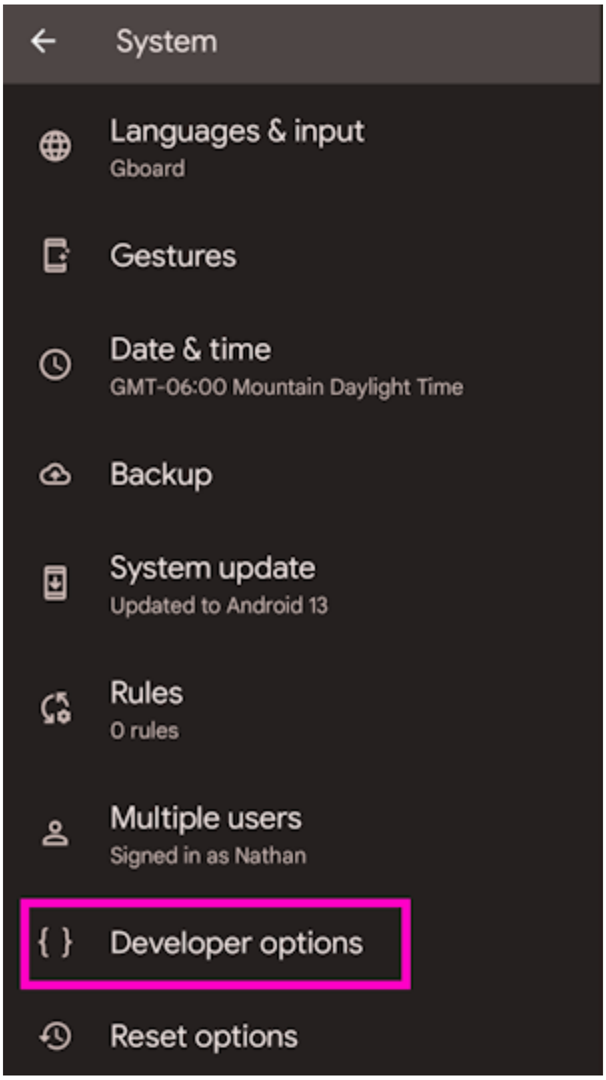


3) Scroll to the bottom and click on *Build number* five or more times. The device will prompt as more taps are required.

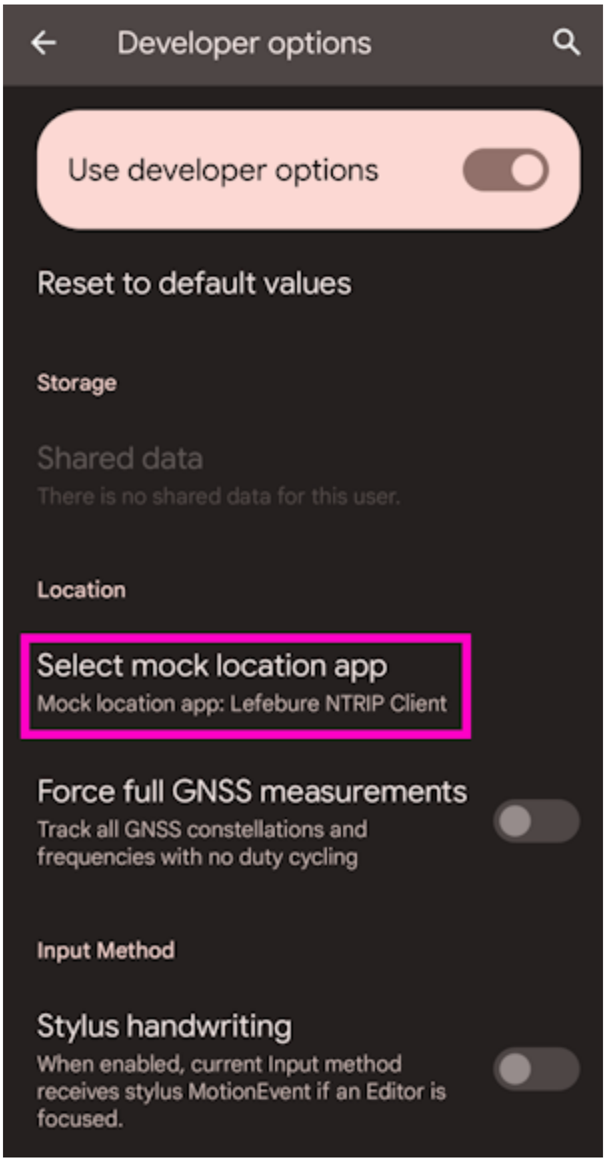
Once Developer Mode is enabled:

1) Open Android settings 

2) Open *System*



3) Open *Developer options*



4) Scroll all the way to the bottom of a very long list of developer options.

5) Select the app to use for Mock Location. This is usually Lefebure or GNSS Master but can be tailored as needed.

3.2 Apple iOS

Please see iOS GIS Software for information about how to connect to individual GIS apps. Some require a BLE connection and some require a WiFi hotspot connection.

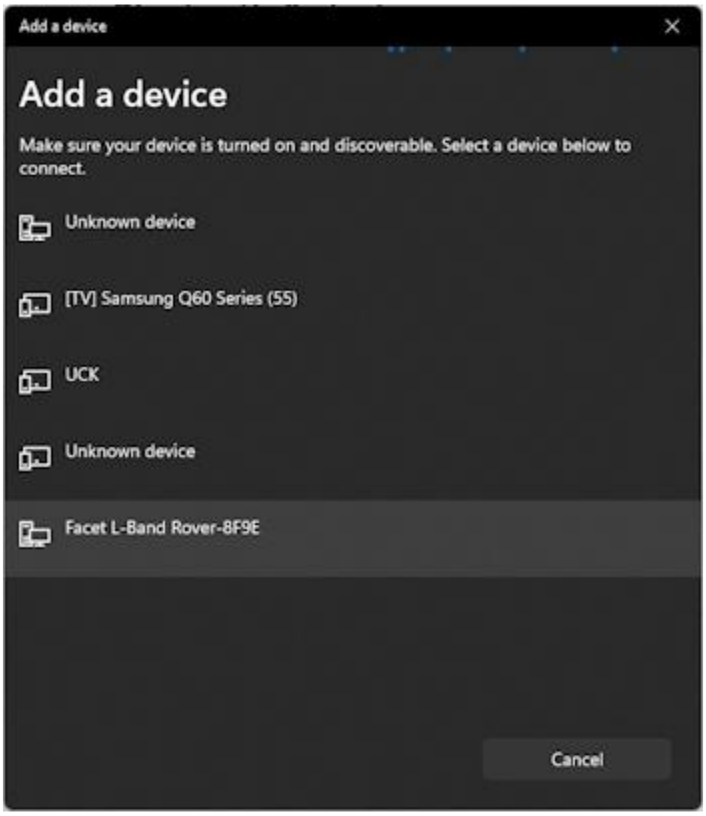
More information is available on the System Menu for switching between Bluetooth SPP and BLE.

3.3 Windows

Open settings and navigate to Bluetooth. Click **Add device**.

Adding Bluetooth Device

Click Bluetooth 'Mice, Keyboards, ...'



Viewing available Bluetooth Devices

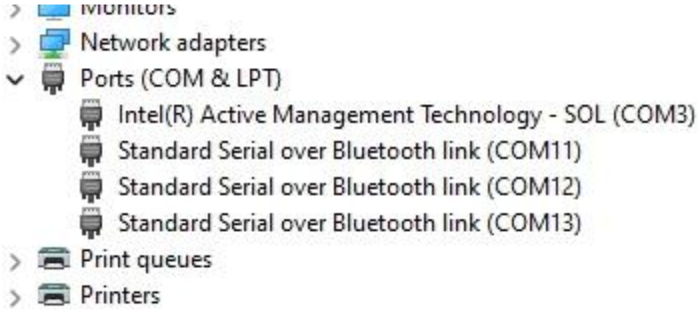
Click on the RTK device. When powered on, the RTK product will broadcast itself as either '[Platform] Rover-5556' or '[Platform] Base-5556' depending on which state it is in. [Platform] is Facet, Express, Surveyor, etc. Discover and pair with this device from your phone or tablet. Once paired, open SW Maps.

Bluetooth MAC address B022 is shown in the upper left corner

Note: *B022* is the last four digits of your unit's MAC address and will be unique to the device in front of you. This is helpful in case there are multiple RTK devices within Bluetooth range.

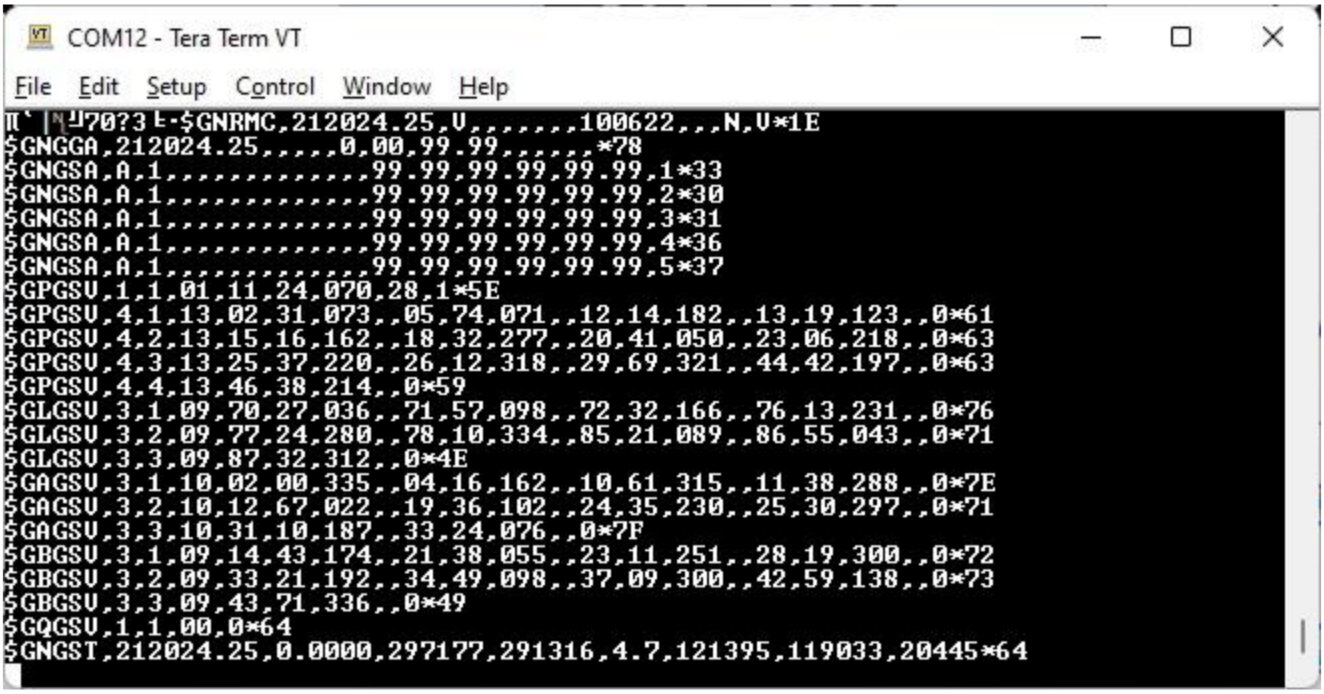
Bluetooth Connection Success

The device will begin pairing. After a few seconds, Windows should report that you are ready to go.



Bluetooth COM ports

The device is now paired and a series of COM ports will be added under 'Device Manager'.



NMEA received over the Bluetooth COM port

If necessary, you can open a terminal connection to one of the COM ports. Because the Bluetooth driver creates multiple COM ports, it's impossible to tell which is the serial stream so it's easiest to just try each port until you see a stream of NMEA sentences (shown above). You're all set! Be sure to close out the terminal window so that other software can use that COM port.

4. GIS Software

4.1 Android

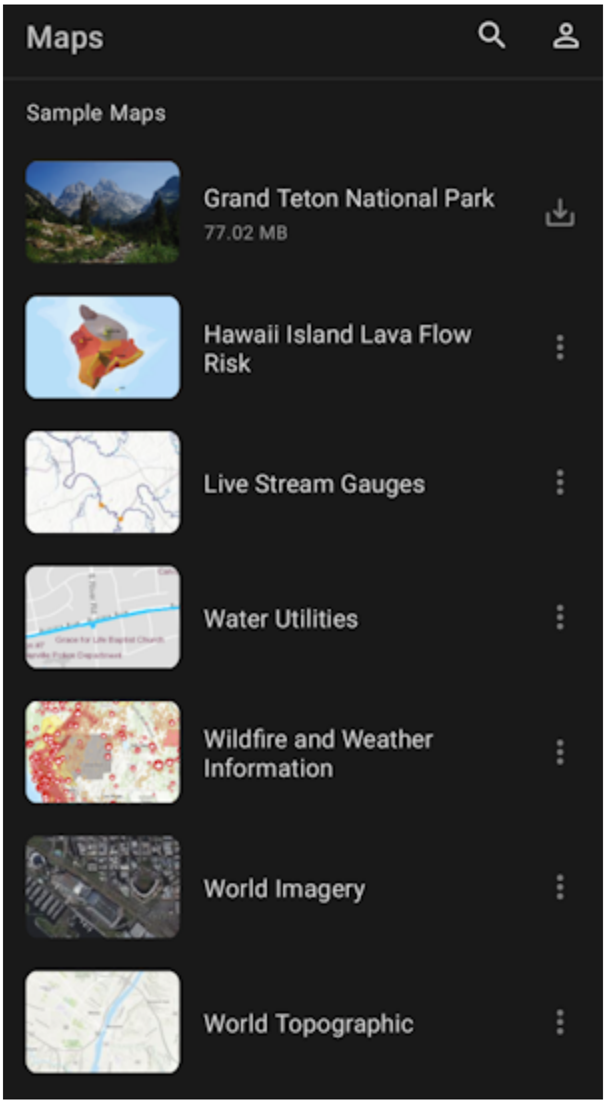
Surveyor: / Express: / Express Plus: / Facet: / Facet L-Band: / Reference Station:

While we recommend SW Maps for Android, there are a variety of 3rd party apps available for GIS and surveying for Android, iOS, and Windows. We will cover a few examples below that should give you an idea of how to get the incoming NMEA data into the software of your choice.

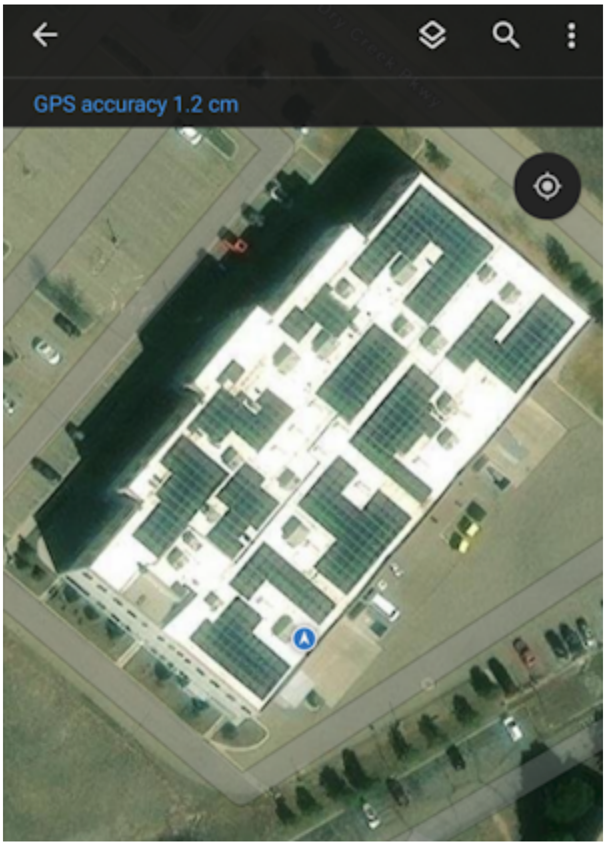
4.1.1 ArcGIS Field Maps

ArcGIS Field Maps by Esri is a popular GIS app. Unfortunately it does not have a built in NTRIP Client to allow high precision corrections down to the RTK device. To enable high-precision, a mock location and an intermediary app such as GNSS Master or Lefebure is needed.

Once a mock location provider is setup, open Field Maps.



Select **World Imagery**.



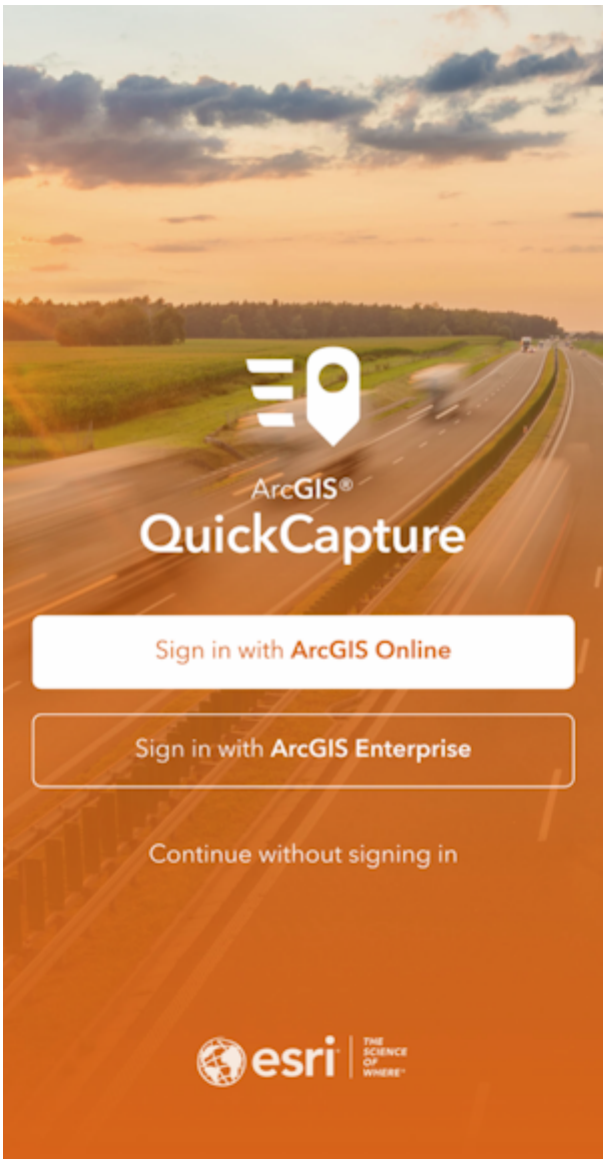
ArcGIS Field Maps with 12mm accuracy

Field Maps will use the device's internal location as its default location provider. With GNSS Master or Lefebure providing the mock location to the phone, Field Maps will have a super precise GNSS location and data collection can begin.

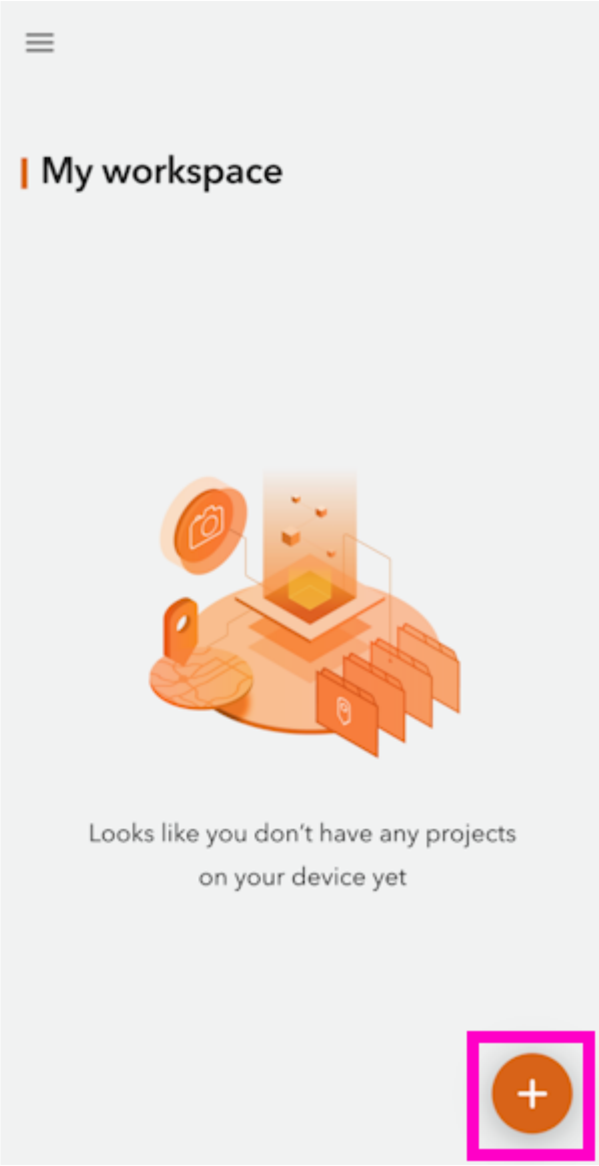
4.1.2 ArcGIS QuickCapture

ArcGIS QuickCapture by Esri is a popular GIS app. Unfortunately it does not allow Bluetooth connections to 3rd party RTK devices. To enable a connection to a SparkFun RTK device, a mock location and an intermediary app such as GNSS Master or Lefebure is needed.

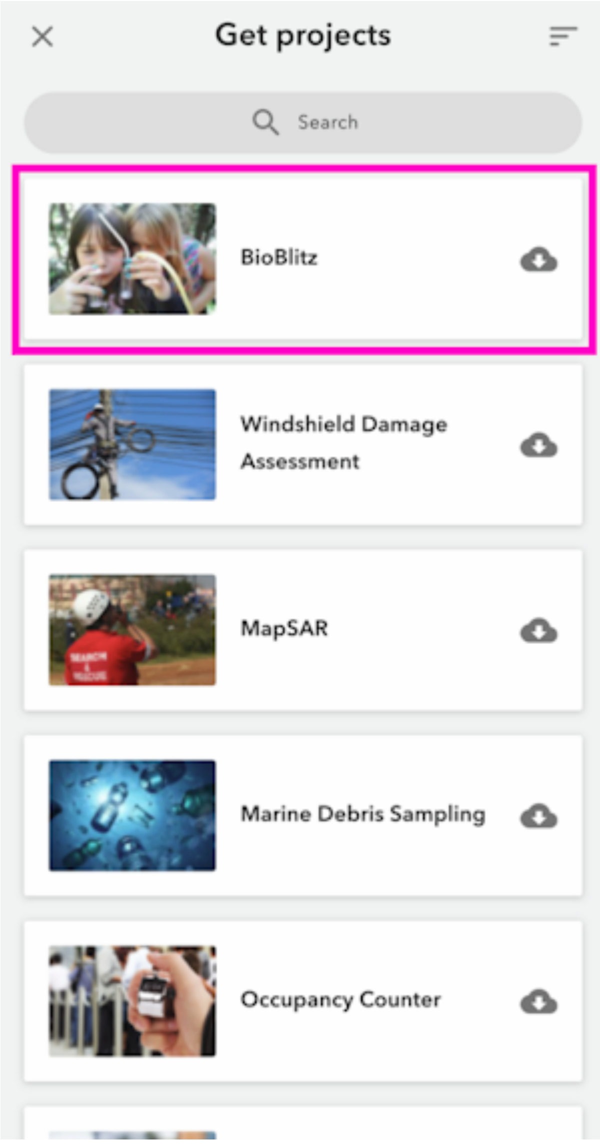
Once a mock location provider is setup, open QuickCapture.



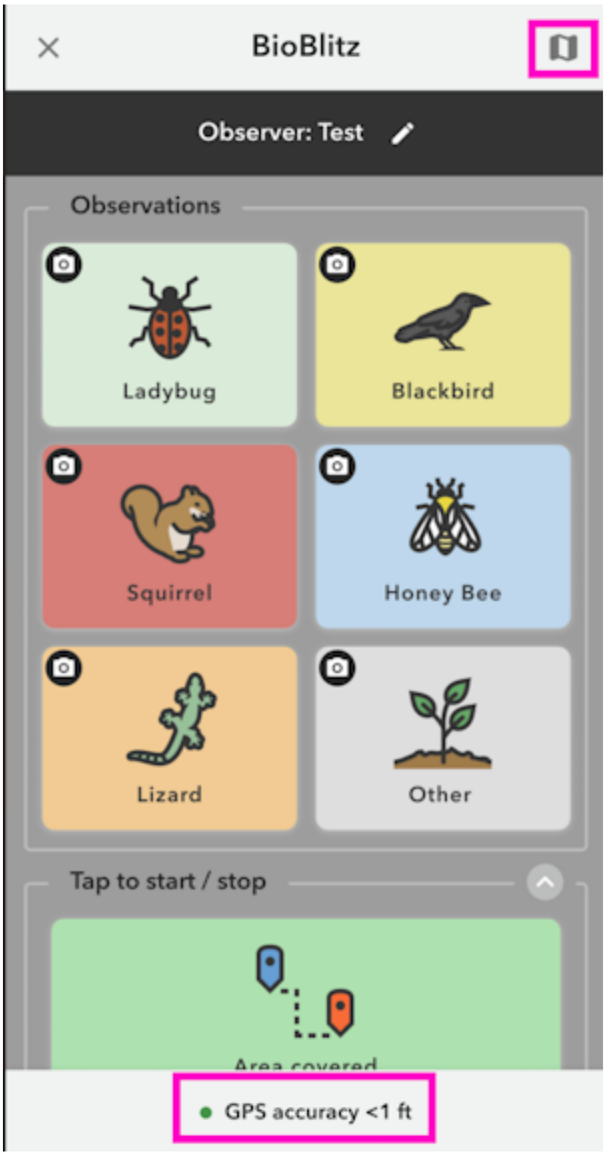
For the purposes of this demonstration, click *Continue without signing in*.



Select the **+** then **Browse Projects**.

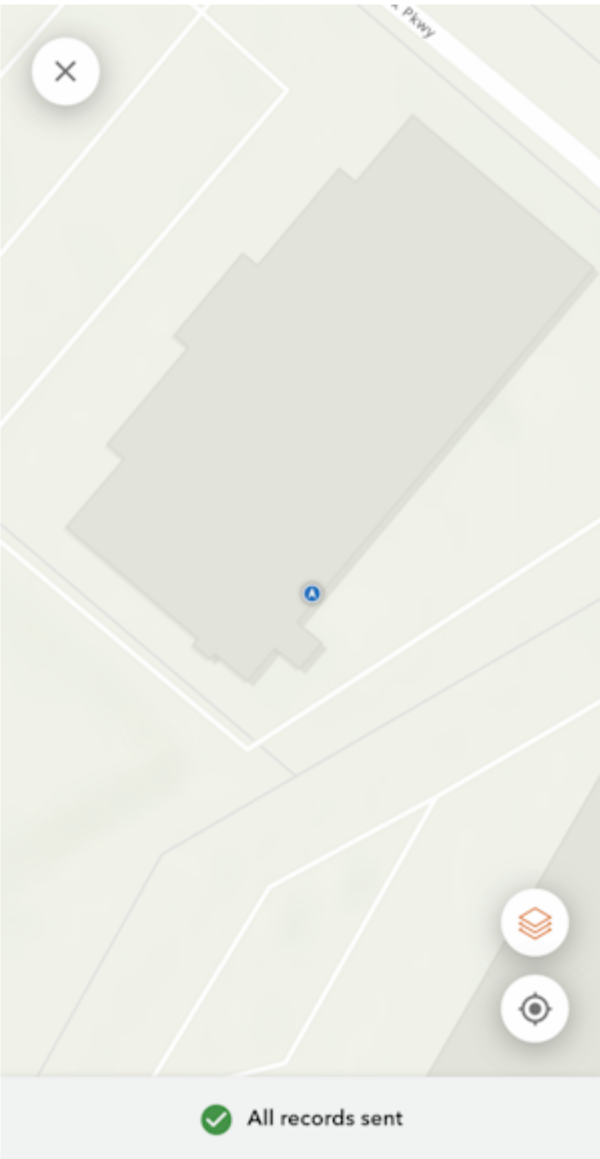


Select a project.



From the BioBlitz project screen we can see we have a GPS accuracy of less than 1 ft. The RTK device has RTK fix and is providing extremely accurate (better than 20mm or 1") positional data.

Click the map icon in the upper right.



The location of the receiver is shown on a map. With GNSS Master or Lefebure providing the mock location to the phone, QuickCapture will have a very precise GNSS location and data collection can begin.

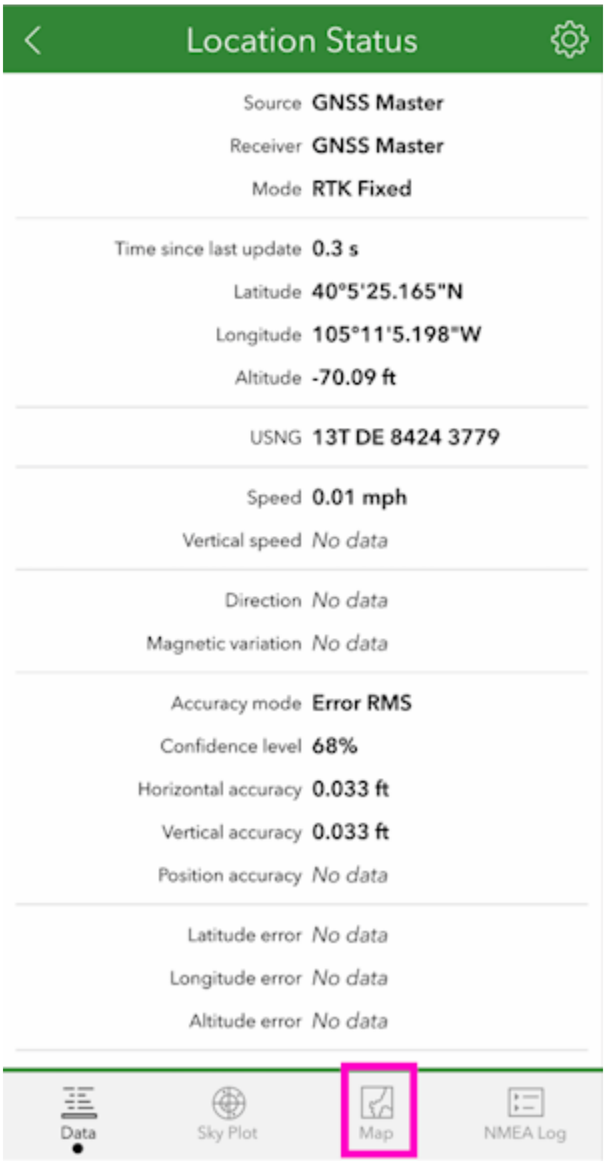
4.1.3 ArcGIS Survey123

ArcGIS Survey123 by Esri is a popular GIS app. Unfortunately it does not allow Bluetooth connections to 3rd party RTK devices. To enable a connection to a SparkFun RTK device, a mock location and an intermediary app such as GNSS Master or Lefebure is needed.

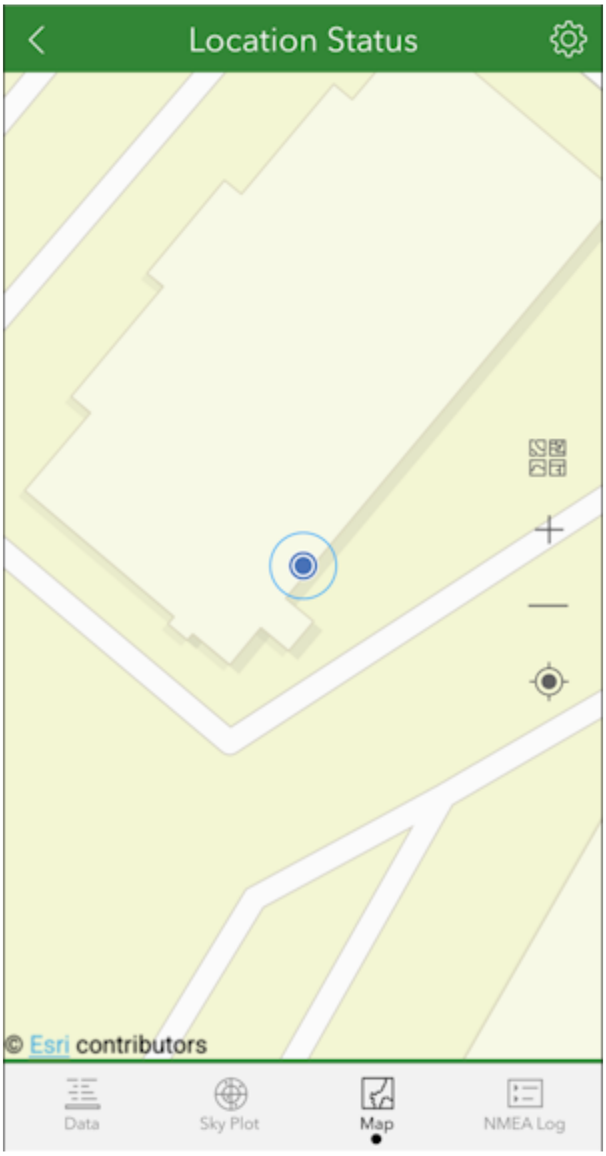
Once a mock location provider is setup, open Survey123.

For the purposes of this demonstration, click *Continue without signing in*.

Select the satellite icon in the upper right corner.



If the mock location provider app is running, you should see the Lat/Lon/Alt from the RTK device. In the above image, RTK Fix is achieved with 0.033ft (10mm) accuracy. Click on the map icon.

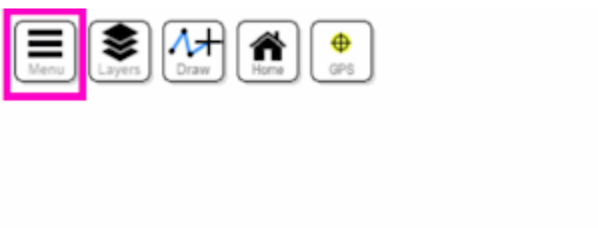


The location of the receiver is shown on a map. With GNSS Master or Lefebure providing the mock location to the phone, Survey123 will have a very precise GNSS location and data collection can begin.

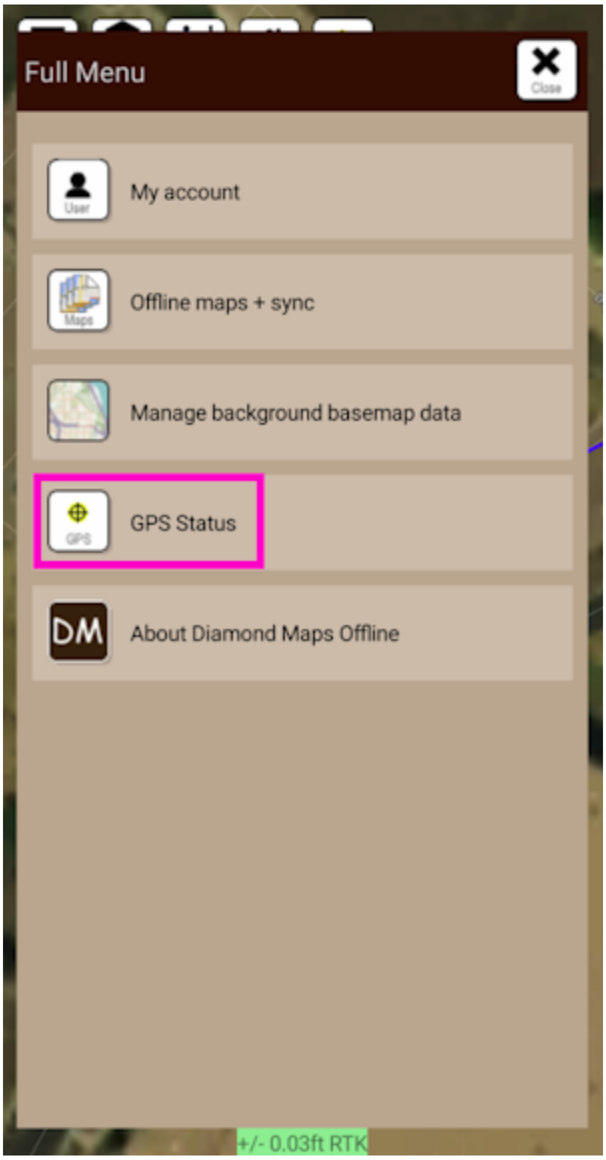
4.1.4 Diamond Maps

Diamond Maps is a great solution for utilities and municipalities. $20/month GIS software with many great features. Get the Android app here.

Be sure your device is paired over Bluetooth.

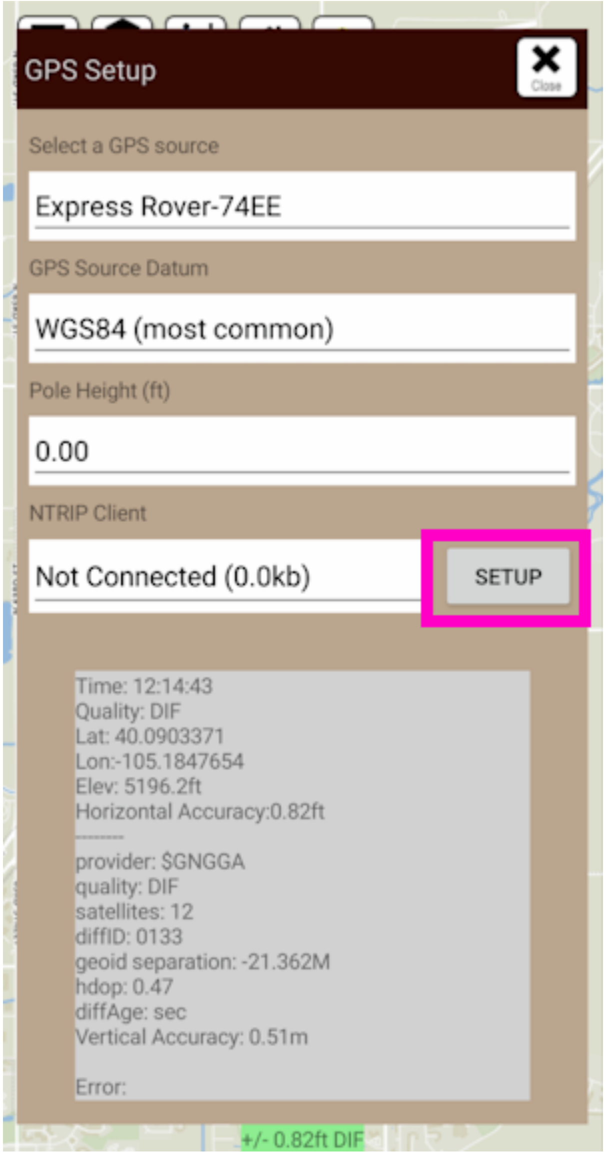


From the Home Screen, click on the 'hamburger' settings button in the top left corner.



Select **GPS Status**.

Click on the **Select a GPS Source** box and select the RTK device that was previously paired with.



Once a receiver is selected, its status will be shown in the GPS Setup window. Additionally, an NTRIP Client is available for corrections.

NTRIP Client

If you’re using a serial radio to connect a Base to a Rover for your correction data, or if you're using the RTK Facet L-Band with built-in corrections, you can skip this part.

From this window, an NTRIP Client can be configured. Enter your NTRIP Caster information then click on **START**. Click *Close* to exit out to the main window.

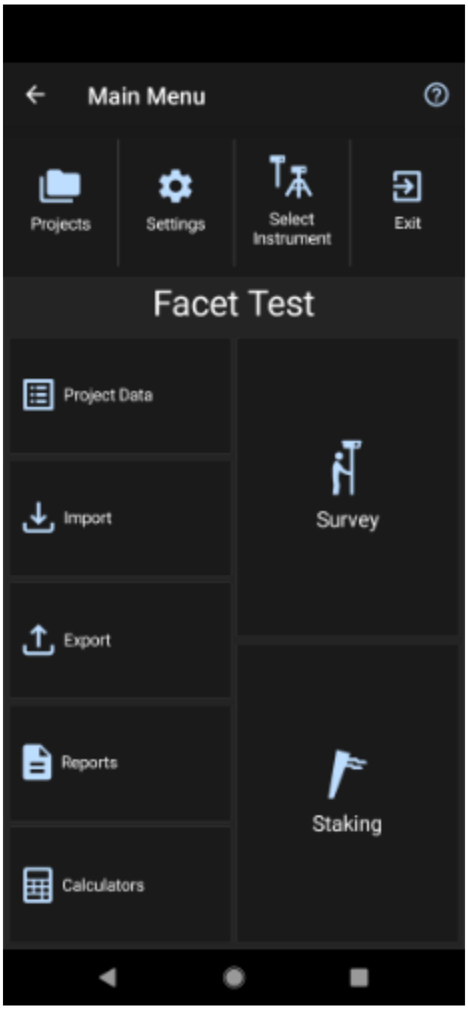
0.03ft accuracy shown in green

Closing the GPS Source window will show the map as well as the relative accuracy in feet.

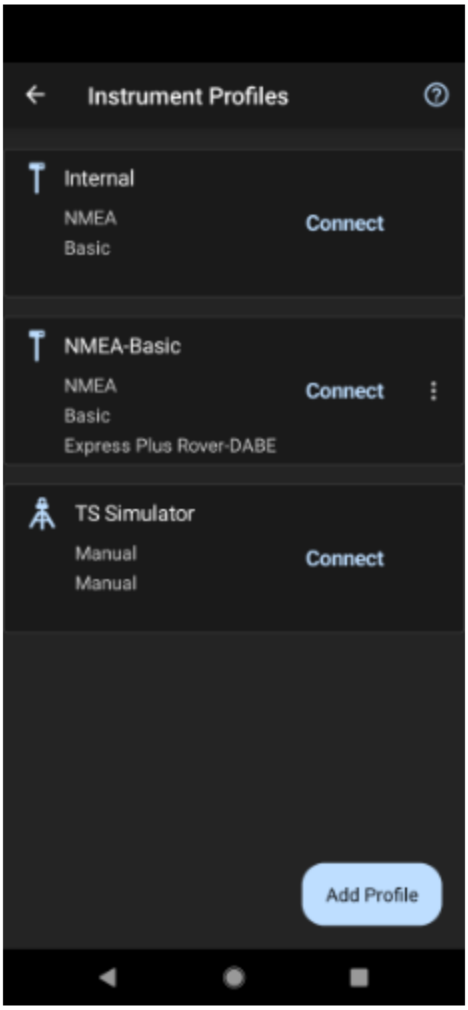
4.1.5 Field Genius

Field Genius for Android is another good solution, albeit a lot more expensive than free.

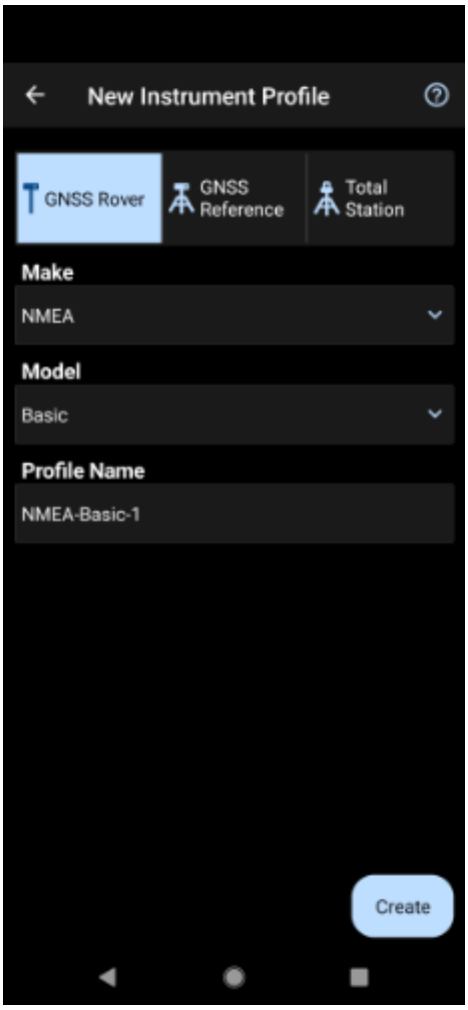
Be sure your device is paired over Bluetooth.



From the Main Menu open `Select Instrument`.

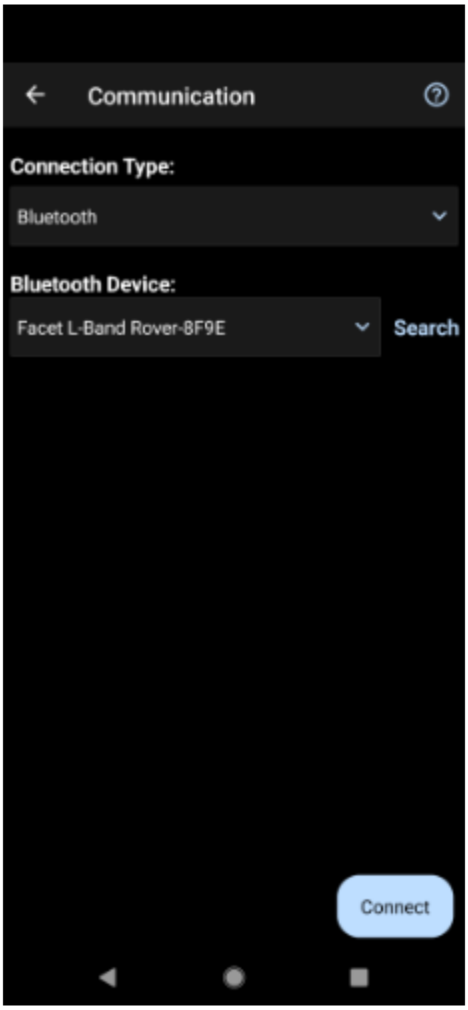


Click the 'Add Profile' button.

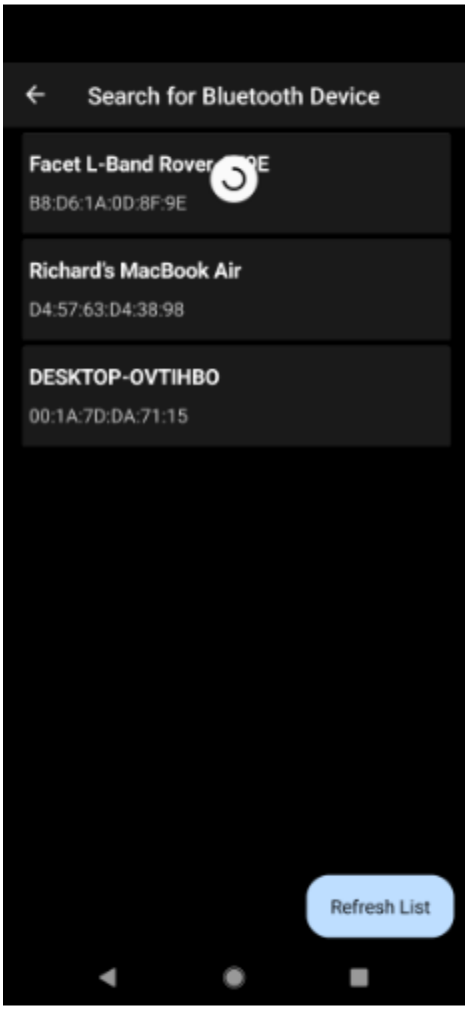


Click `GNSS Rover` and select *NMEA* as the Make. Set your Profile Name to something memorable like 'RTK-Express' then click the 'Create' button.

Click on 'SET UP COMMUNICATION'.



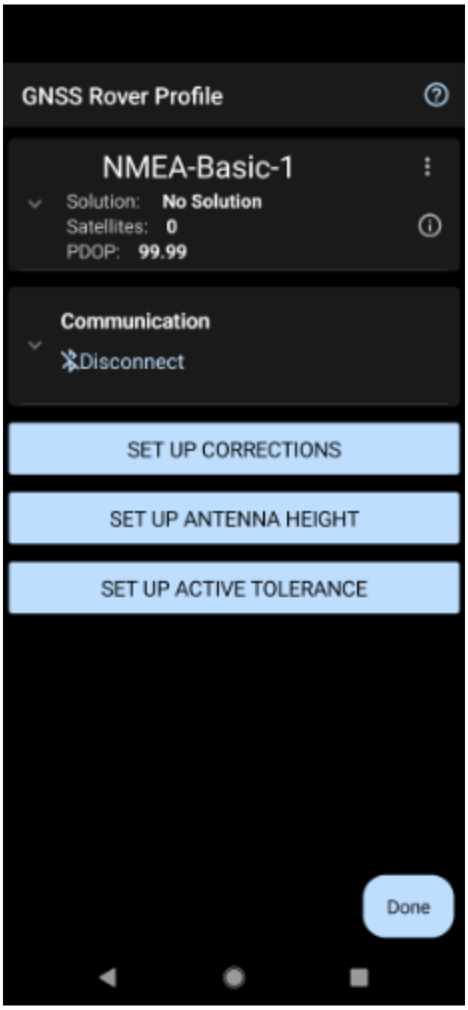
From the Bluetooth communication page, click the 'Search' button.



You will be shown a list of paired devices. Select the RTK device you'd like to connect to then click 'Connect'. The RTK device will connect and the MAC address shown on the RTK device OLED will change to the Bluetooth icon indicating a link is open.

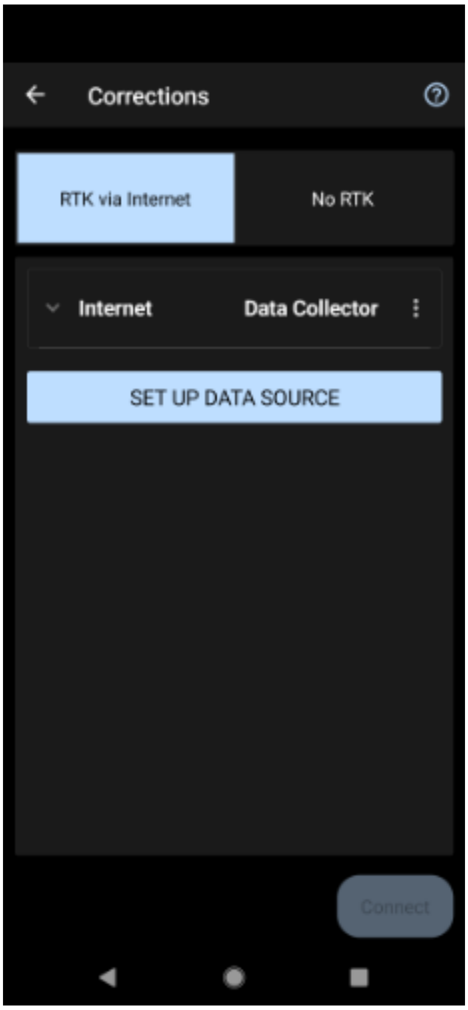
NTRIP Client

If you’re using a serial radio to connect a Base to a Rover for your correction data, or if you're using the RTK Facet L-Band with built-in corrections, you can skip this part.



We need to send RTCM correction data from the phone back to the RTK device so that it can improve its fix accuracy. Your phone can be the radio link! Click on 'SET UP CORRECTIONS'.

Click on 'RTK via Internet' then 'SET UP INTERNET', then 'Done'.



Click on 'SET UP DATA SOURCE'.

Click 'Add New Source'.

Enter your NTRIP Caster credentials and click 'DONE'.

What's an NTRIP Caster? In a nutshell, it's a server that is sending out correction data every second. There are thousands of sites around the globe that calculate the perturbations in the ionosphere and troposphere that decrease the accuracy of GNSS accuracy. Once the inaccuracies are known, correction values are encoded into data packets in the RTCM format. You, the user, don't need to know how to decode or deal with RTCM, you simply need to get RTCM from a source within 10km of your location into the RTK device. The NTRIP client logs into the server (also known as the NTRIP caster) and grabs that data, every second, and sends it over Bluetooth to the RTK device.

Don't have access to an NTRIP Caster? You can use a 2nd RTK product operating in Base mode to provide the correction data. Check out Creating a Permanent Base. If you're the DIY sort, you can create your own low-cost base station using an ESP32 and a ZED-F9P breakout board. Check out How to Build a DIY GNSS Reference Station. If you'd just like a service, Syklark provides RTCM coverage for $49 a month (as of writing) and is extremely easy to set up and use. Remember, you can always use a 2nd RTK device in *Base* mode to provide RTCM correction data but it will be less accurate than a fixed position caster.

Click 'My NTRIP1' then 'Done' and 'Connect'.

You will then be presented with a list of Mount Points. Select the mount point you'd like to use then click 'Select' then 'Confirm'.

Select 'Done' then from the main menu select 'Survey' to begin using the device.

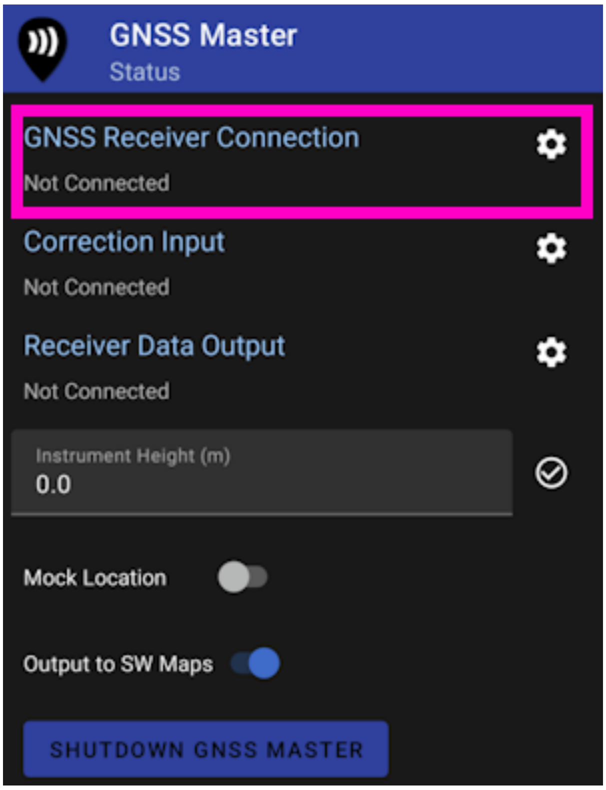
Now you can begin using the SparkFun RTK device with Field Genius.

4.1.6 GNSS Master

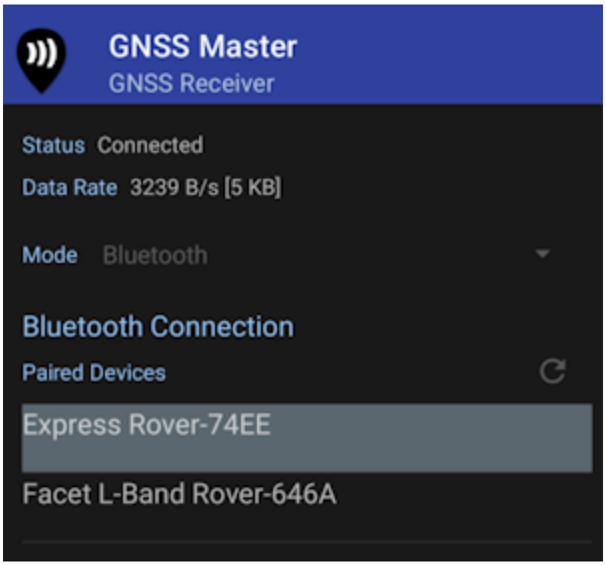
GNSS Master is a great utility when a given GIS app does not have an NTRIP Client or a way to connect over Bluetooth. GNSS Master connects to a RTK device over Bluetooth (or Bluetooth BLE) as well as any correction source (NTRIP, PointPerfect, even USB Serial), and then acts as the phone's location using Mock Location.

Note: Most GIS apps will not need GNSS Master or Mock Location enabled and this section can be skipped.

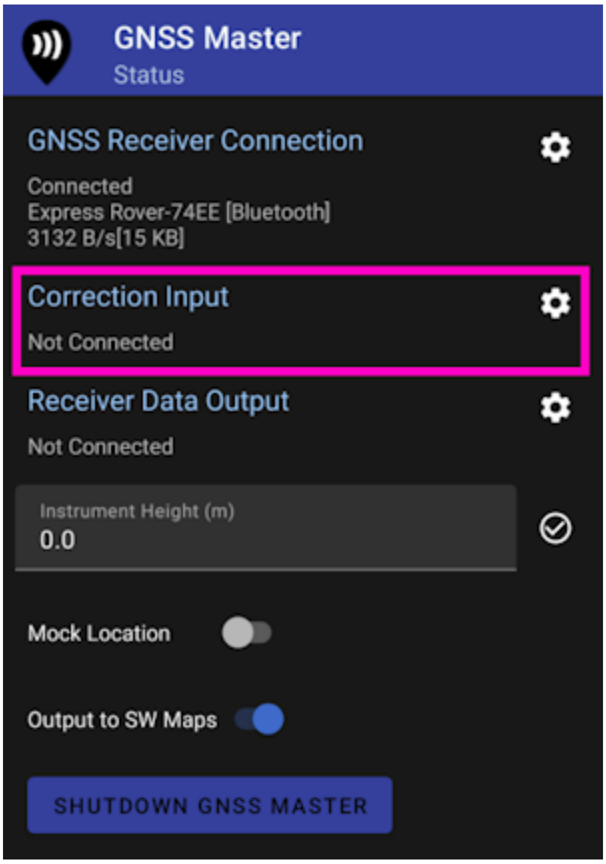
Read how to Enable Mock Location.



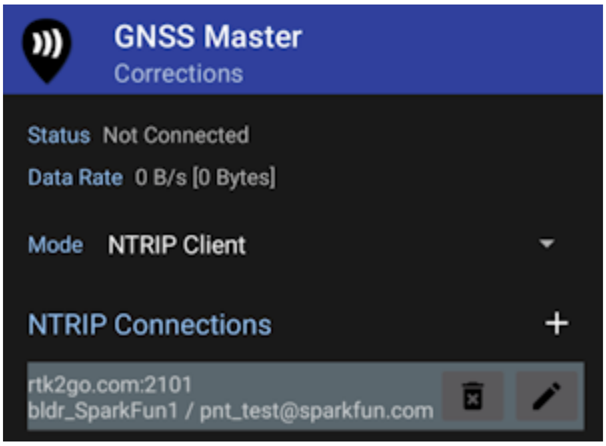
From the GNSS Master main screen, select **GNSS Receiver Connection**.



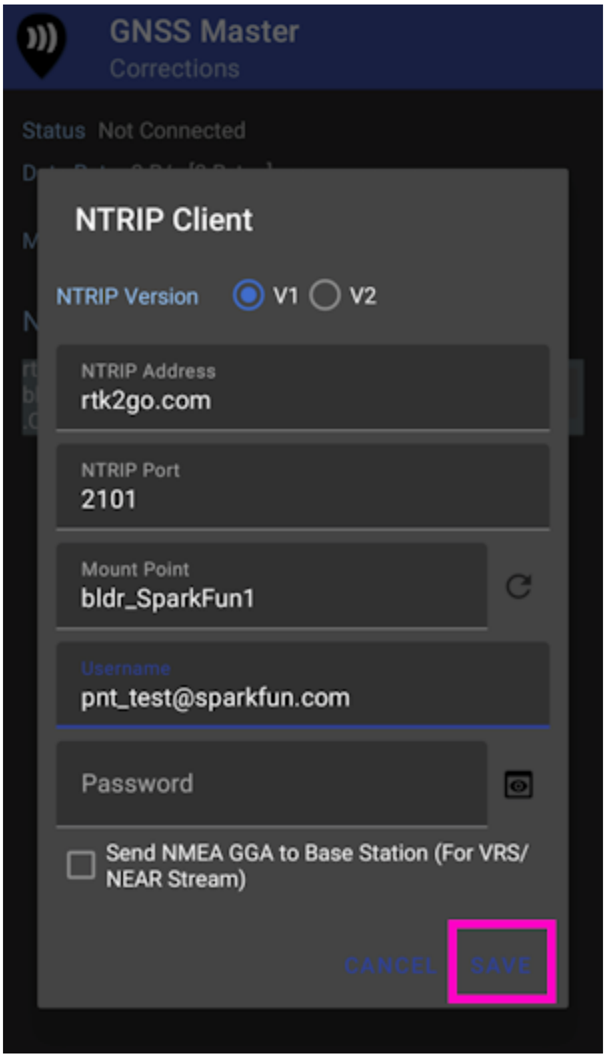
Pick the RTK device to connect to from the list, then click *Connect*. The **Data Rate** should increase indicating data flowing from the RTK device to the GNSS Master app. Click the back button to return to the main screen.



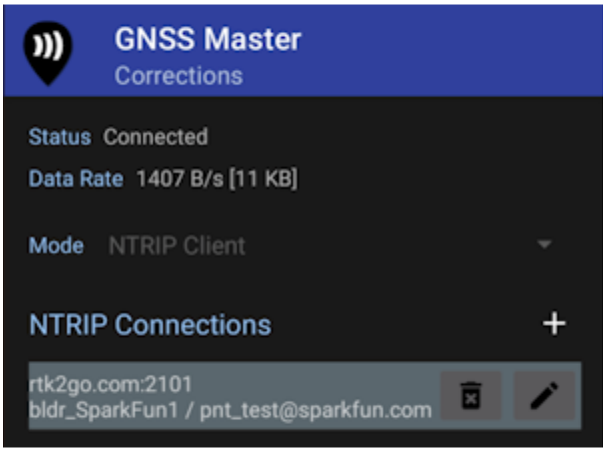
Select **Correction Input** to setup an NTRIP Client.



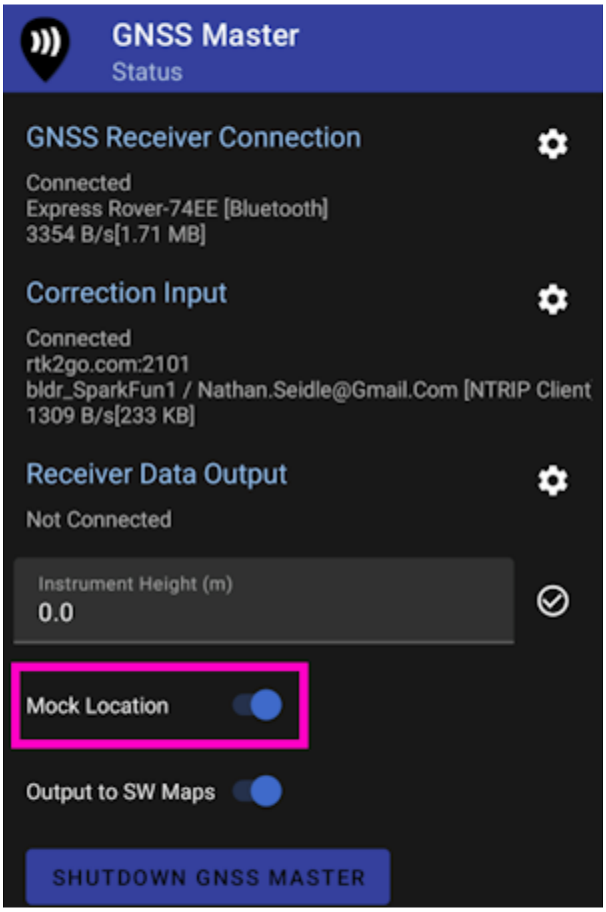
This is one of the powerful features of GNSS Master - multiple connections can be entered. This is helpful if you regularly switch between locations or NTRIP Casters and your GIS software only allows entry of a single NTRIP source. GNSS Master supports corrections from NTRIP Casters but also PointPerfect and a direct serial connection to a GNSS receiver. This can be really helpful in advanced setups.



Enter your NTRIP Client information then click **SAVE**.



Once connected the *Data Rate* should increase above 0 bytes per second. Return to the home screen by hitting the back button.



Enable mock location. If GNSS Master throws an error, re-enable GNSS Master as your Mock Location provider in Developer Options.

Once enabled, any GIS app that selects 'Internal' or 'Phone Location' as its source will instead be fed the high precision NMEA being generated by the RTK device connected over Bluetooth.

4.1.7 Lefebure

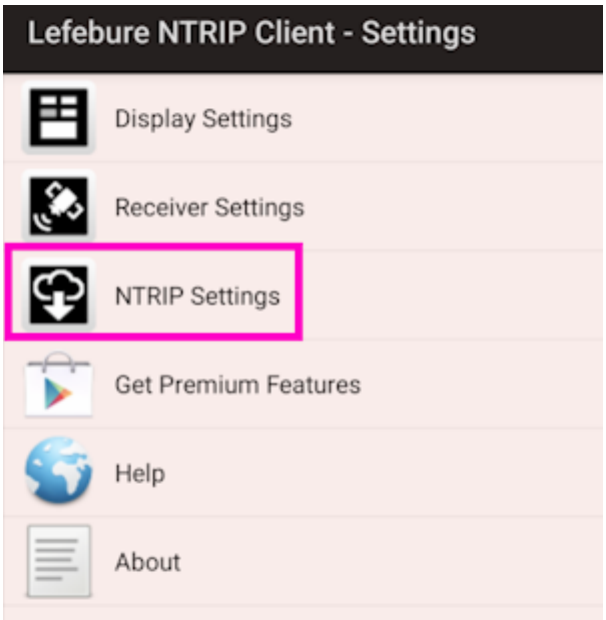
Lefebure NTRIP Client is the *original* app for getting correction from an NTRIP caster and down over Bluetooth. It's an oldie but a goodie.

Note: Most GIS apps will not need Lefebure or Mock Location enabled and this section can be skipped.

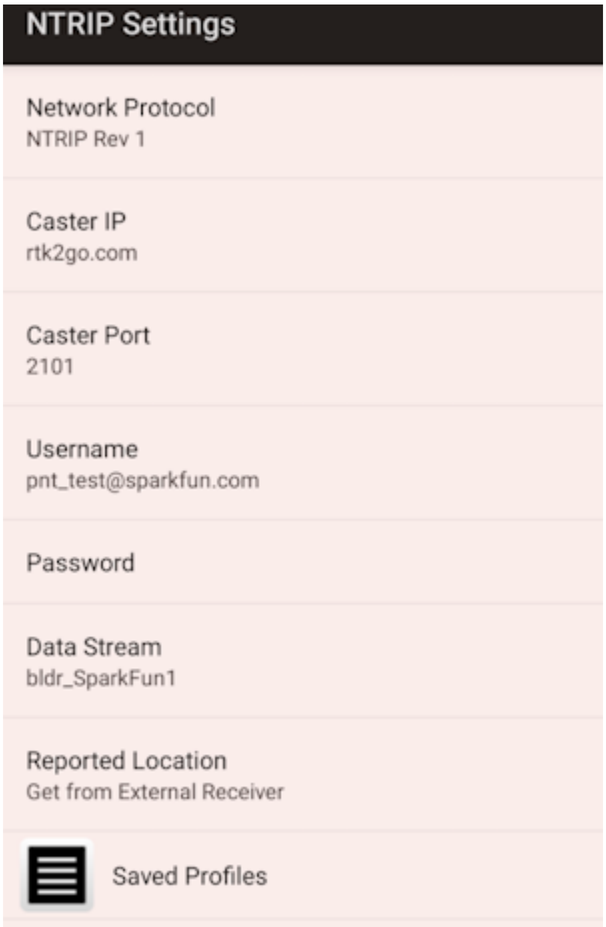
The problem is that if Lefebure is connected to the RTK device providing RTCM corrections over Bluetooth, then other GIS applications cannot use the same Bluetooth connection at the same time. That's where mock locations save the day. Lefebure can be setup to take over or 'mock' the GPS location being reported by the phone. Nearly all GIS apps can use the phone's GPS location. So if the phone's location is magically super precise, then Lefebure can be the NTRIP Client and data provide, and your GIS app is none the wiser, and uses the phone's location.

Read how to Enable Mock Location.

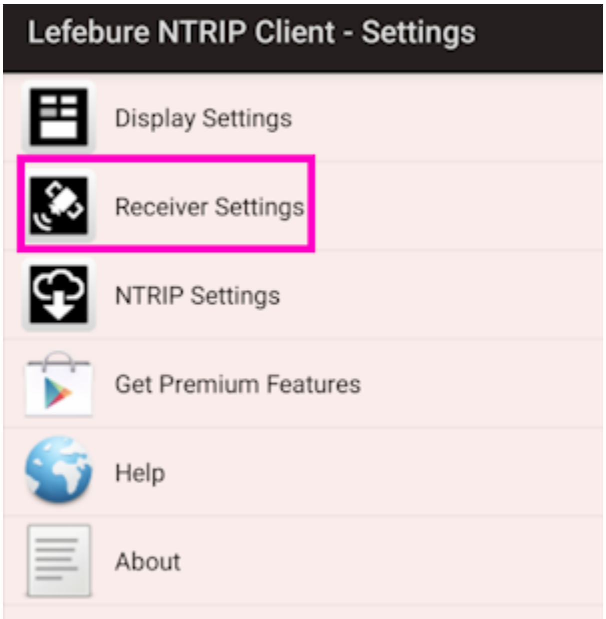
Once mock locations are enabled, click on the *Settings* gear in the top left corner.



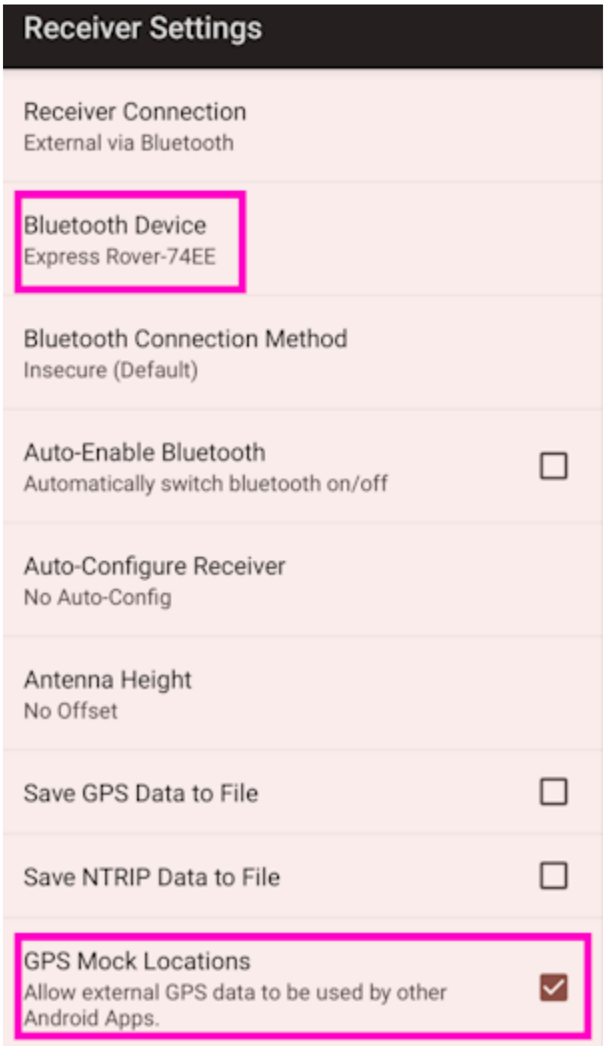
If needed, an NTRIP Client can be setup to provide corrections over Bluetooth to the RTK device.



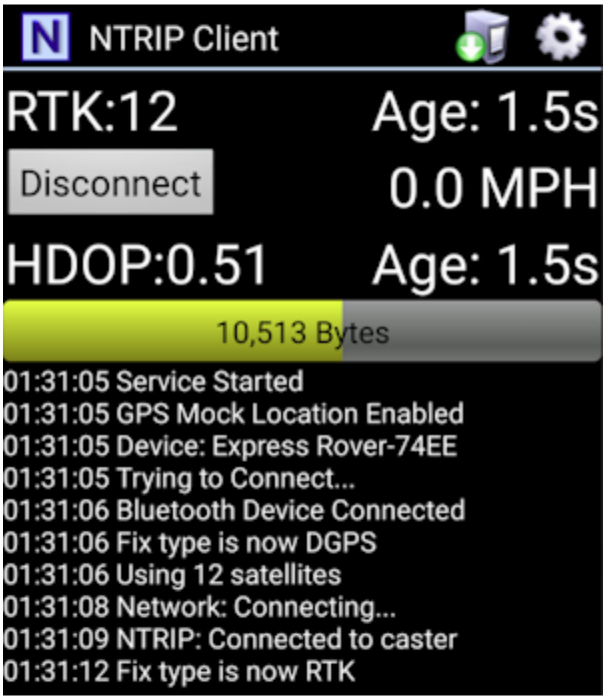
Enter the Caster information and hit the back button.



Select *Receiver Settings*.

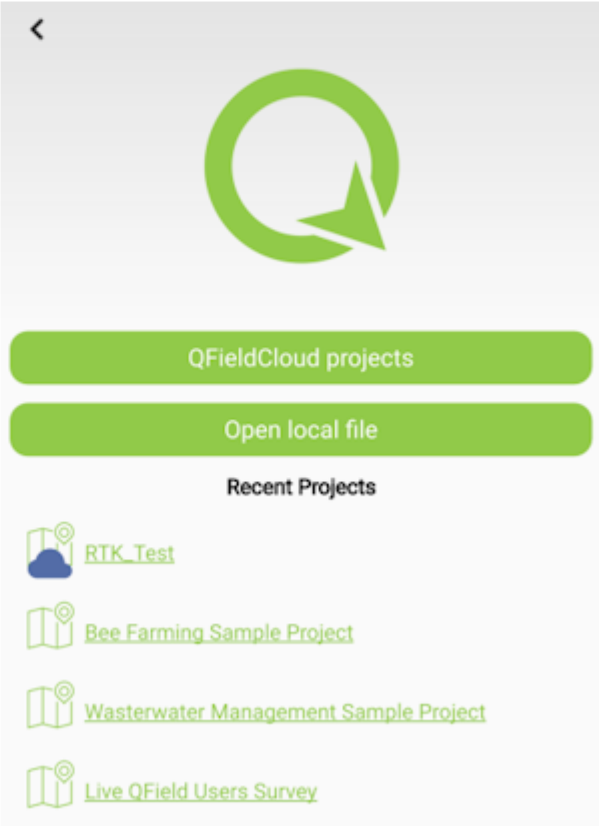


Select the RTK device that has been paired over Bluetooth. Also enable Mock Locations. Hit the back button to return to the main screen.

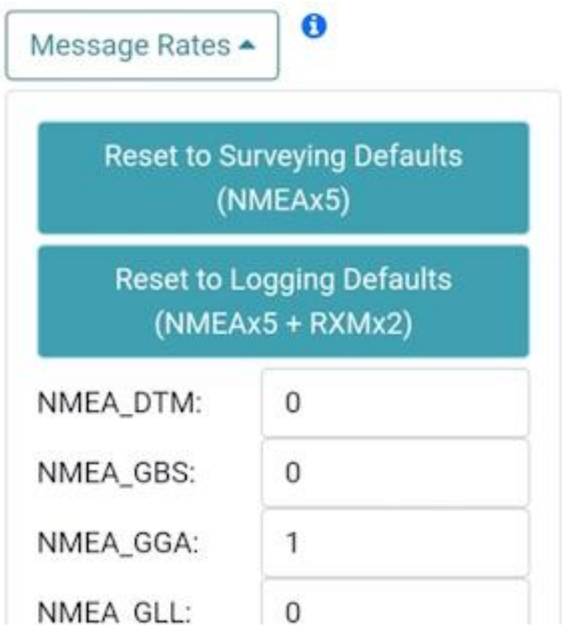


Press the **Connect** button. The app will connect to the NTRIP Caster. Now, any GIS app that selects 'Internal' or 'Phone Location' as its source will instead be fed the high precision NMEA being generated by the RTK device connected over Bluetooth.

4.1.8 QField



QField is a free GIS Android app that runs QGIS.

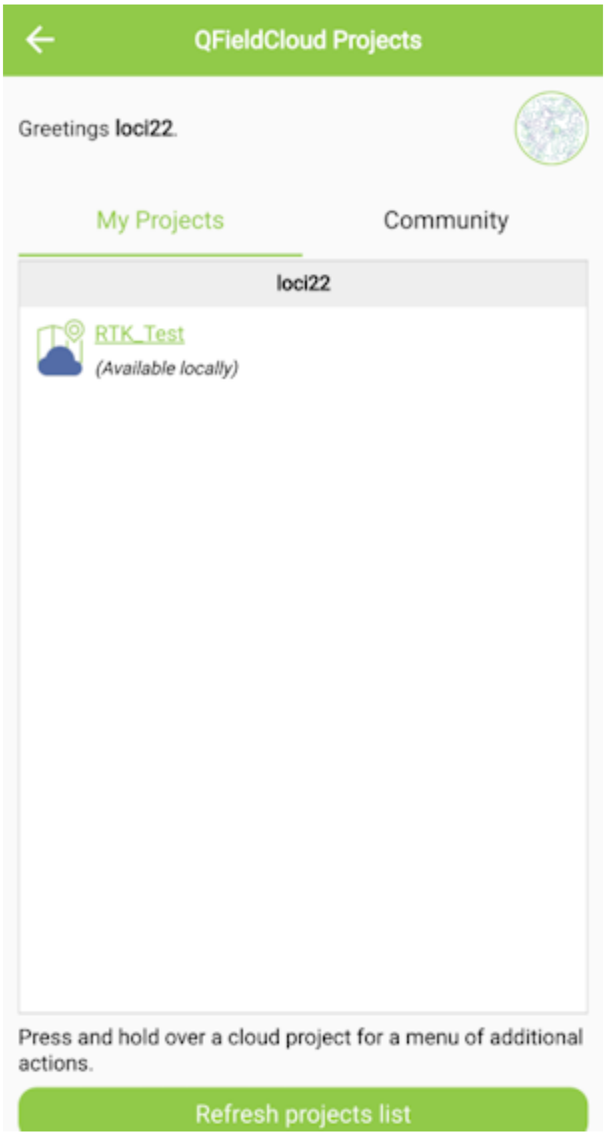


The 'Reset to Surveying Defaults' button

First, configure the RTK device to output *only* NMEA messages. QField currently does not correctly parse other messages such as RAWX or RTCM so these will interfere with communication if they are enabled.

These RTK device settings can be found under the Messages menu through the WiFi config page or through the Serial Config menu.

Create an account and project on QFieldCloud. This project will be synchronized and viewable on the QField app.

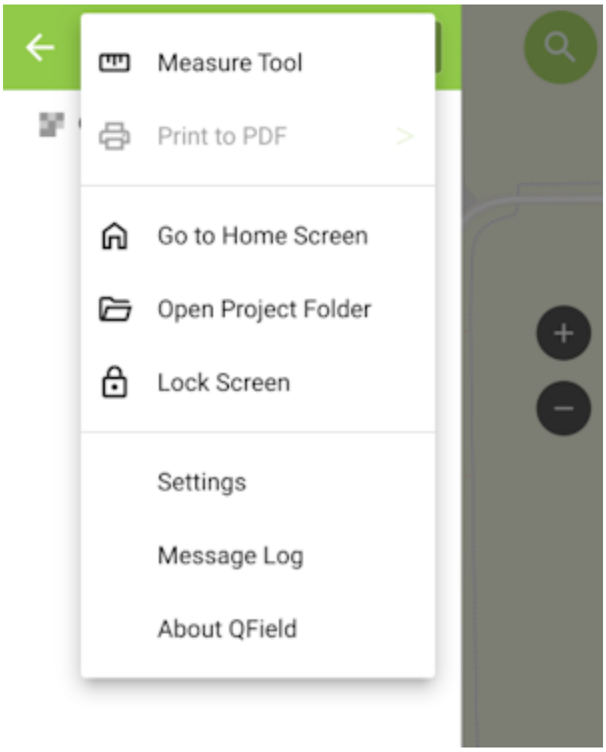


Refresh Projects button

Once the project is created, press the Refresh projects list button to update the list. Then select your project.

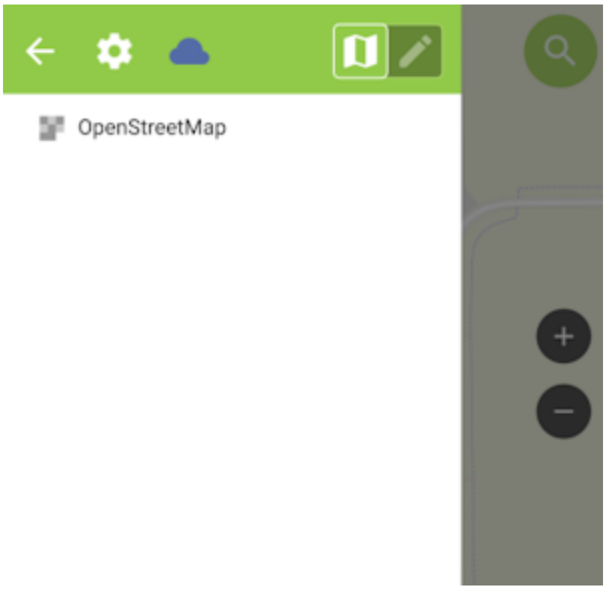
'*Hamburger*' *menu in upper right corner*

Press the icon in the top left corner of the app to open the project settings.



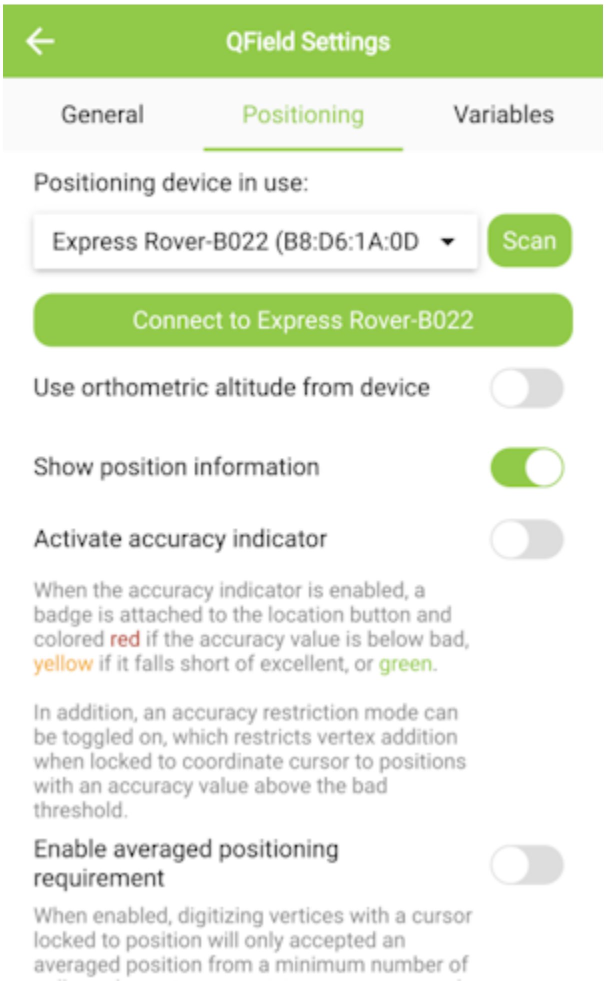
Project settings

From the project settings menu, press the gear icon to open the device settings dropdown menu.



Project settings submenu

From the submenu, select 'Settings'.



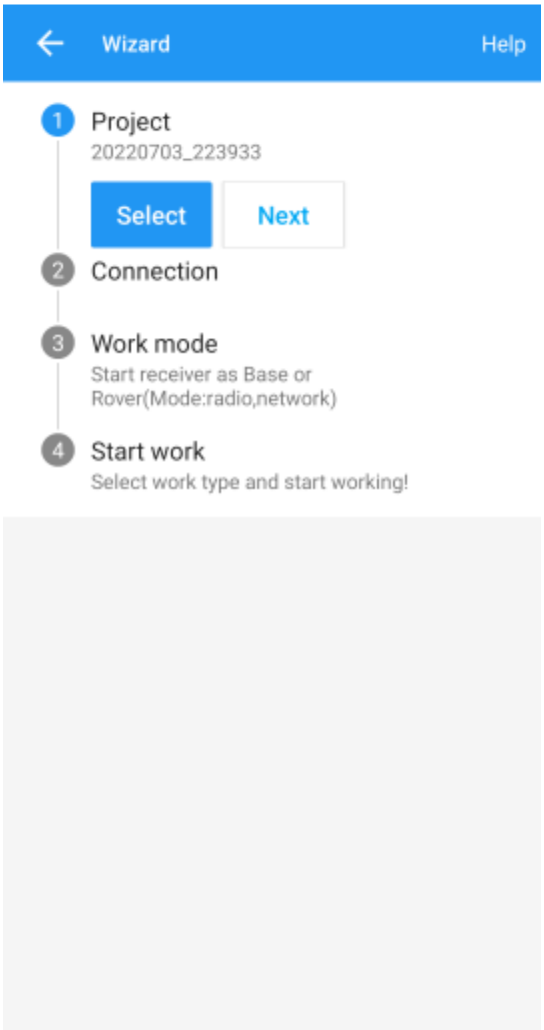
Positioning Menu

Select the Positioning Menu. Then, with your RTK device on and in normal mode (not AP Config) press the Scan button in the QField app to update the dropdown list of available Bluetooth devices. If your device is not detected, be sure you've paired your cellphone or laptop with Bluetooth.

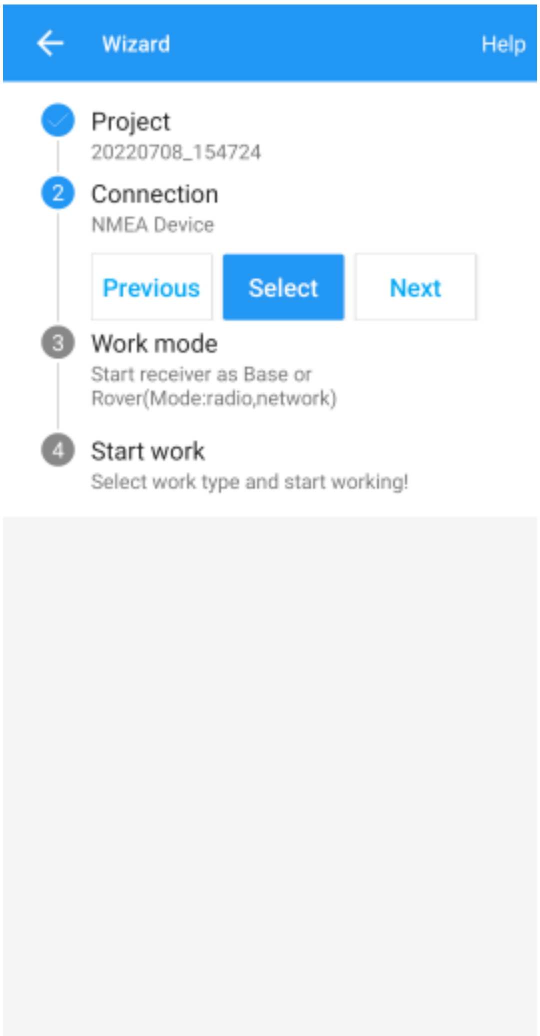
Once connected exit out of the menus and see position information within your project.

4.1.9 Survey Master

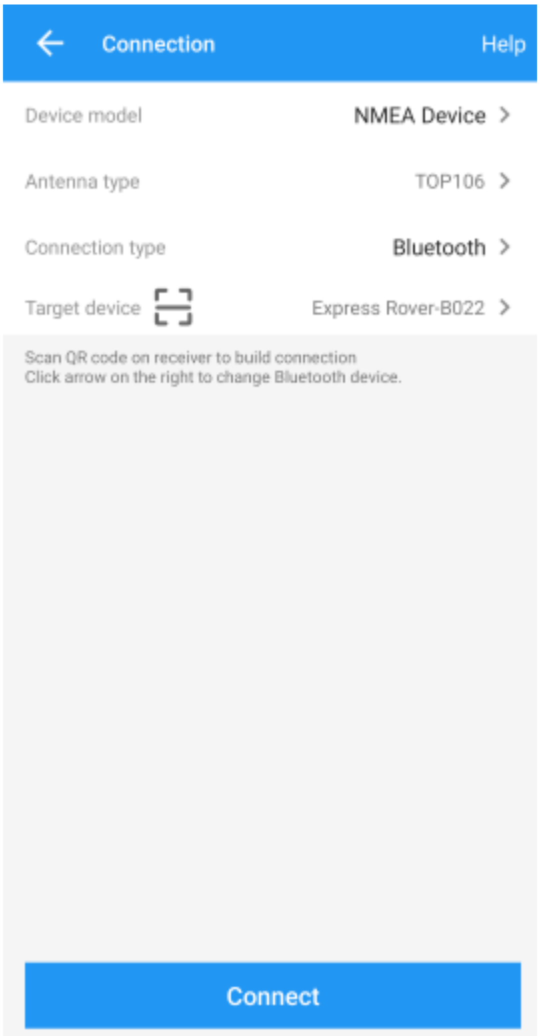
Survey Master by ComNam / SinoGNSS is an Android-based option. The download location can vary so google 'Survey Master ComNav Download' if the link above fails. Download the zip file, send the APK file to a phone and install the program.



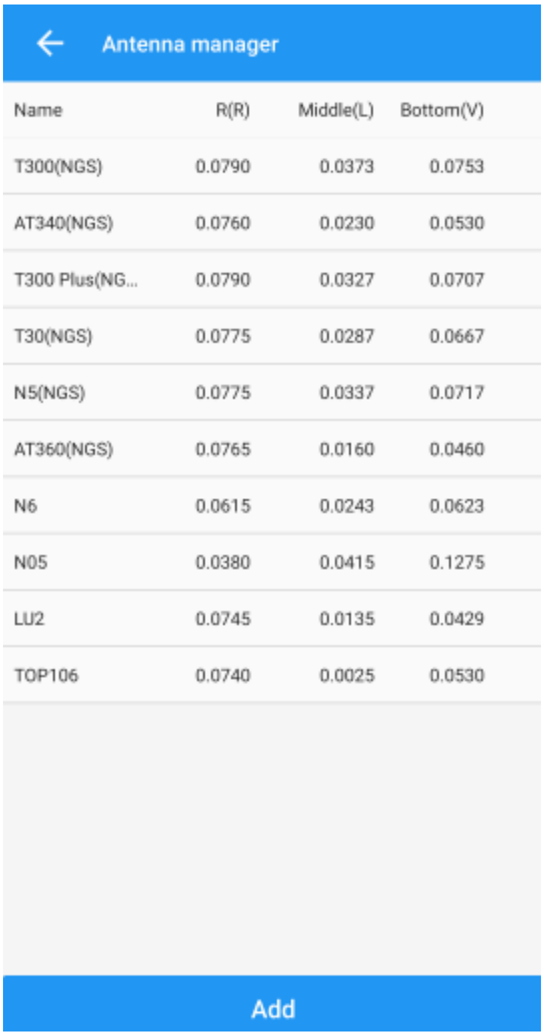
By default, a wizard will guide you through the setup. The Project step will ask you for the name of the project, the datum, etc.



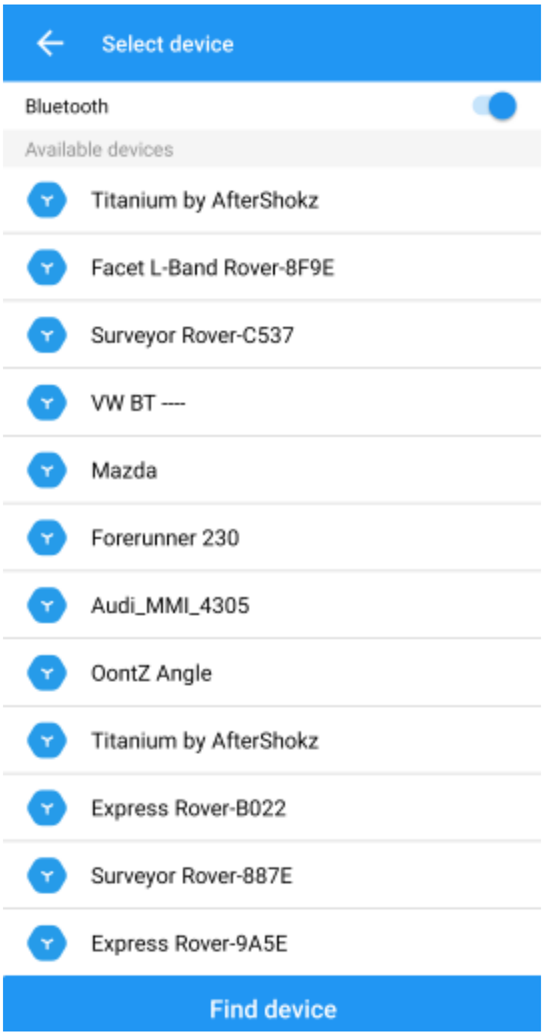
Next select your connection.



For the Device Model select 'NMEA Device'.

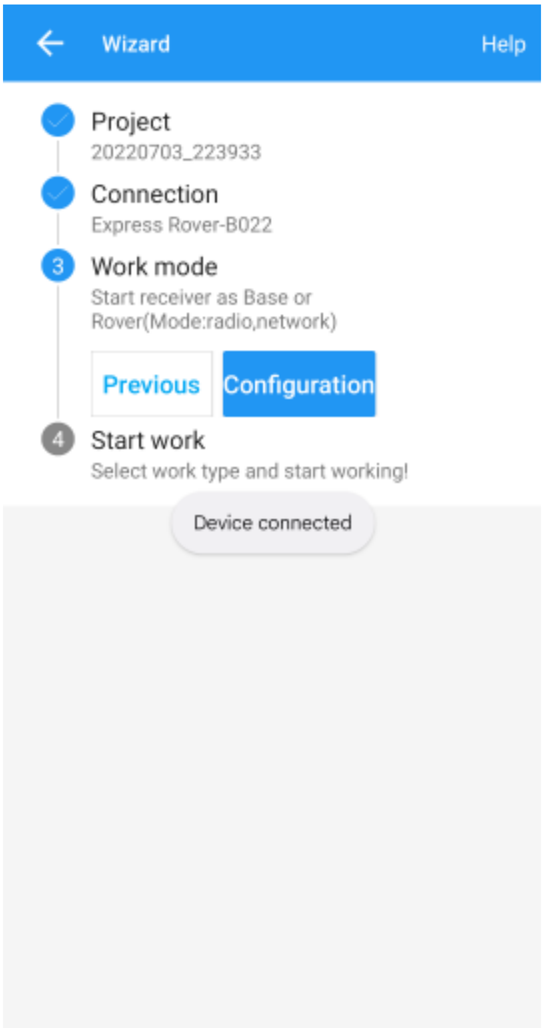


If you are just getting started, use one of the default antenna types. If you are attempting to get sub-centimeter accuracy, enter the parameters of your antenna and add it. Above are the NGS-certified parameters for the TOP106 antenna.

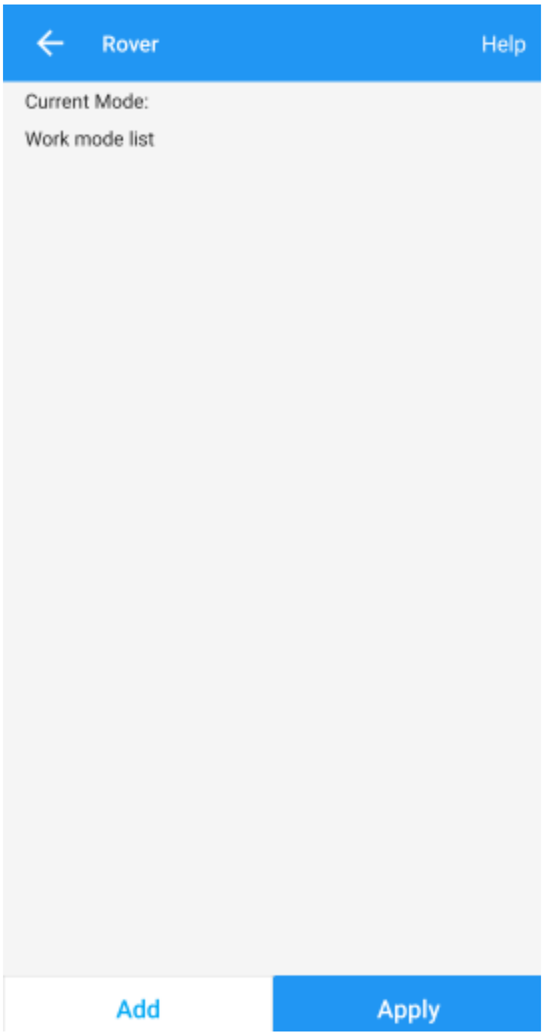


Click the 'Target Device' option to get a list of available Bluetooth devices. Make sure your RTK product is on and you should see the device. In this example 'Express Rover-B022' was chosen.

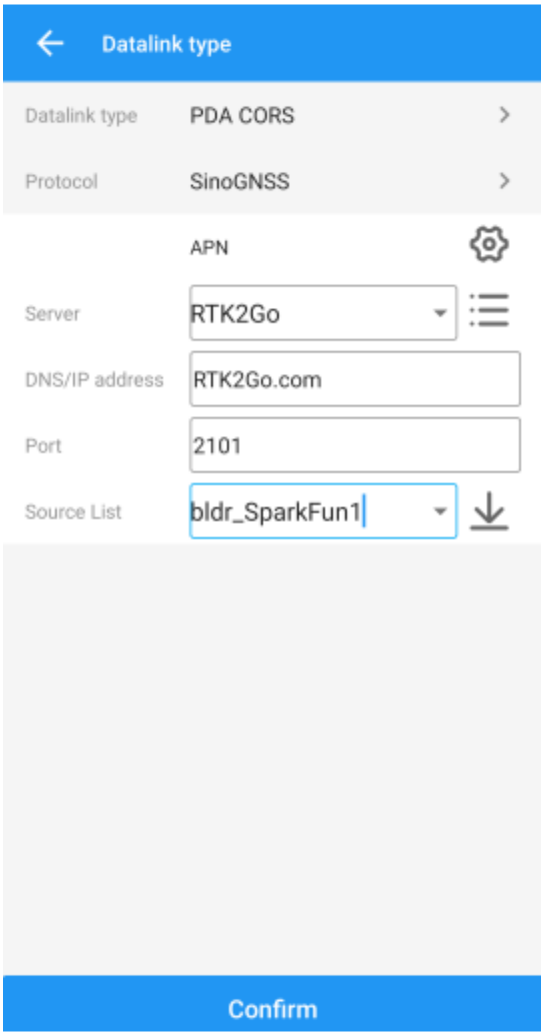
To finish, click 'Connect'. You should see the Bluetooth MAC address on your RTK product change to the Bluetooth icon indicating a connection is established.



Next is configuring the 'Work mode' of the device. The step is where we set up our NTRIP correction source.

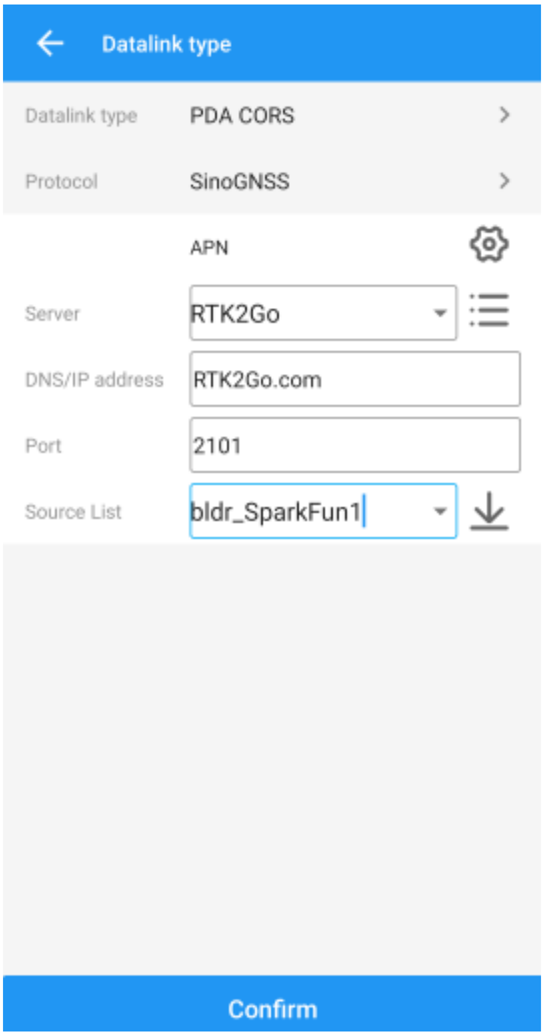


Click 'Add' to create a new work mode.

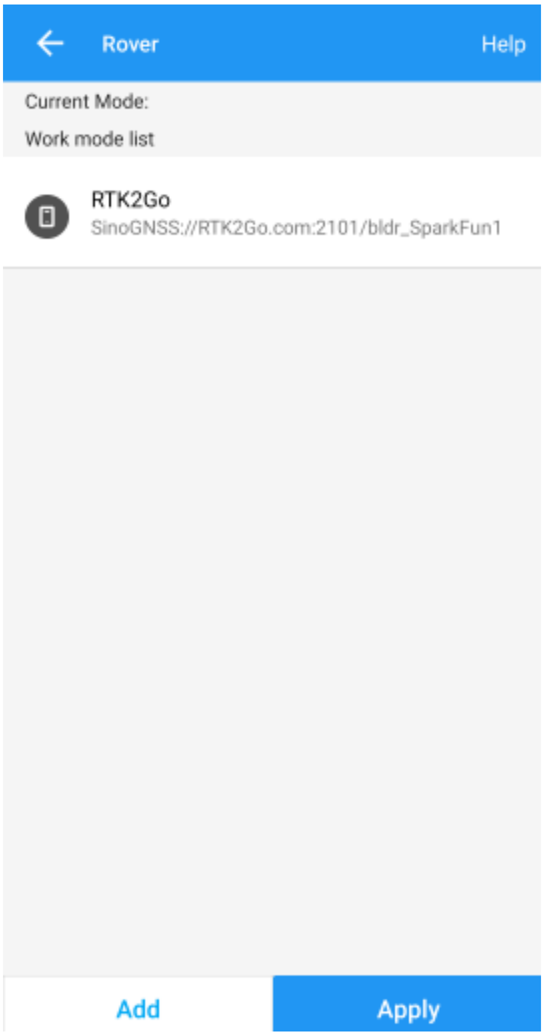


Shown above, we configure the NTRIP Client. Survey Master calls this the 'SinoGNSS' Protocol. Click on the three bars to the right of 'Server' to enter a new NTRIP connection.

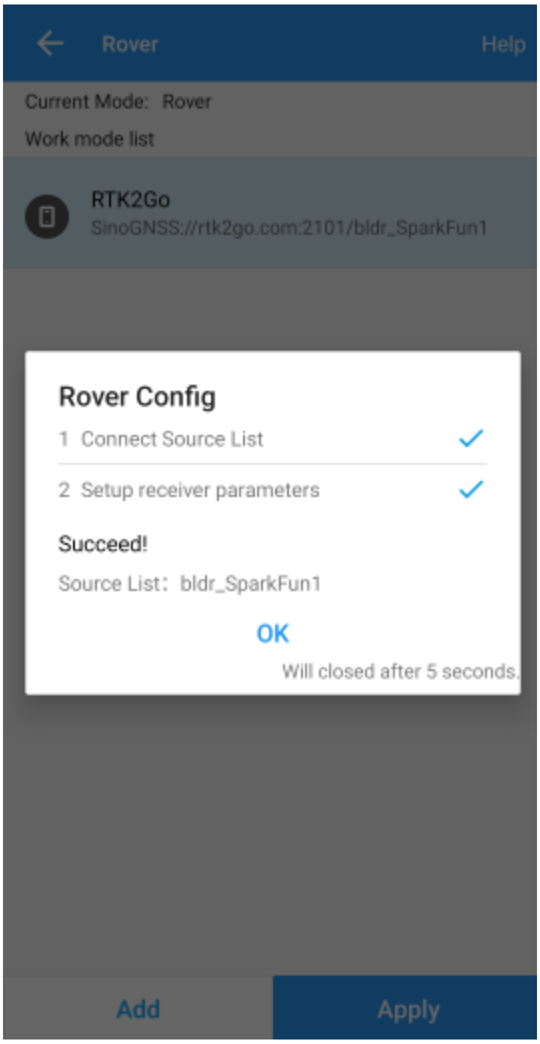
Here you can add different NTRIP Caster providers. If you're using RTK2Go be sure to enter your contact email into the user name.



Return to the 'Datalink type' window and select the Server you just entered. Re-enter the server address and port for your NTRIP Caster. Once complete, click on the down-pointing arrow. This will ping the Caster and obtain the mount point table. Select your mount point.

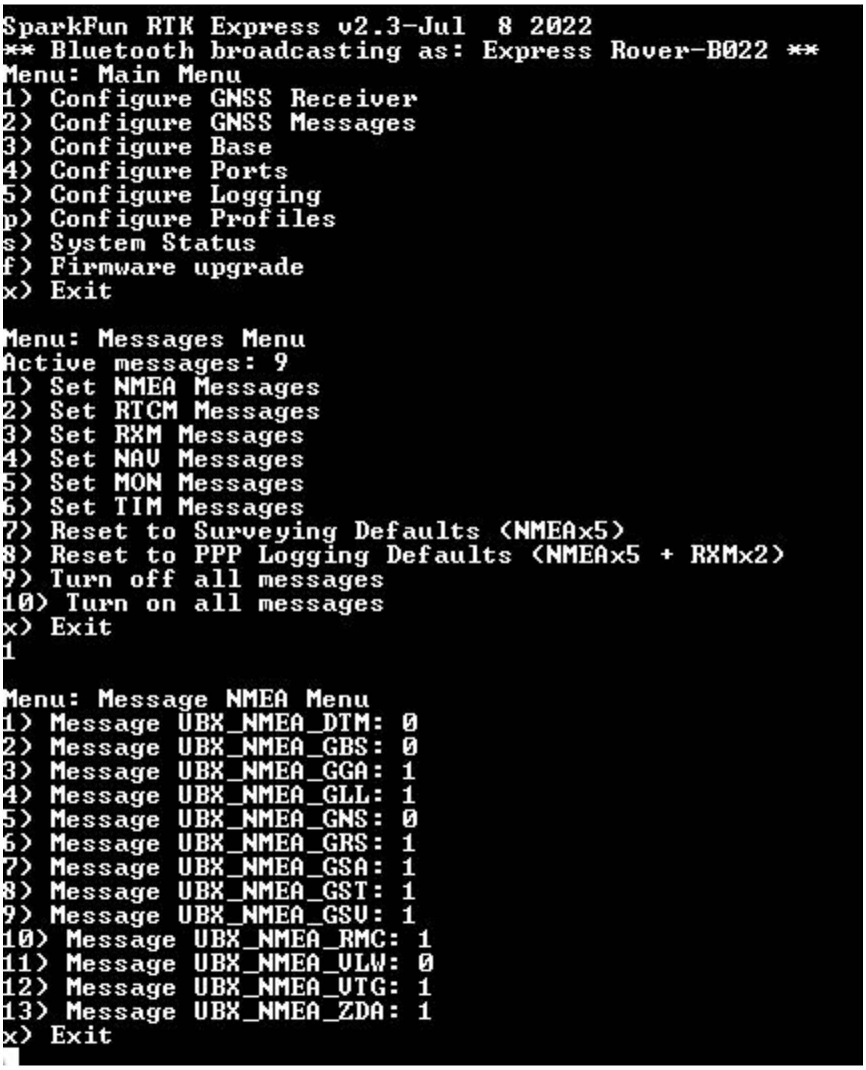


Select the newly created work mode and press the 'Apply' button.



Survey Master will attempt to connect to your specified RTK corrections source (NTRIP Caster). Upon success, you will be located on the Project menu.

Survey Master expects many more NMEA sentences than most GIS software. We must enable some additional messages on the RTK device to correctly communicate with Survey Master.



Note above: There are 9 enabled messages and GSV is set to '1'.

Connect to the RTK device either over WiFi AP config or via Serial. Above is shown the serial method.

Open a terminal at 115200bps and press a key to open the serial configuration menu. Press '2' for GNSS Messages, press '1' for NMEA messages, now be sure to enable 9 messages to a rate of 1:

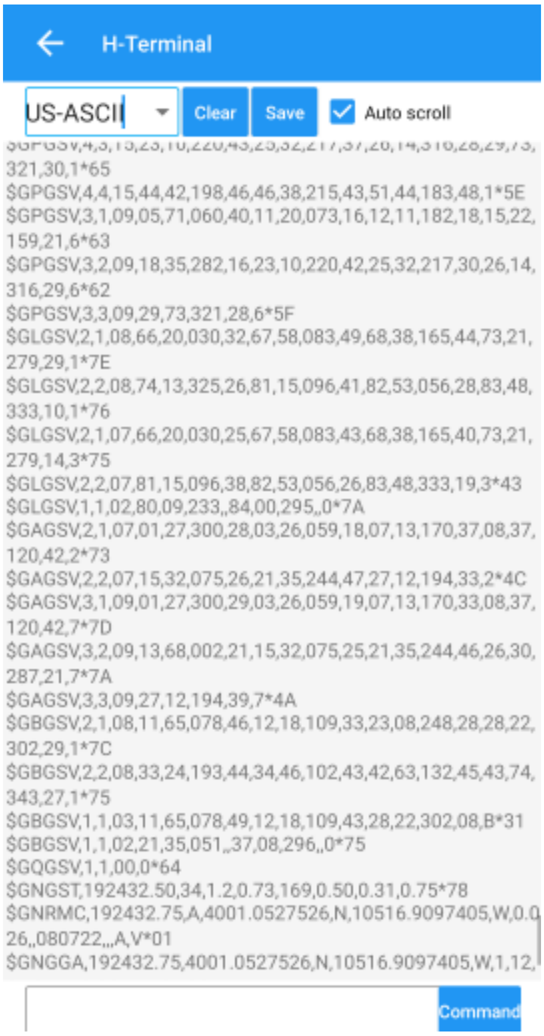
- GGA
- GLL
- GRS
- GSA
- GST
- GSV
- RMC
- VTG
- ZDA

Once complete, press x until you exit the serial menus. Now we may return to Survey Master.

Click on the 'Survey' menu and then 'Topo Survey'. Above we can see a device with RTK float, and 117mm horizontal positional accuracy.

Known Issues:

- Survey Master parses the GxGSV sentence improperly and will only indicate GPS satellites even though the fix solution is using all satellites.

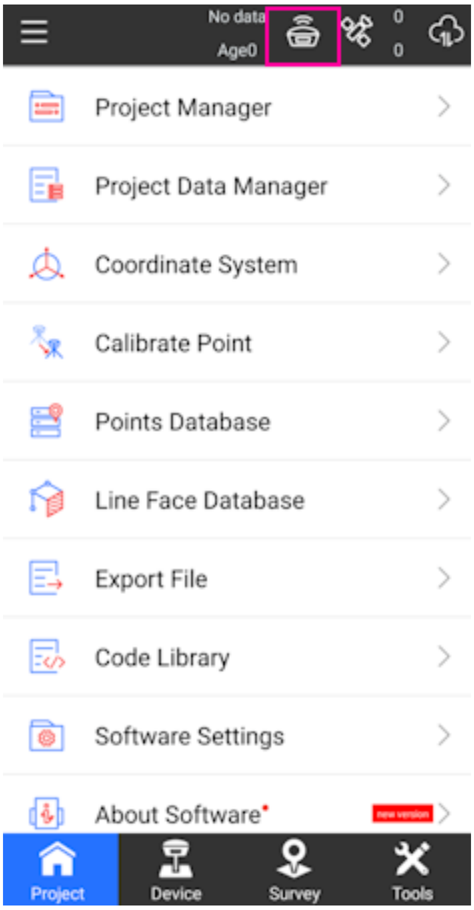


To verify the NMEA sentences are being delivered correctly, Survey Master has a built-in tool. Select the Device->Rover->More->'H-Terminal'.

4.1.10 SurPad

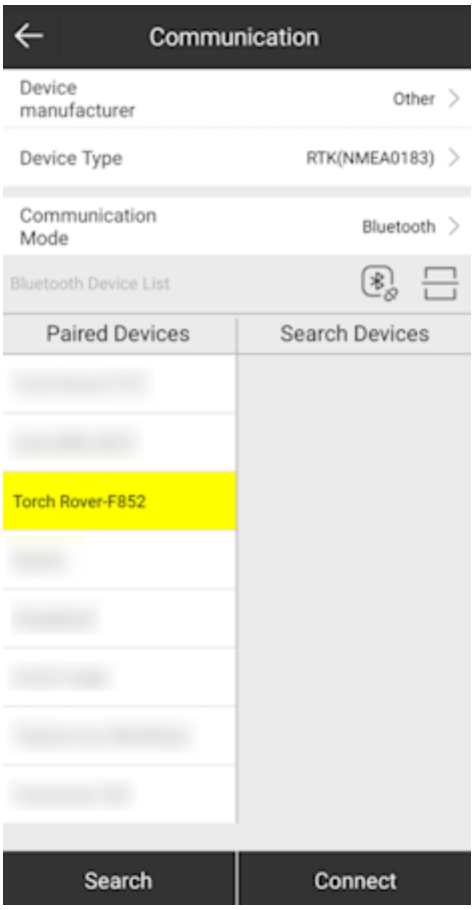
SurPad is an Android app available as a free trial for 30-days. It's loaded as an APK (rather than through Google Play).

Be sure your RTK device has been paired over Bluetooth to your phone.



SurPad Home Screen

Create a project and get to the home screen. Shown above, click on the GNSS receiver icon.

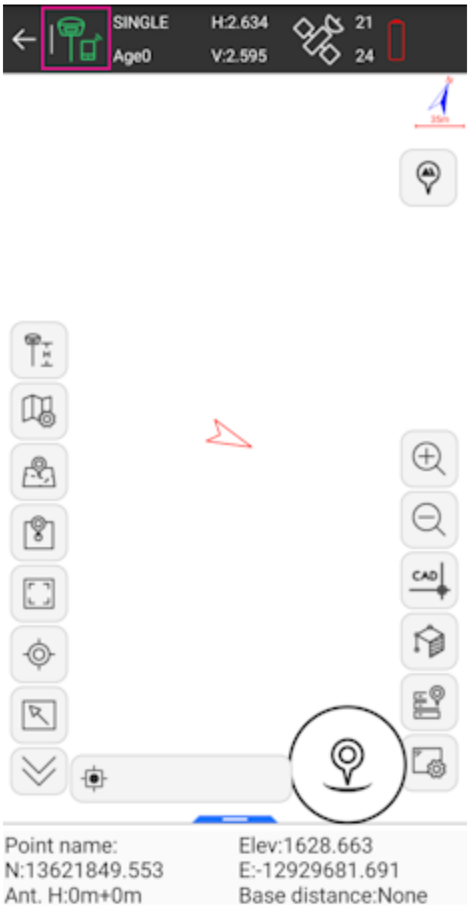


SurPad connecting over Bluetooth

Set the **Device manufacturer** to *Other*, **Device type** to *RTK(NMEA0183)*, and **Communication Mode** to *Bluetooth*. Select the SparkFun RTK device that you would like to connect to on the **Paired Devices** list and then click *Connect*.

Once connected to the device a *Debug* button will appear. This is one of the nice features of SurPad: Running debug will allow you to inspect the NMEA coming across the link.

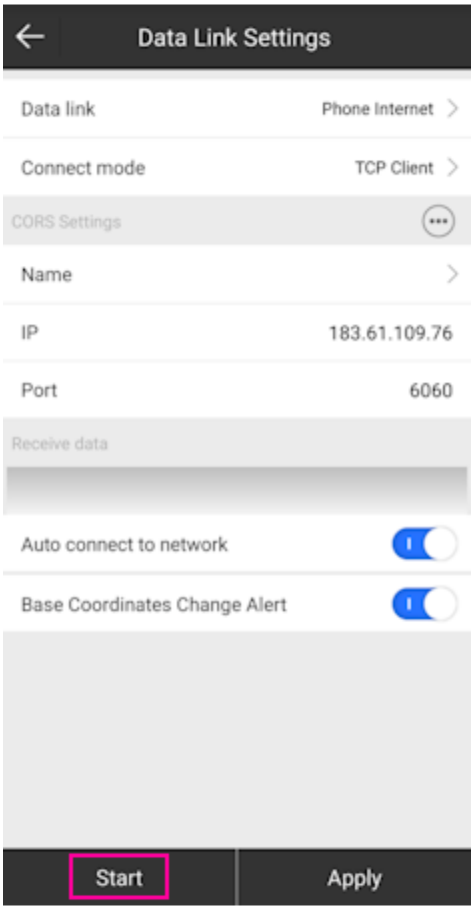
Once done, press the back arrow (top left corner) to return to the home screen.



SurPad Point Survey map

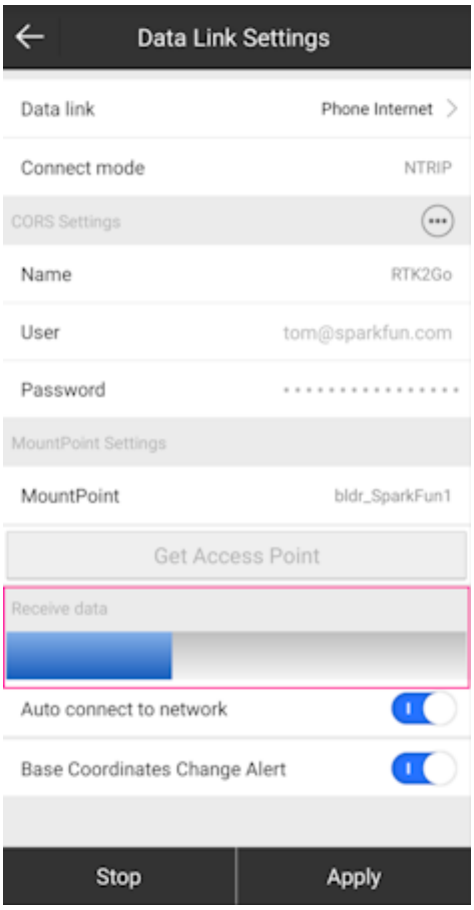
Above: From the home screen press the **Survey** button at the bottom, then **Point Survey** to bring up the map.

In the top left corner, press the green hamburger + cell phone icon. This will open the NTRIP settings.



SurPad Data Link NTRIP Configuration

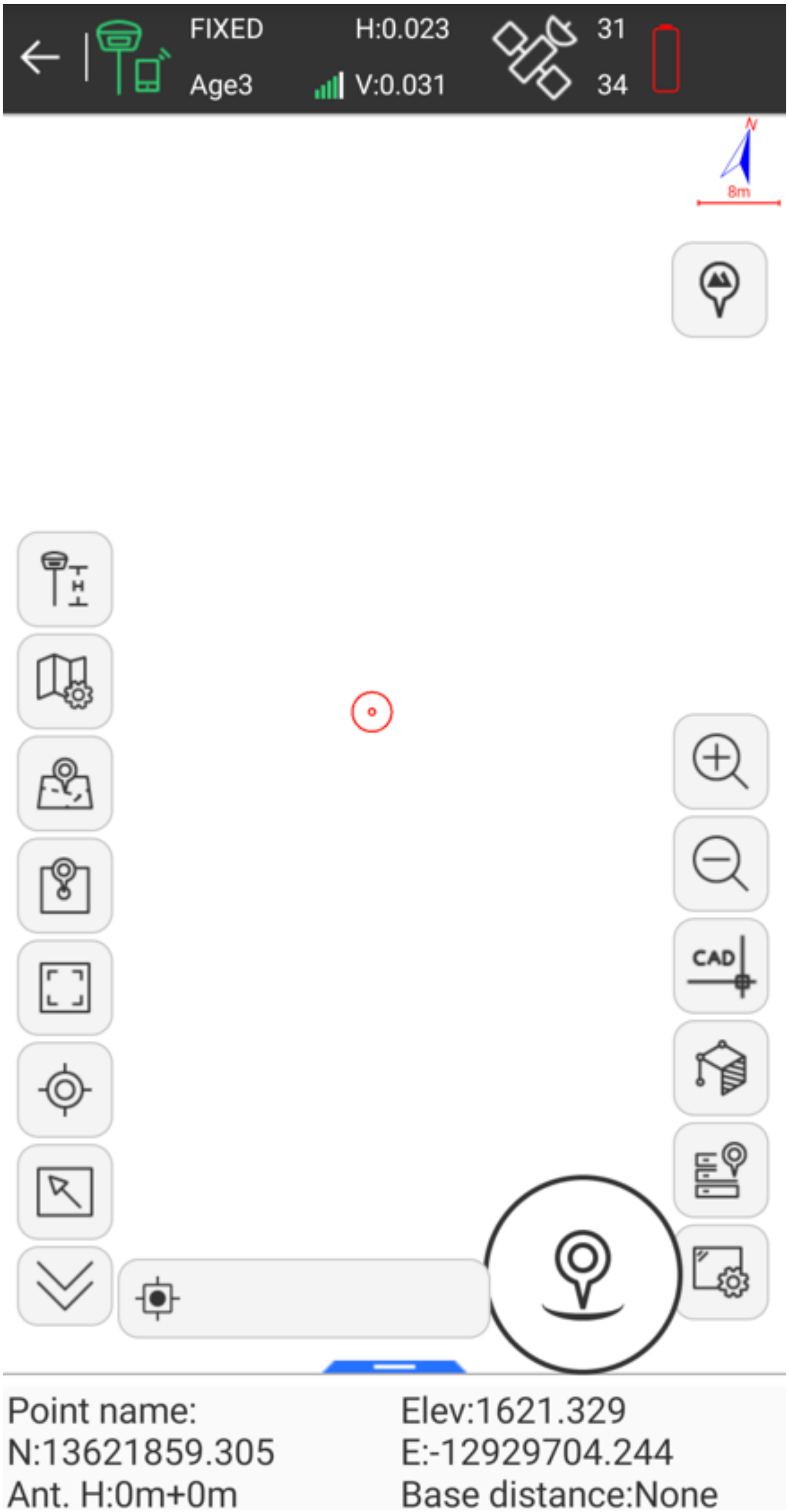
Change the **Connect Mode** from *TCP Client* to *NTRIP*. If you are unable to edit or change the **Connect Mode** from TCP Client be sure the TCP Client is stopped by pressing the *Stop* button in the lower left corner (located in the same spot as the highlighted *Start*).



SurPad NTRIP Connection

Enter the information for your NTRIP caster. In the above example, we are connected to the SparkFun base station on RTK2Go. For RTK2Go you will need to enter a valid email address for a user name but a password is not required.

Click on *Start* and you should see the 'Receive data' progress bar (highlighted above) increase each second indicating a connection. Once complete, press 'Apply' to return to the map.

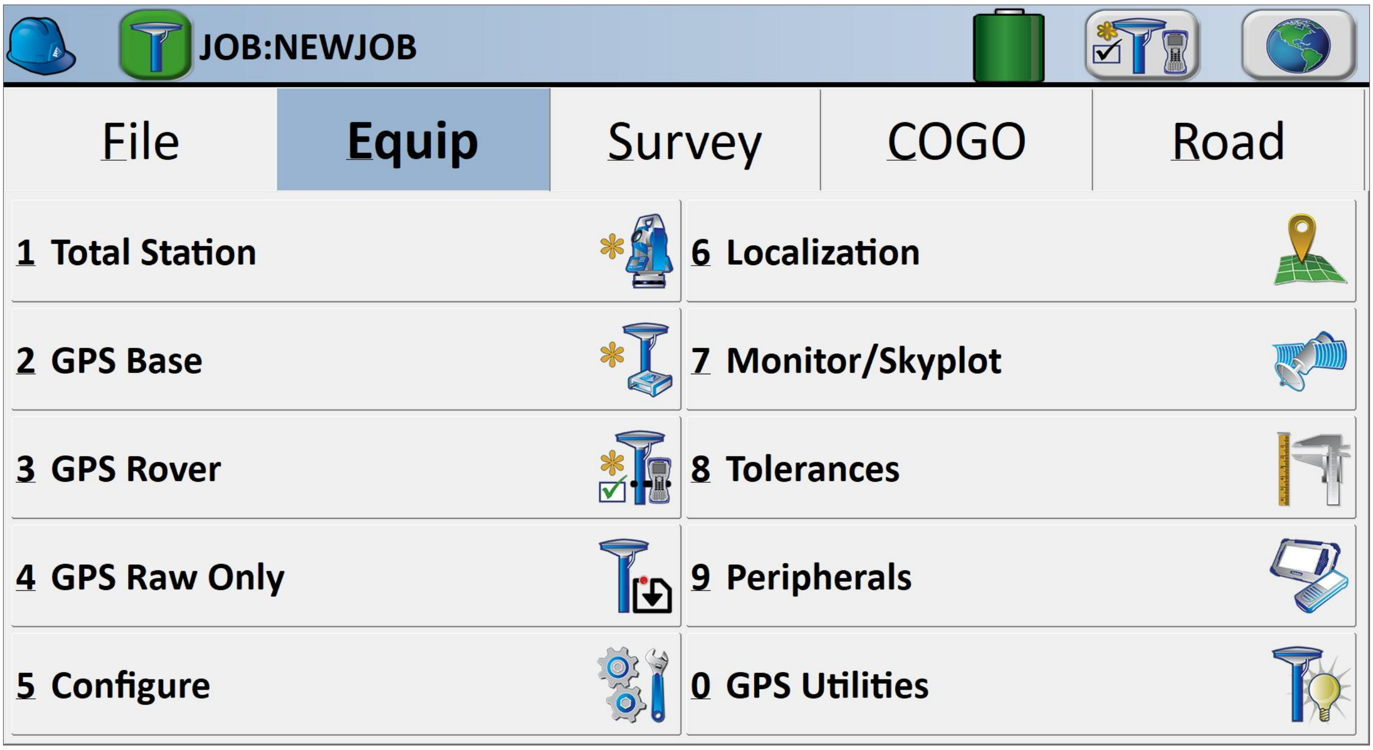


SurPad with RTK Fix

Above: After a few moments, the RTK device should move to RTK Float, then RTK Fix. You can see the age of the RTCM data in the upper bar, along with the horizontal (23mm) and vertical (31mm) accuracy estimates. Now you can begin taking points.

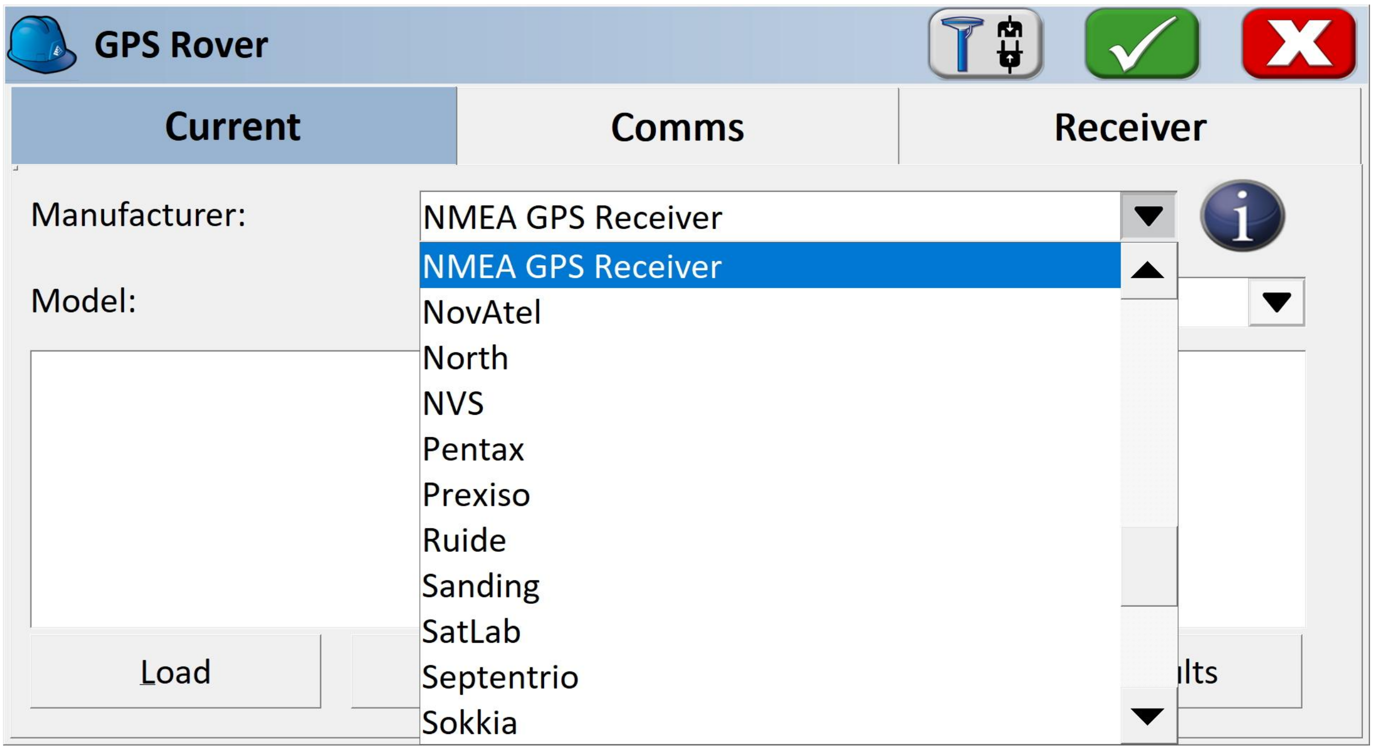
4.1.11 SurvPC

Be sure your device is paired over Bluetooth.



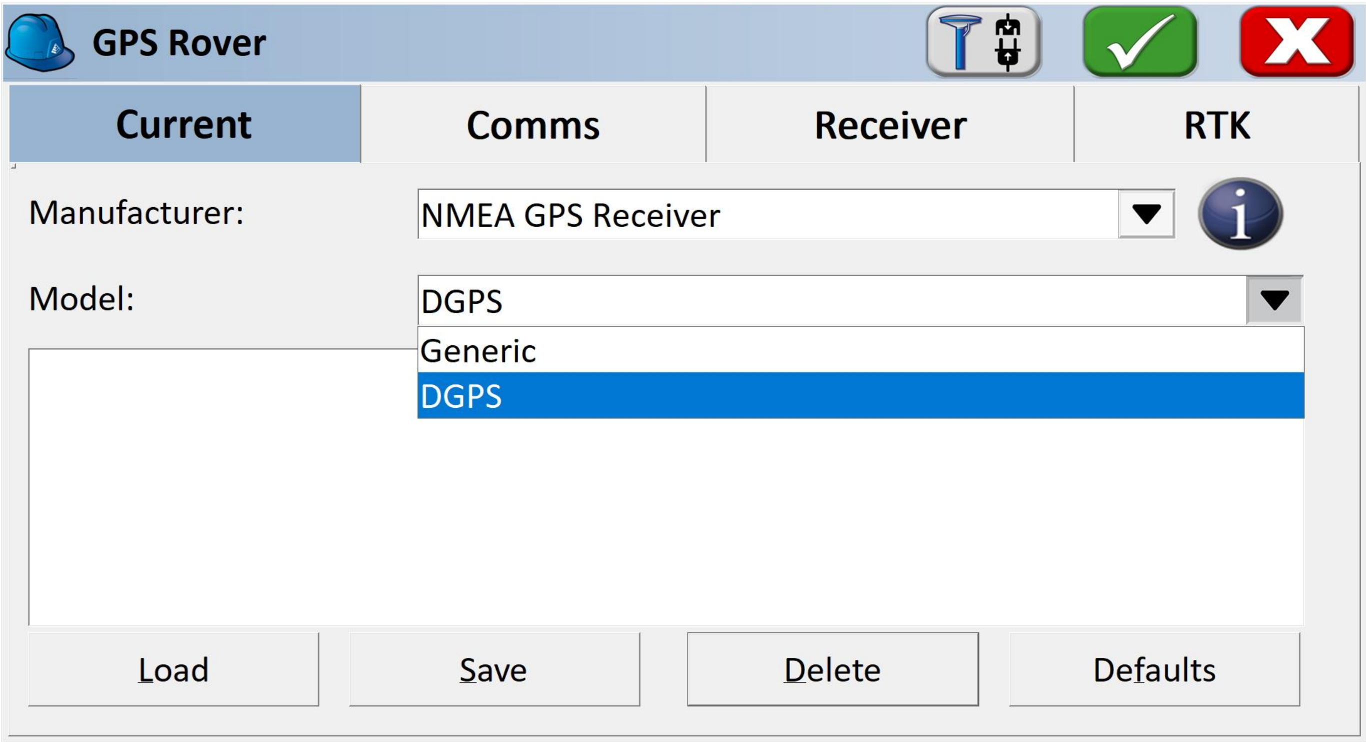
Equip Sub Menu

Select the *Equip* sub menu then `GPS Rover`



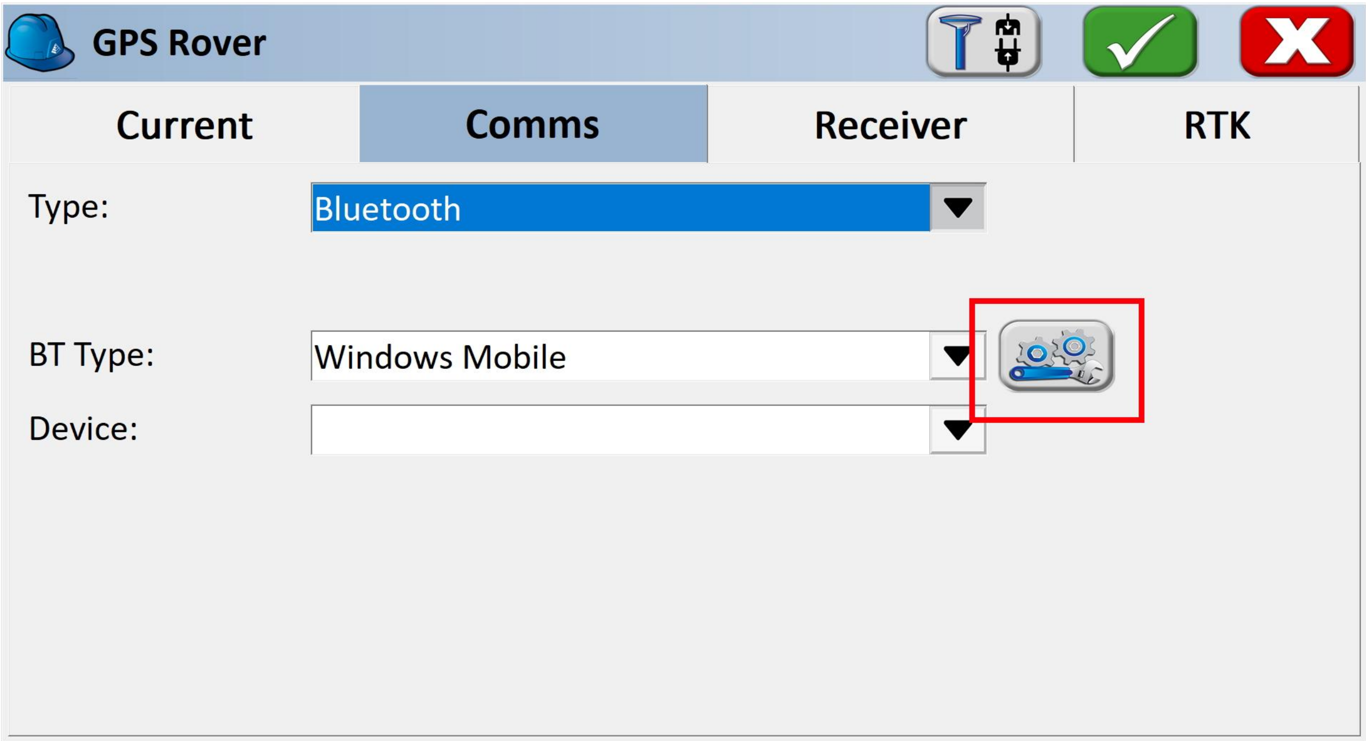
Select NMEA GPS Receiver

From the drop down, select `NMEA GPS Receiver`.



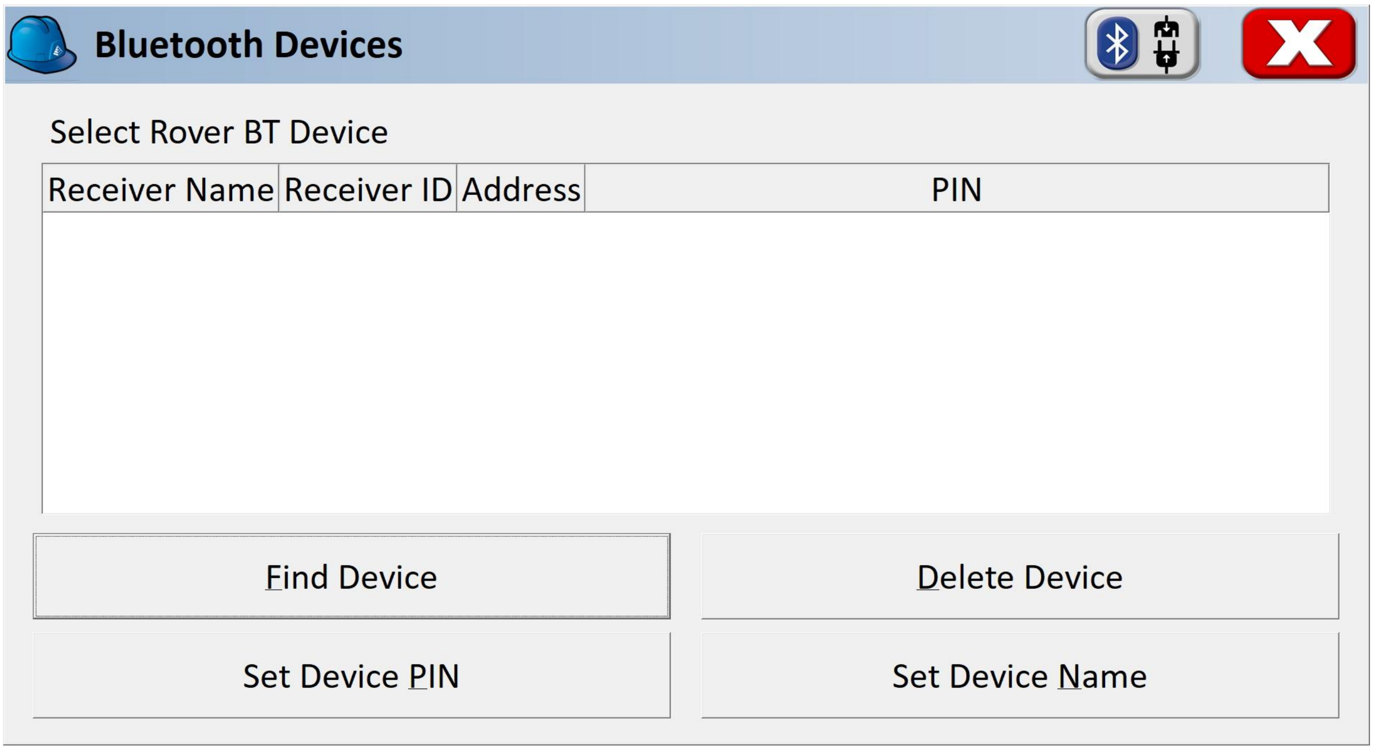
Select Model: DGPS

Select DGPS if you'd like to connect to an NTRIP Caster. If you are using the RTK Facet L-Band, or do not need RTK fix type precision, leave the model as Generic.



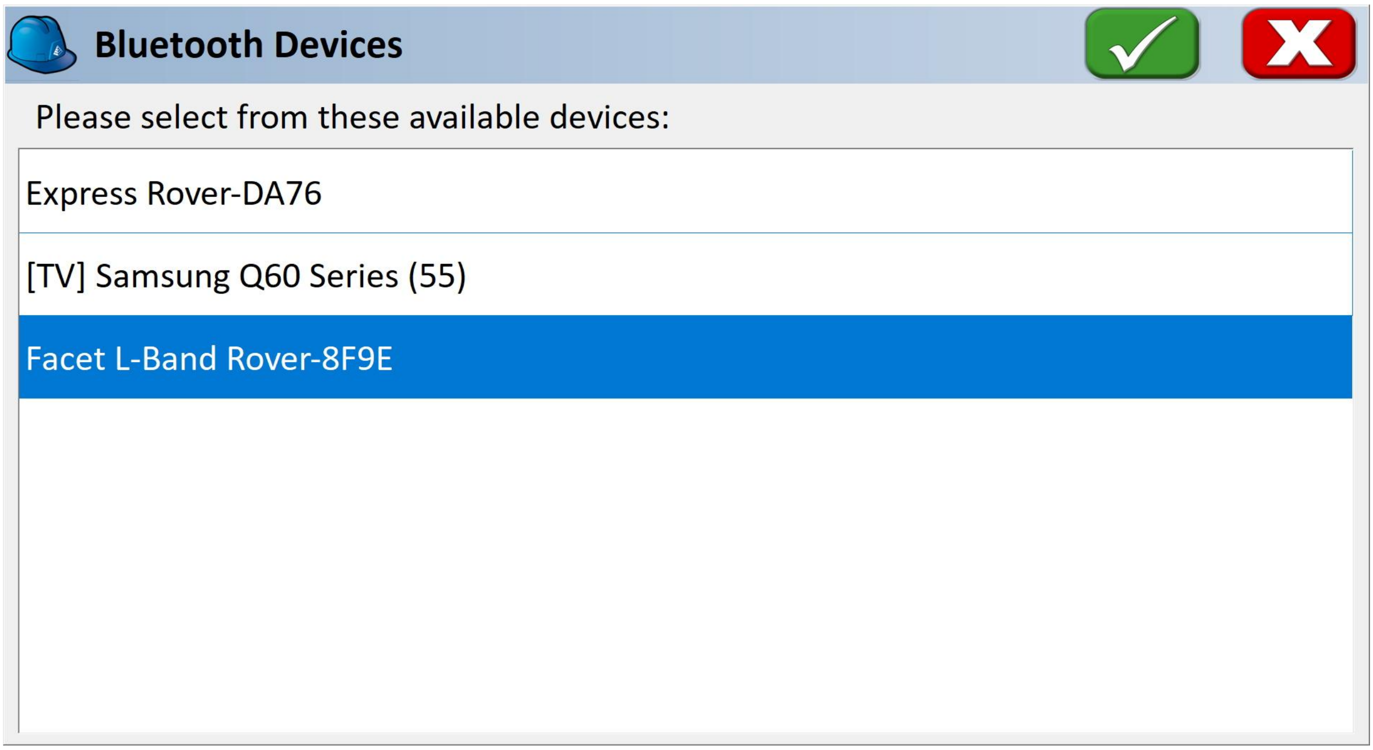
Bluetooth Settings Button

From the `Comms` submenu, click the Blueooth settings button.



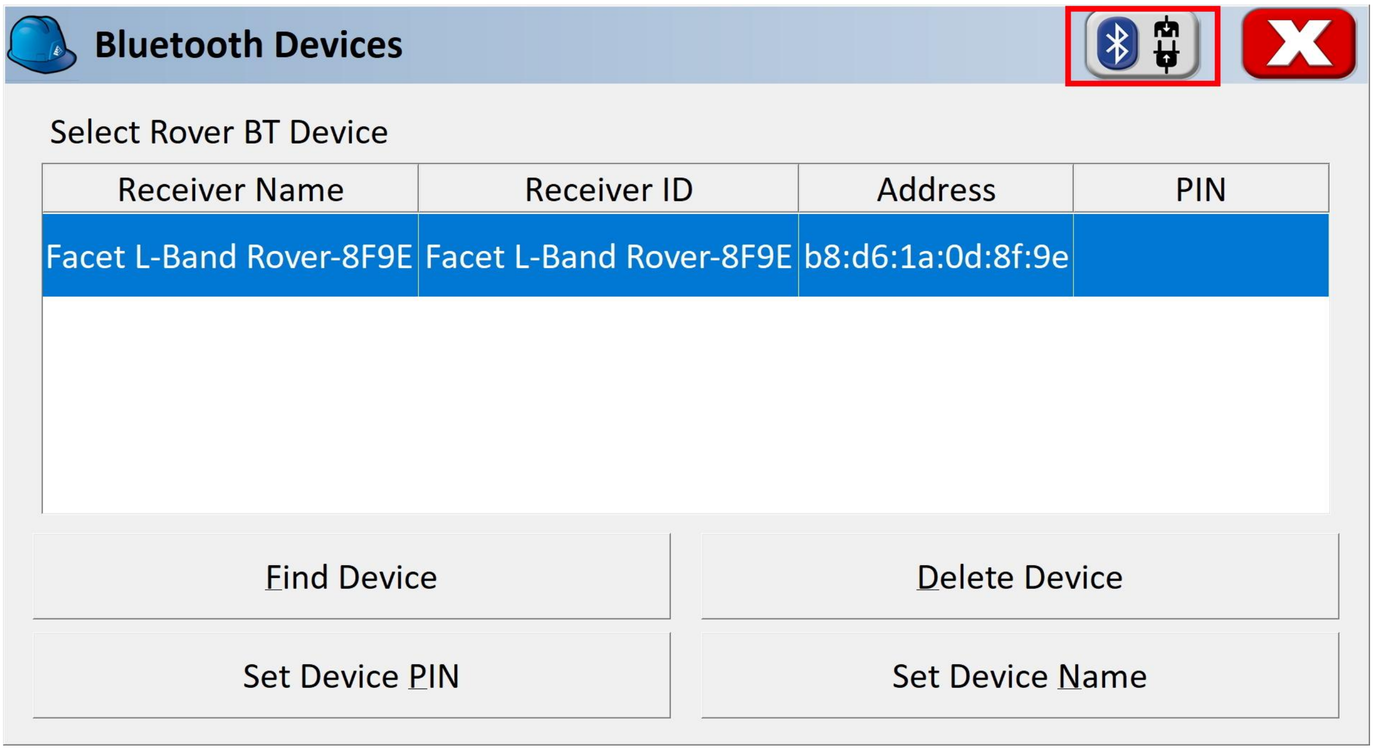
SurvPC Bluetooth Devices

Click `Find Device`.



List of Paired Bluetooth Devices

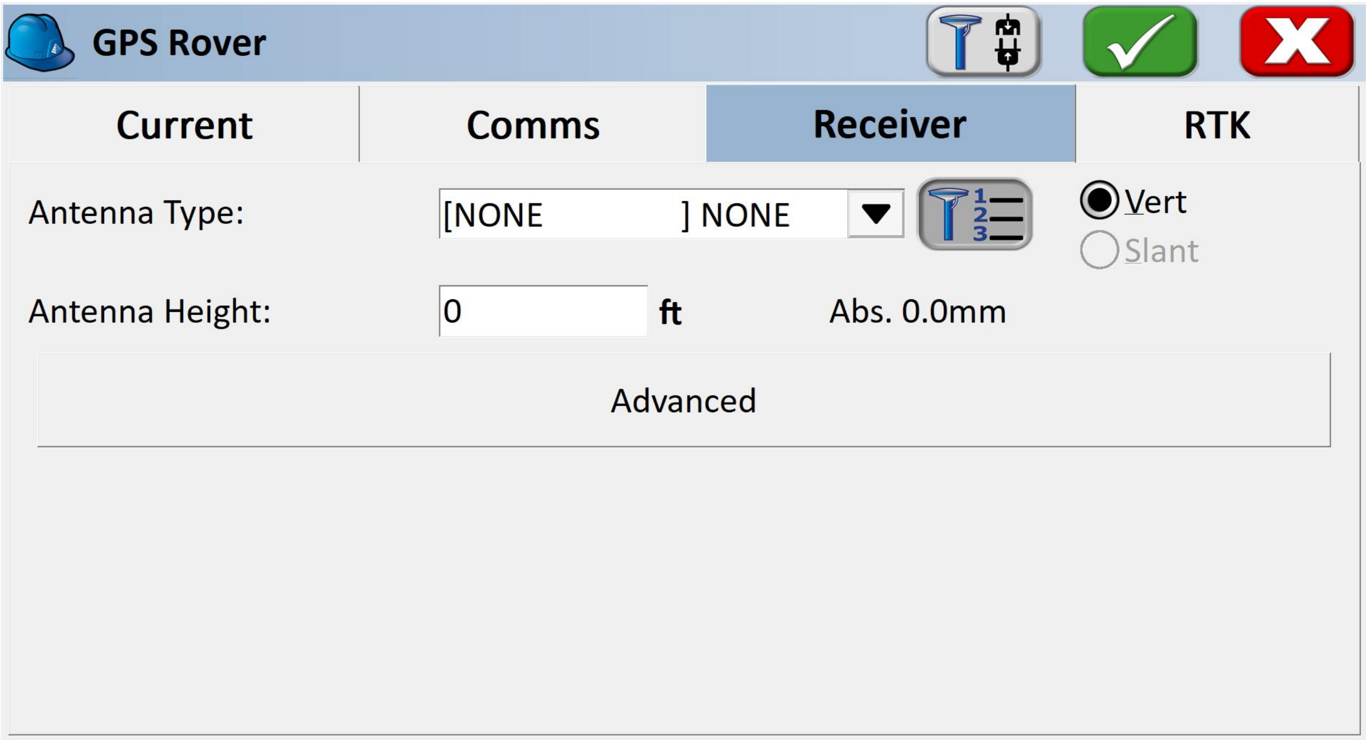
You will be shown a list of devices that have been paired. Select the RTK device you want to connect to.



Connect to Device

Click the `Connect Bluetooth` button, shown in red in the top right corner. The software will begin a connection to the RTK device. You'll see the MAC address on the RTK device changes to the Bluetooth icon indicating it's connected.

If SurvPC detects NMEA, it will report a successful connection.

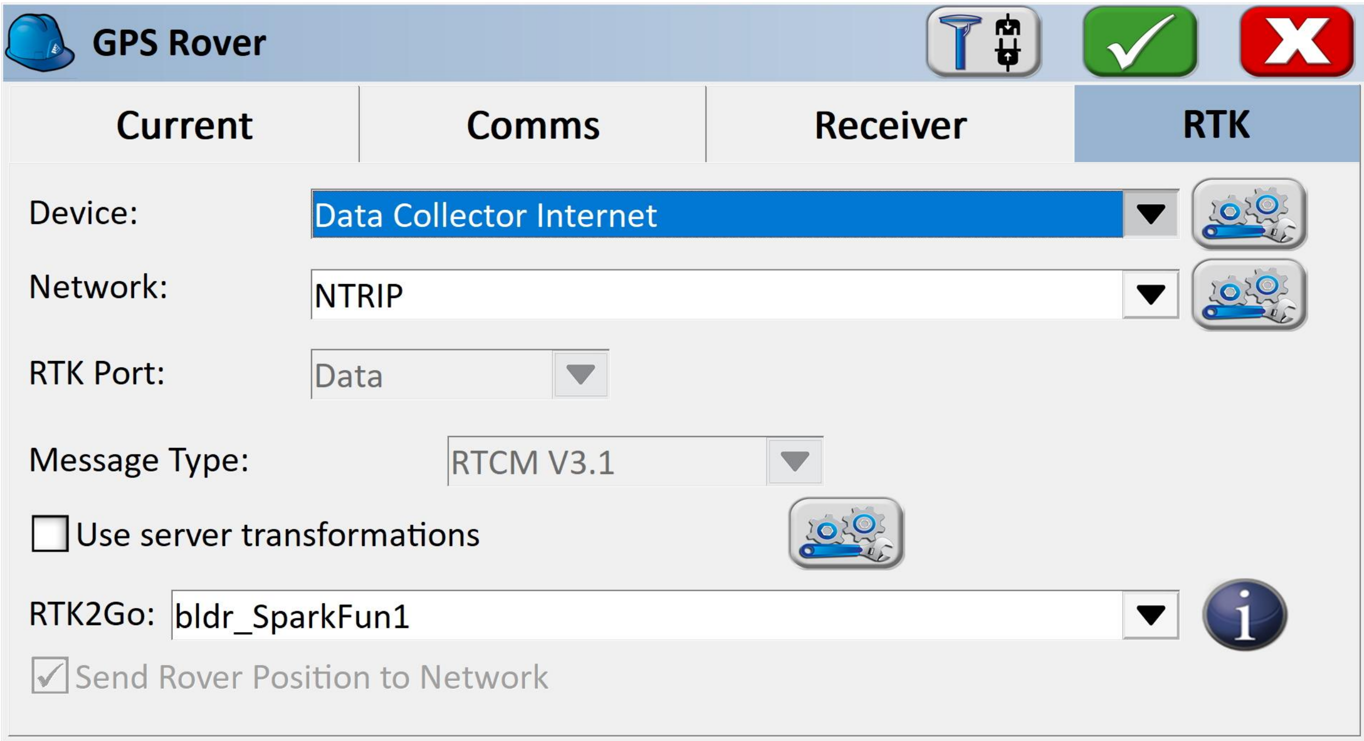


Receiver Submenu

You are welcome to enter the ARP (antenna reference point) and surveying stick length for your particular setup.

NTRIP Client

Note: If you are using a radio to connect Base to Rover, or if you are using the RTK Facet L-Band you do not need to set up NTRIP; the device will achieve RTK fixes and output extremely accurate location data by itself. But if L-Band corrections are not available, or you are not using a radio link, the NTRIP Client can provide corrections to this Rover.



RTK Submenu

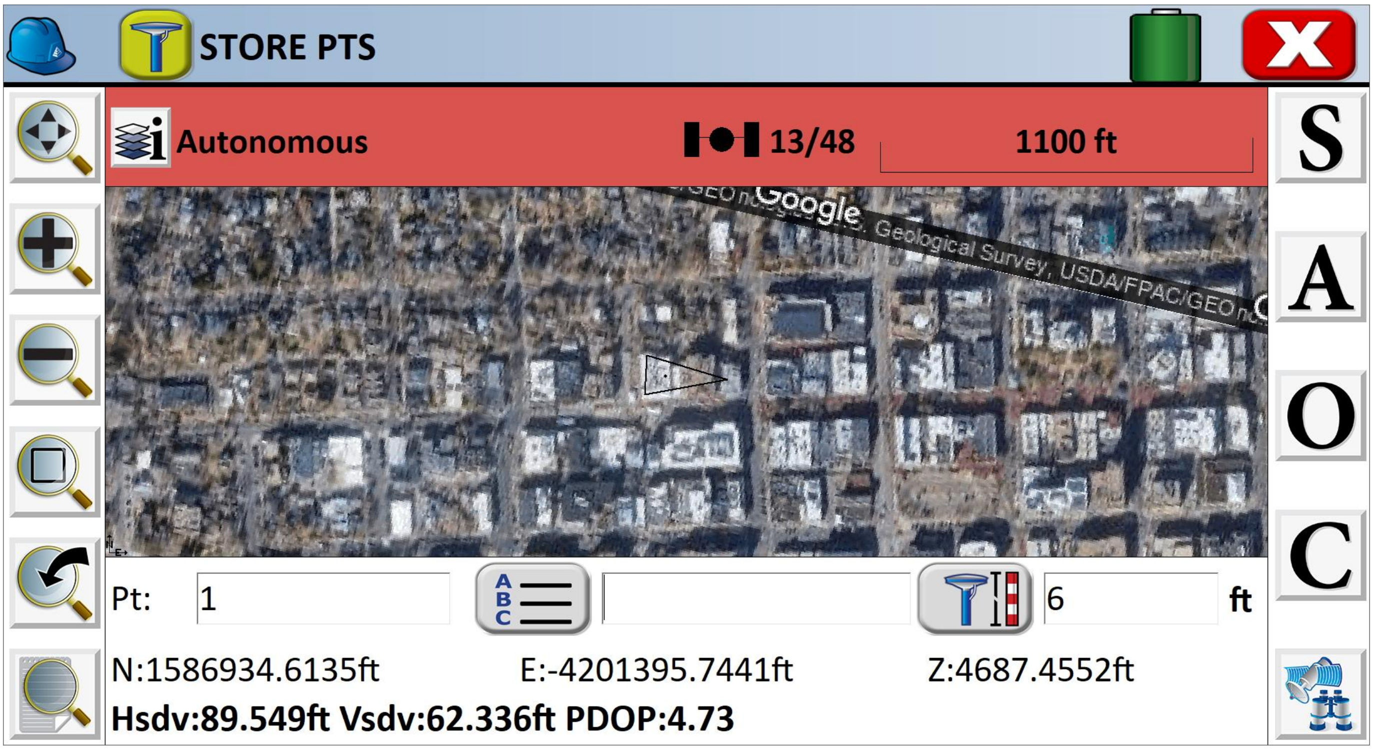
If you selected 'DGPS' as the Model type, the RTK submenu will be shown. This is where you give the details about your NTRIP Caster such as your mount point, user name/pw, etc. For more information about creating your own NTRIP mount point please see Creating a Permanent Base

Enter your NTRIP Caster credentials and click connect. You will see bytes begin to transfer from your phone to the RTK device. Within a few seconds, the RTK device will go from ~300mm accuracy to 14mm. Pretty nifty, no?

What's an NTRIP Caster? In a nutshell, it's a server that is sending out correction data every second. There are thousands of sites around the globe that calculate the perturbations in the ionosphere and troposphere that decrease the accuracy of GNSS accuracy. Once the inaccuracies are known, correction values are encoded into data packets in the RTCM format. You, the user, don't need to know how to decode or deal with RTCM, you simply need to get RTCM from a source within 10km of your location into the RTK device. The NTRIP client logs into the server (also known as the NTRIP caster) and grabs that data, every second, and sends it over Bluetooth to the RTK device.

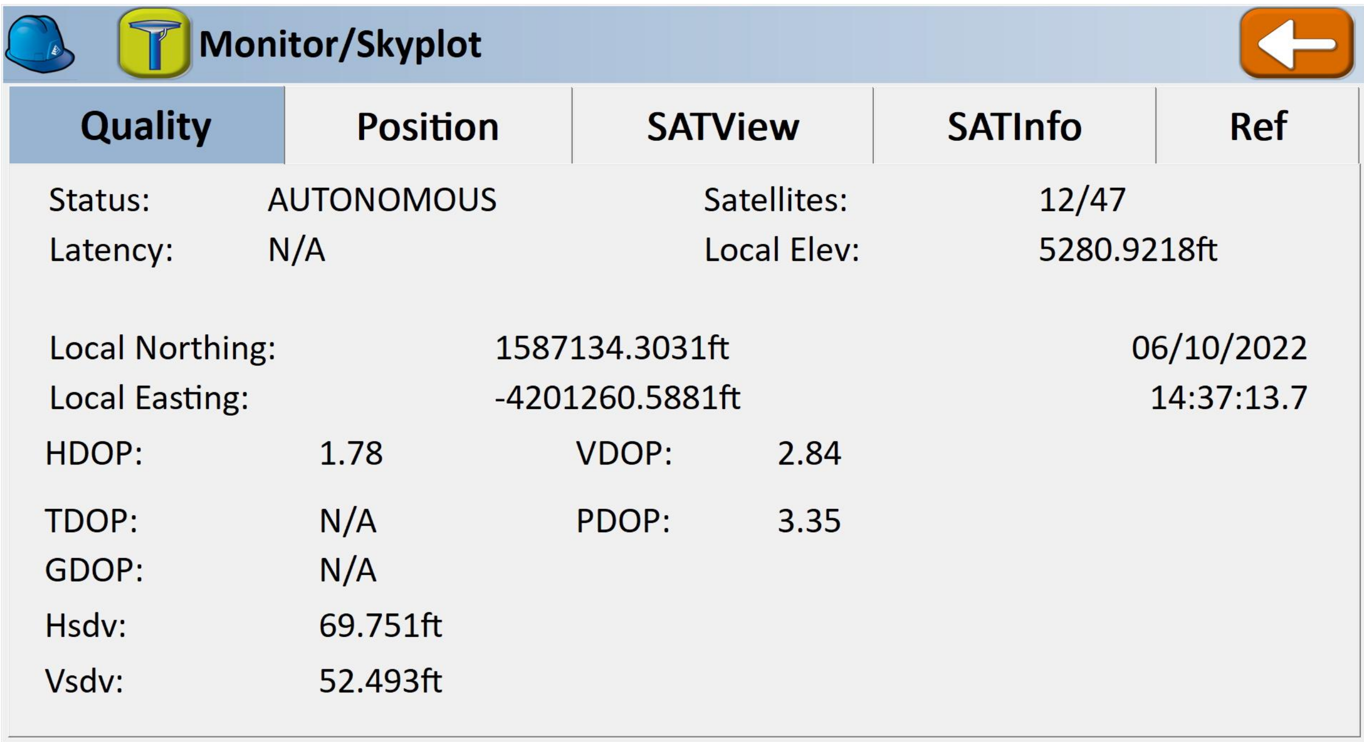
Don't have access to an NTRIP Caster? You can use a 2nd RTK product operating in Base mode to provide the correction data. Checkout Creating a Permanent Base. If you're the DIY sort, you can create your own low-cost base station using an ESP32 and a ZED-F9P breakout board. Check out How to Build a DIY GNSS Reference Station](https://learn.sparkfun.com/tutorials/how-to-build-a-diy-gnss-reference-station). If you'd just like a service, Syklark provides RTCM coverage for $49 a month (as of writing) and is extremely easy to set up and use. Remember, you can always use a 2nd RTK device in *Base* mode to provide RTCM correction data but it will be less accurate than a fixed position caster.

Once everything is connected up, click the Green check in the top right corner.



Storing Points

Now that we have a connection, you can use the device, as usual, storing points and calculating distances.



SurvPC Skyplot

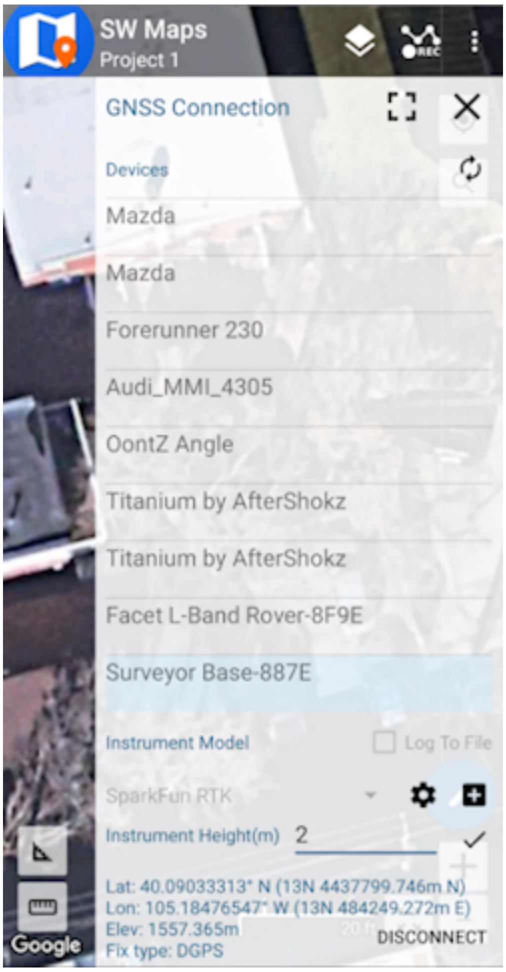
Opening the Skyplot will allow you to see your GNSS details in real-time.

If you are a big fan of SurvPC please contact your sales rep and ask them to include SparkFun products in their Manufacturer drop-down list.

4.1.12 SW Maps

The best mobile app that we've found is the powerful, free, and easy-to-use SW Maps by Softwel. It is compatible with Android and iOS, either phone or tablet with Bluetooth. What makes SW Maps truly powerful is its built-in NTRIP client. This is a fancy way of saying that we'll be showing you how to get RTCM correction data over the cellular network.

Be sure your device is paired over Bluetooth.



List of available Bluetooth devices

From SW Map's main menu, select *Bluetooth GNSS*. This will display a list of available Bluetooth devices. Select the Rover or Base you just paired with. If you are taking height measurements (altitude) in addition to position (lat/long) be sure to enter the height of your antenna off the ground including any ARP offsets of your antenna (this should be printed on the side).

Click on 'CONNECT' to open a Bluetooth connection. Assuming this process takes a few seconds, you should immediately have a location fix.

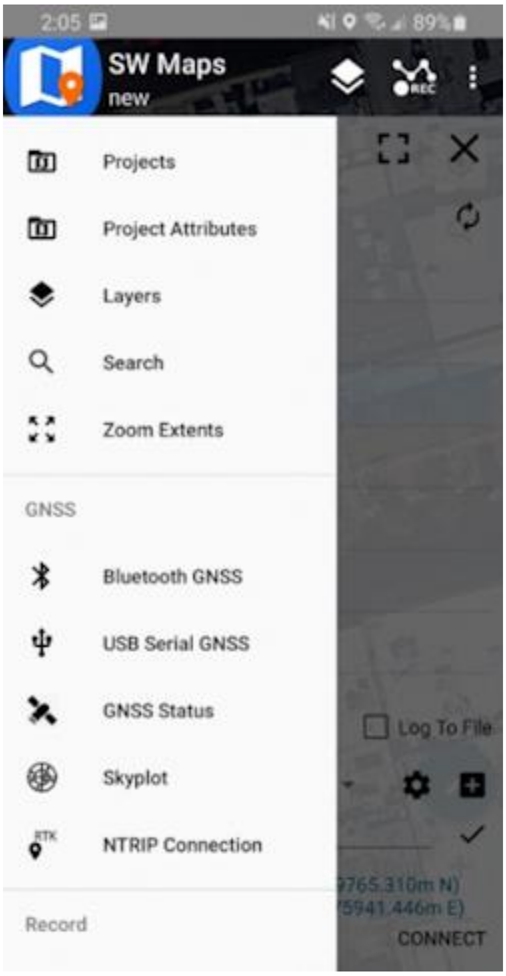
SW Maps with RTK Fix

You can open the GNSS Status sub-menu to view the current data.

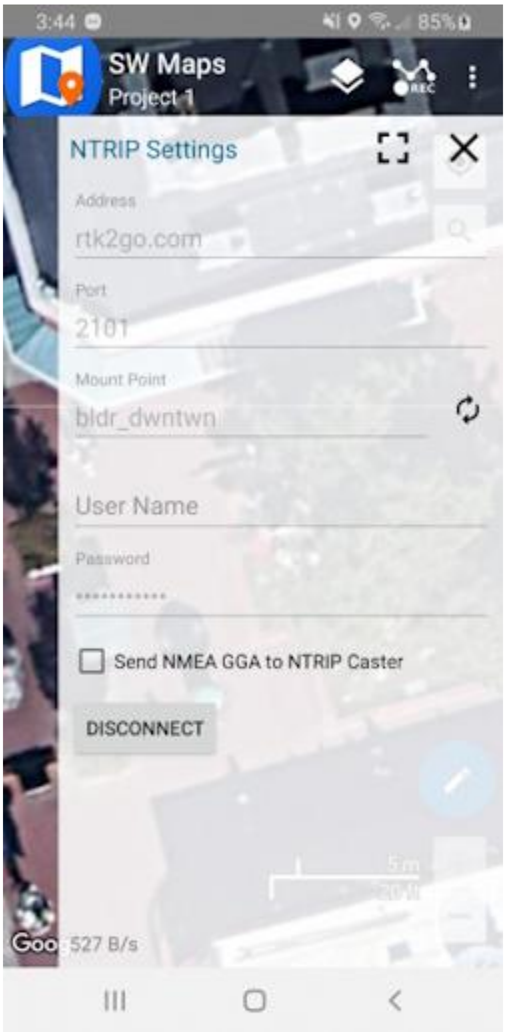
NTRIP Client

If you’re using a serial radio to connect a Base to a Rover for your correction data, or if you're using the RTK Facet L-Band with built-in corrections, you can skip this part.

We need to send RTCM correction data from the phone back to the RTK device so that it can improve its fix accuracy. This is the amazing power of the SparkFun RTK products and SW Maps. Your phone can be the radio link! From the main SW Maps menu select NTRIP Client. Not there? Be sure the 'SparkFun RTK' instrument was automatically selected connecting. Disconnect and change the instrument to 'SparkFun RTK' to enable the NTRIP Connection option.



NTRIP Connection - Not there? Be sure to select 'SparkFun RTK' was selected as the instrument



Connecting to an NTRIP Caster

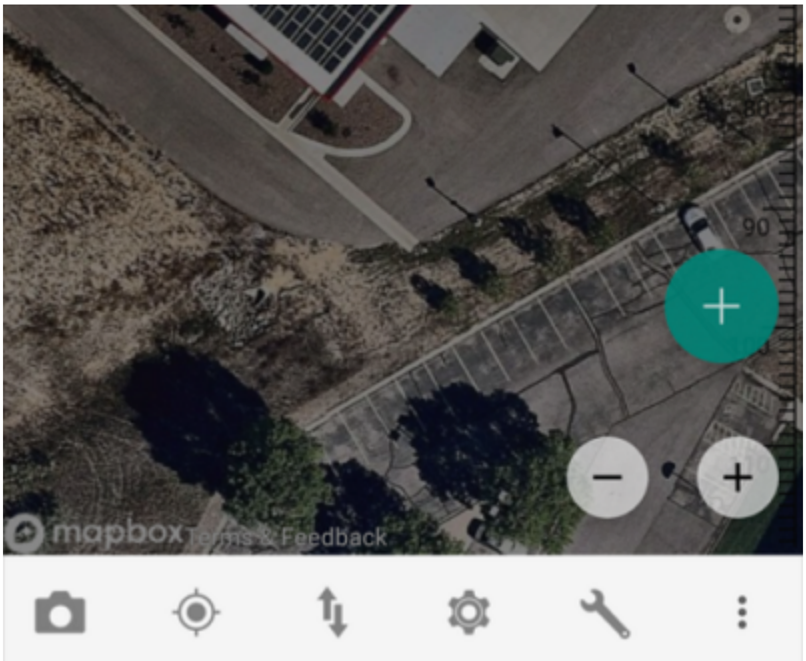
Enter your NTRIP Caster credentials and click connect. You will see bytes begin to transfer from your phone to the RTK device. Within a few seconds, the RTK device will go from ~300mm accuracy to 14mm. Pretty nifty, no?

Once you have a full RTK fix you'll notice the location bubble in SW Maps turns green. Just for fun, rock your rover monopole back and forth on a fixed point. You'll see your location accurately reflected in SW Maps. Millimeter location precision is a truly staggering thing.

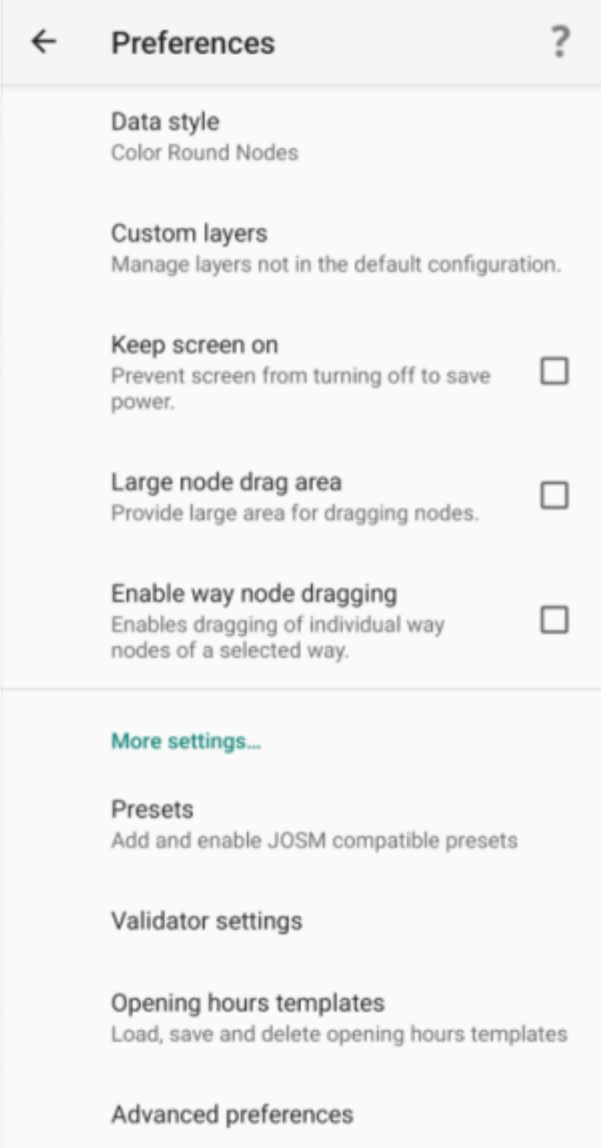
4.1.13 Vespucci

Vespucci is an Open Street Map editor for Android.

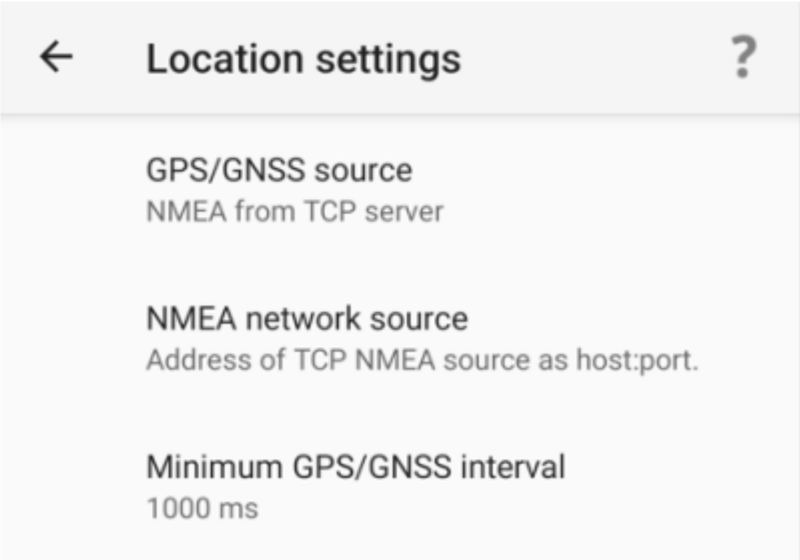
This software requires the RTK device to connect over TCP. Be sure you have a local WiFi network entered into the WiFi Config menu, have a TCP Client or Server enabled, and have noted the TCP port (it's 2948 by default).



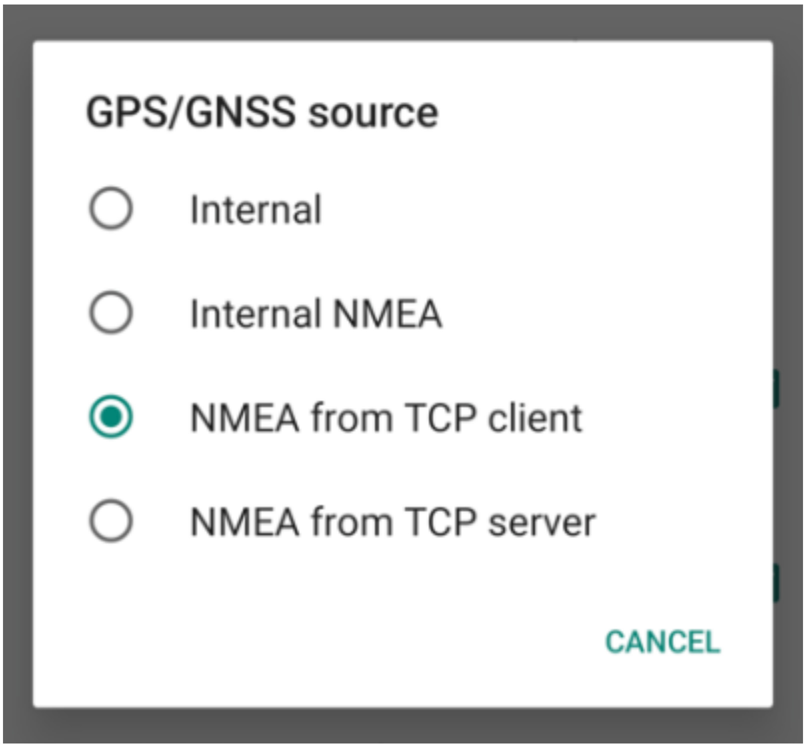
With a map open, select the gear icon on the bottom bar.



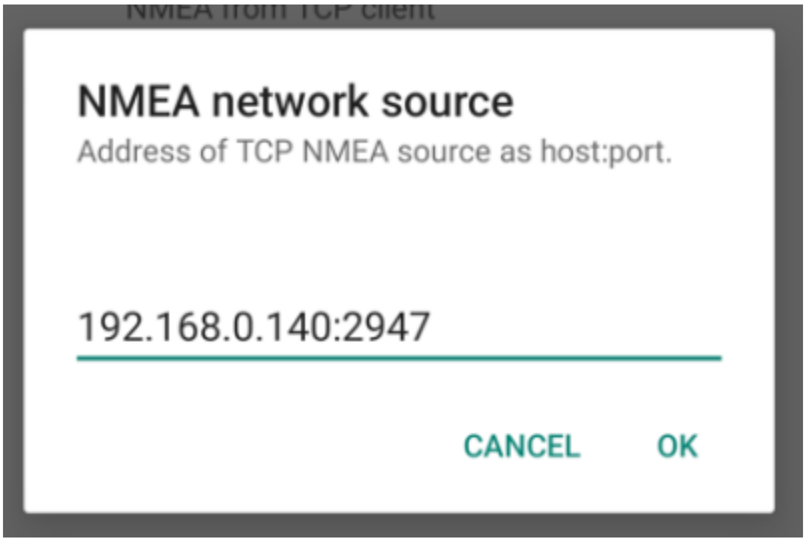
From the Preferences menu, scroll to the bottom and select 'Advanced Preferences'.



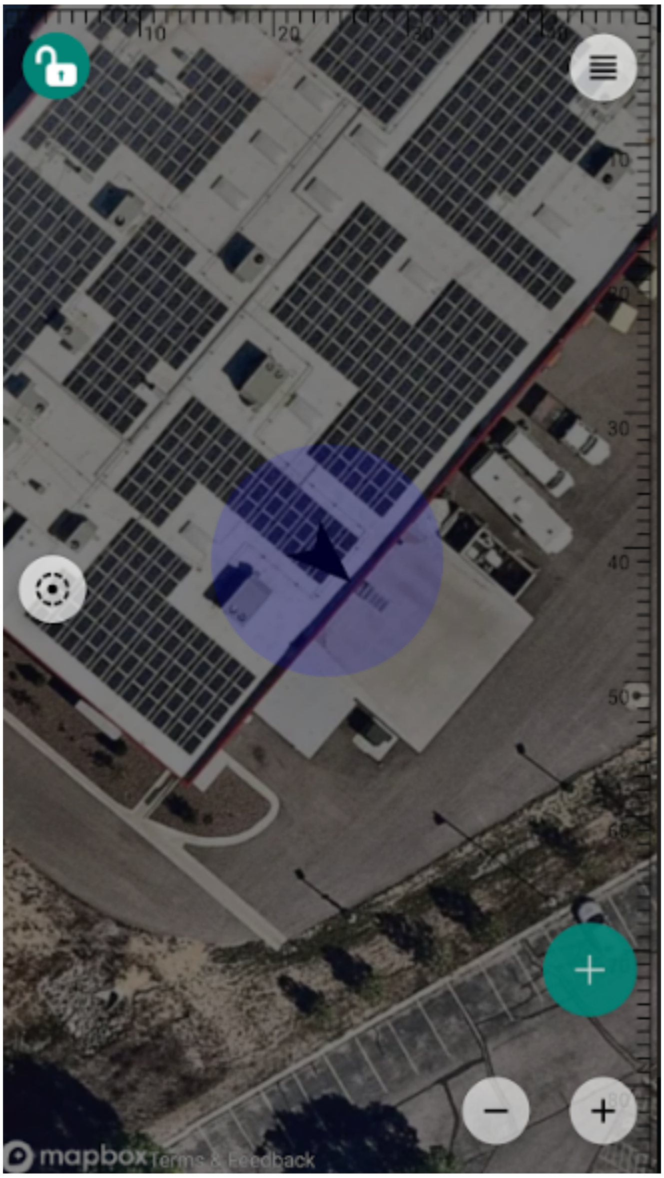
Select **Location settings**.



Select **GPS/GNSS source**. Select **NMEA from TCP client**. TCP server is also supported.



Select **NMEA network source**. Enter the IP address and TCP port of the RTK device. The IP address can be found by opening a serial terminal while connected to WiFi (it is reported every few seconds). The TCP port is entered into the WiFi Config menu.



Close all menus and you should see your location within Vespucci.

4.1.14 Other GIS Packages

Hopefully, these examples give you an idea of how to connect the RTK product line to most any GIS software. If there is other GIS software that you'd like to see configuration information about, please open an issue on the RTK Firmware repo and we'll add it.

4.1.15 What's an NTRIP Caster?

In a nutshell, it's a server that is sending out correction data every second. There are thousands of sites around the globe that calculate the perturbations in the ionosphere and troposphere that decrease the accuracy of GNSS accuracy. Once the inaccuracies are known, correction values are encoded into data packets in the RTCM format. You, the user, don't need to know how to decode or deal with RTCM, you simply need to get RTCM from a source within 10km of your location into the RTK device. The NTRIP client logs into the server (also known as the NTRIP caster) and grabs that data, every second, and sends it over Bluetooth to the RTK device.

4.1.16 Where do I get RTK Corrections?

Be sure to see Correction Sources.

Don't have access to an NTRIP Caster or other RTCM correction source? There are a few options.

The SparkFun RTK Facet L-Band gets corrections via an encrypted signal from geosynchronous satellites. This device gets RTK Fix without the need for a WiFi or cellular connection.

Also, you can use a 2nd RTK product operating in Base mode to provide the correction data. Check out Creating a Permanent Base.

If you're the DIY sort, you can create your own low-cost base station using an ESP32 and a ZED-F9P breakout board. Check out How to Build a DIY GNSS Reference Station.

There are services available as well. Syklark provides RTCM coverage for $49 a month (as of writing) and is extremely easy to set up and use. Point One also offers RTK NTRIP service with a free 14 day trial and easy to use front end.

4.2 iOS

Surveyor: / Express: / Express Plus: / Facet: / Facet L-Band: / Reference Station:

There are a variety of 3rd party apps available for GIS and surveying for Android, iOS, and Windows. We will cover a few examples below that should give you an idea of how to get the incoming NMEA data into the software of your choice.

The software options for Apple iOS are much more limited because Apple products do not support Bluetooth SPP. That's ok! The SparkFun RTK products support additional connection options including TCP and Bluetooth Low Energy (BLE).

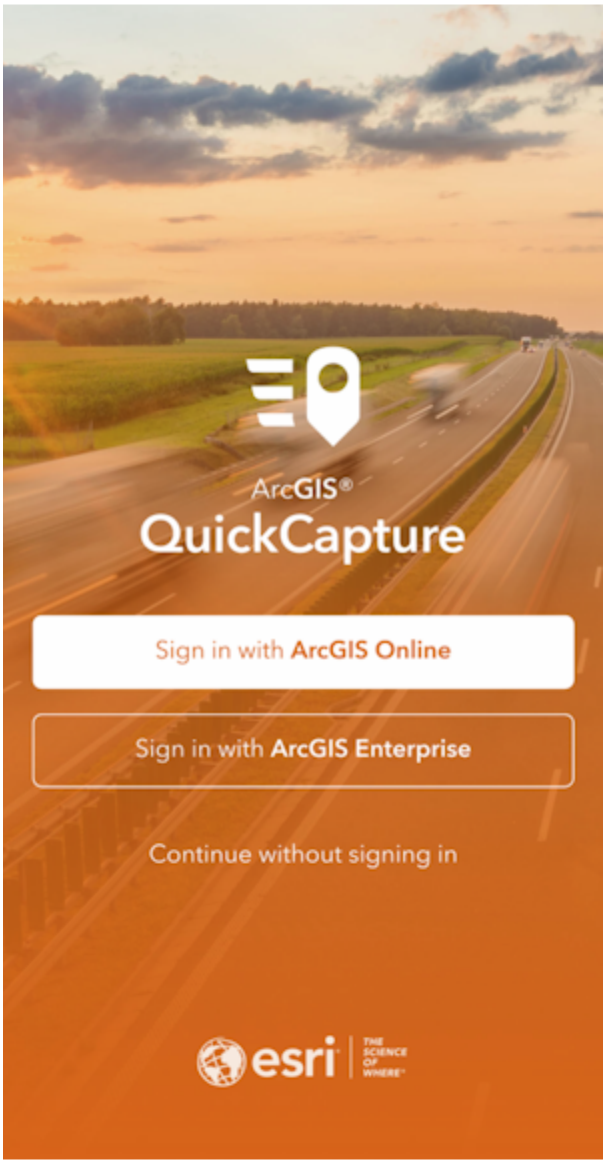
4.2.1 ArcGIS Field Maps

For reasons unknown, Esri removed TCP support from Field Maps for iOS and is therefore not usable by SparkFun RTK devices at this time.

If you must use iOS, checkout SW Maps, ArcGIS QuickCapture, or ArcGIS Survey123.

Field Maps for Android is supported.

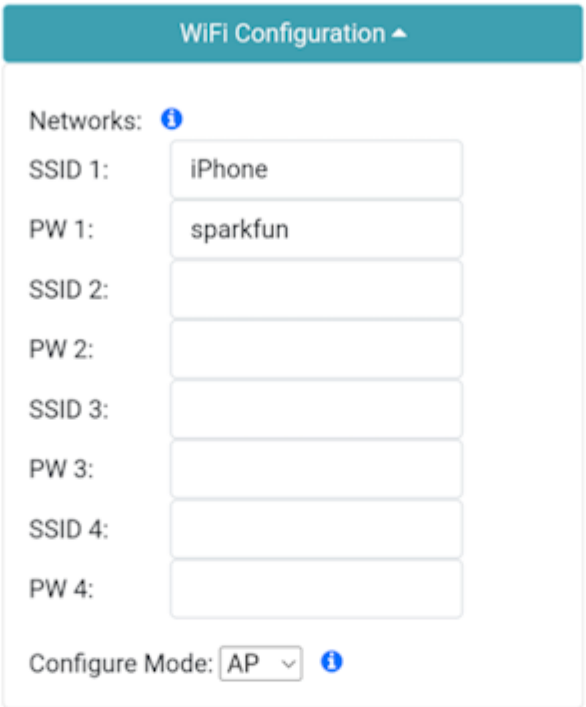
4.2.2 ArcGIS QuickCapture



ArcGIS QuickCapture is a popular offering from Esri that works well with SparkFun RTK products.

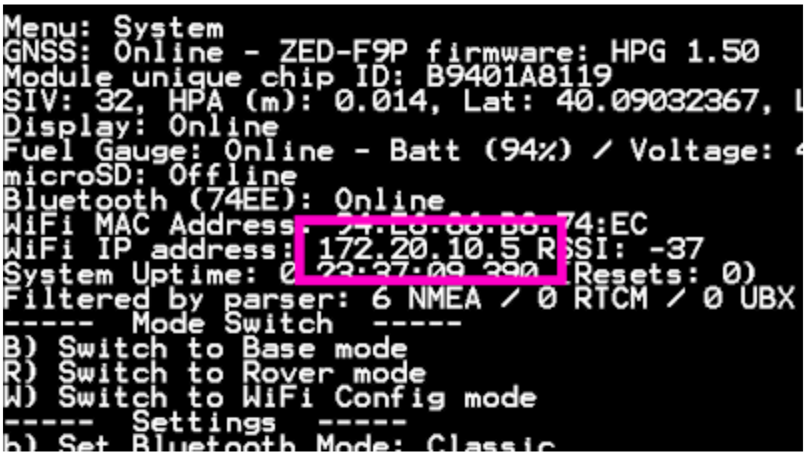
ArcGIS QuickCapture connects to the RTK device over TCP. In other words, the RTK device needs to be connected to the same WiFi network as the device running QuickCapture. Generally, this is an iPhone or iPad operating as a hotspot.

Note: The iOS hotspot defaults to 5.5GHz. This must be changed to 2.4GHz. Please see Hotspot Settings.

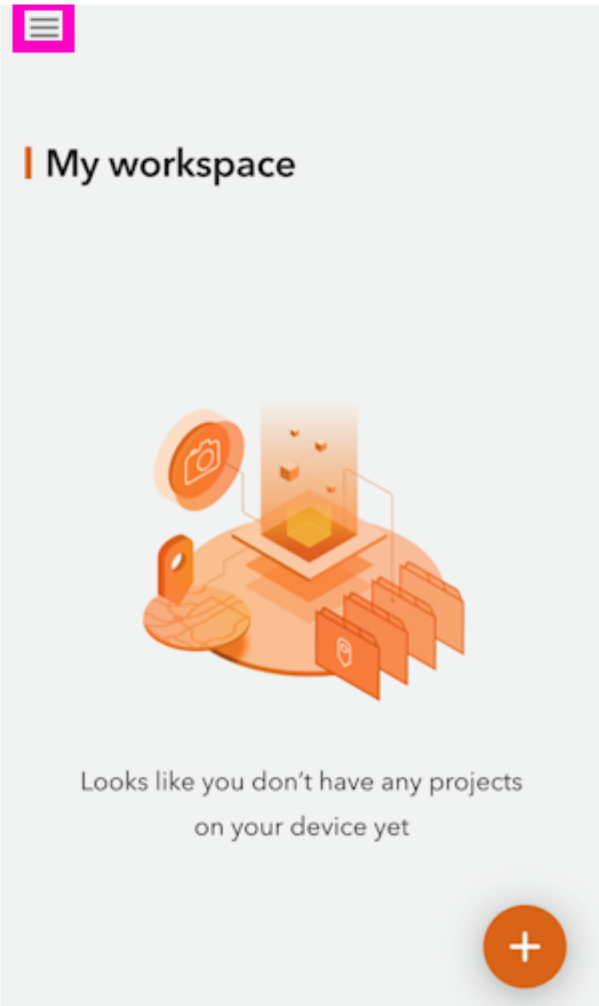


The RTK device must use WiFi to connect to the iPhone or iPad. In the above image, the device will attempt to connect to *iPhone* (a cell phone hotspot) when WiFi is needed.

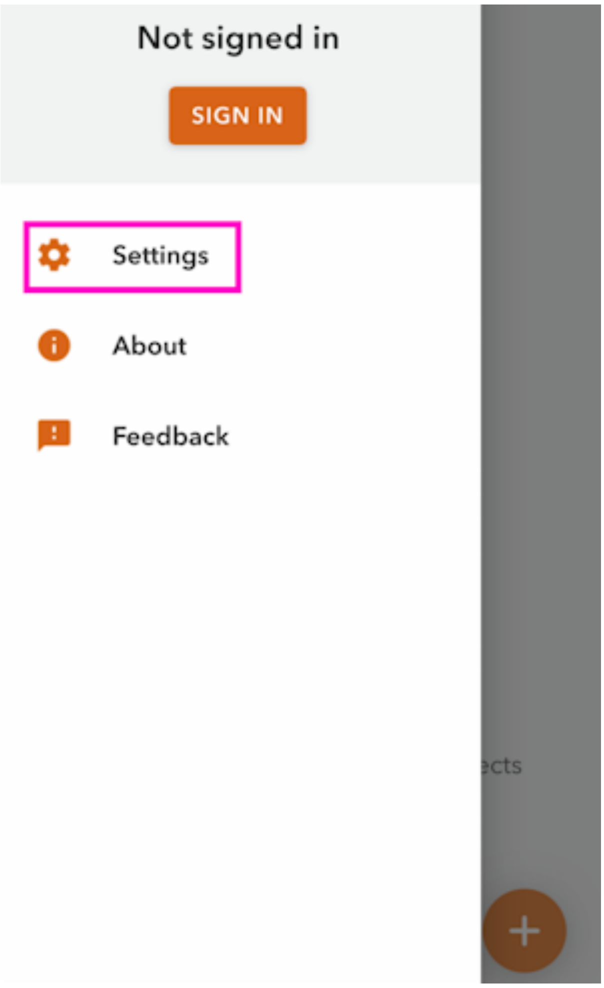
Next, the RTK device must be configured as a *PVT Server*. The default port of 2948 works well. See TCP/UDP Menu for more information.



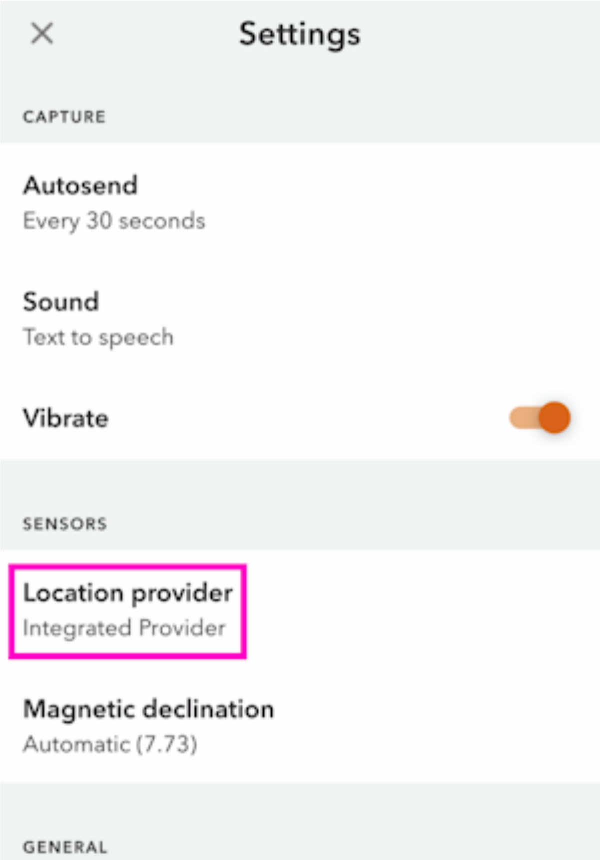
Once the RTK device connects to the WiFi hotspot, its IP address can be found in the System Menu. This is the number that needs to be entered into QuickCapture. You can now proceed to the QuickCapture app to set up the software connection.



From the main screen, press the hamburger icon in the top left corner.

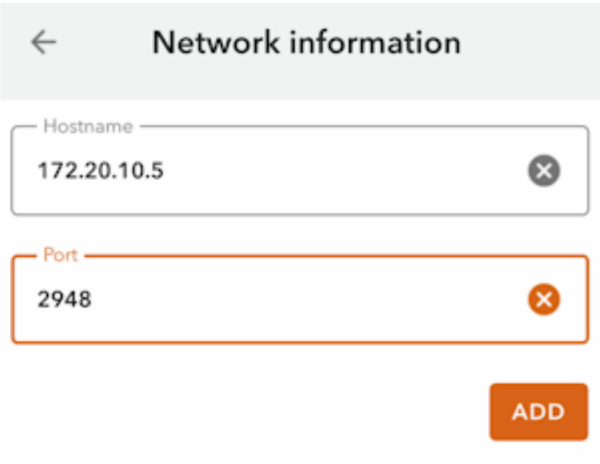


Press the **Settings** button.

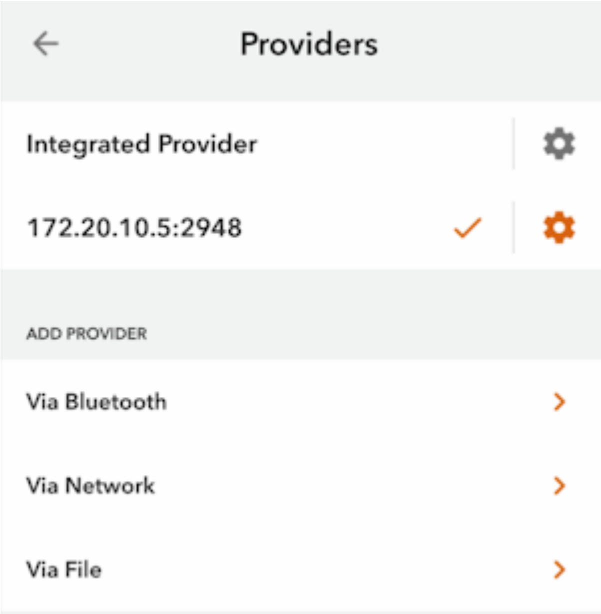


Select the **Location Provider** option.

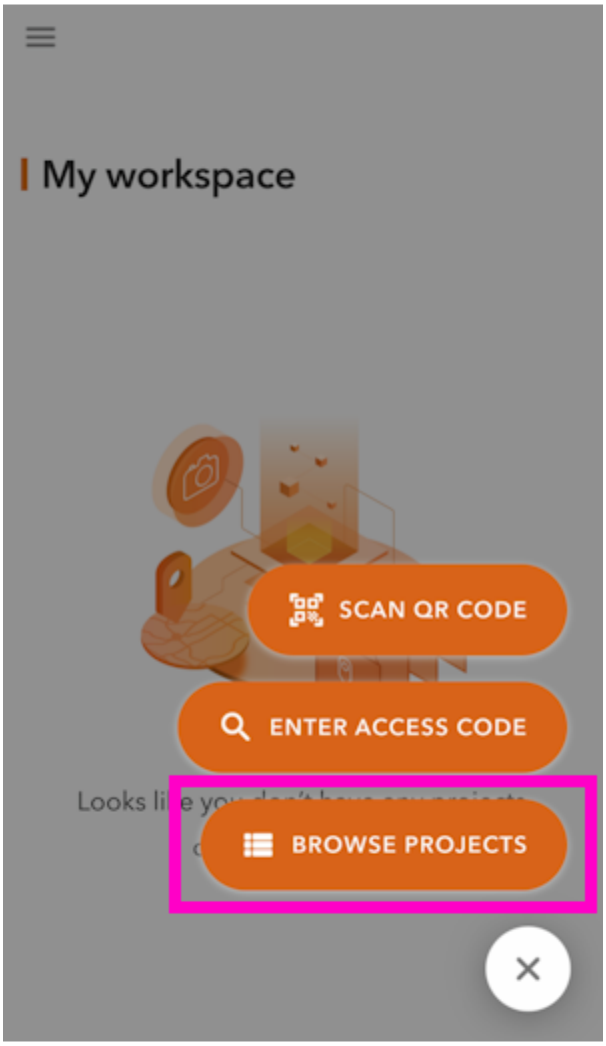
Select **Via Network**.



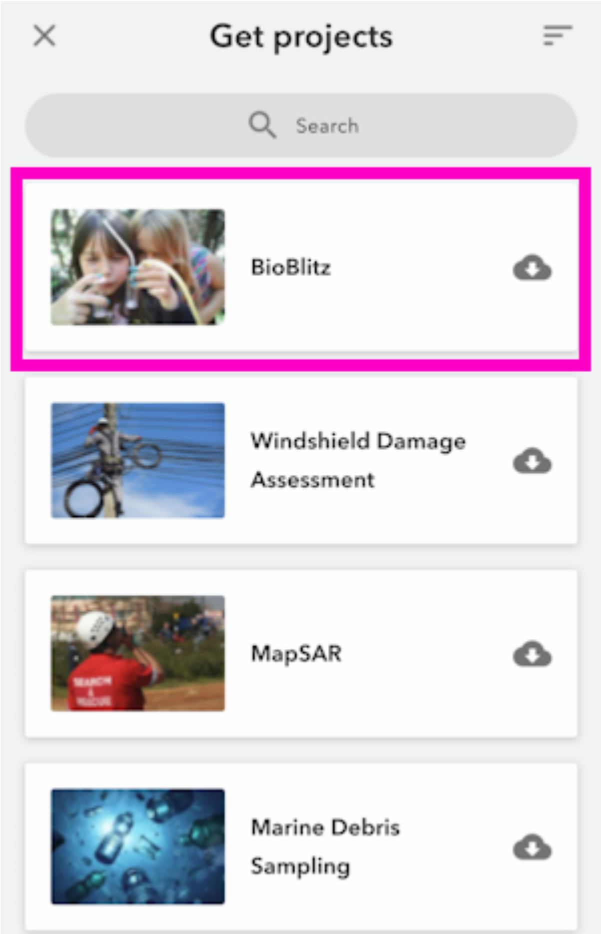
Enter the IP address and port previously obtained from the RTK device and click **ADD**.



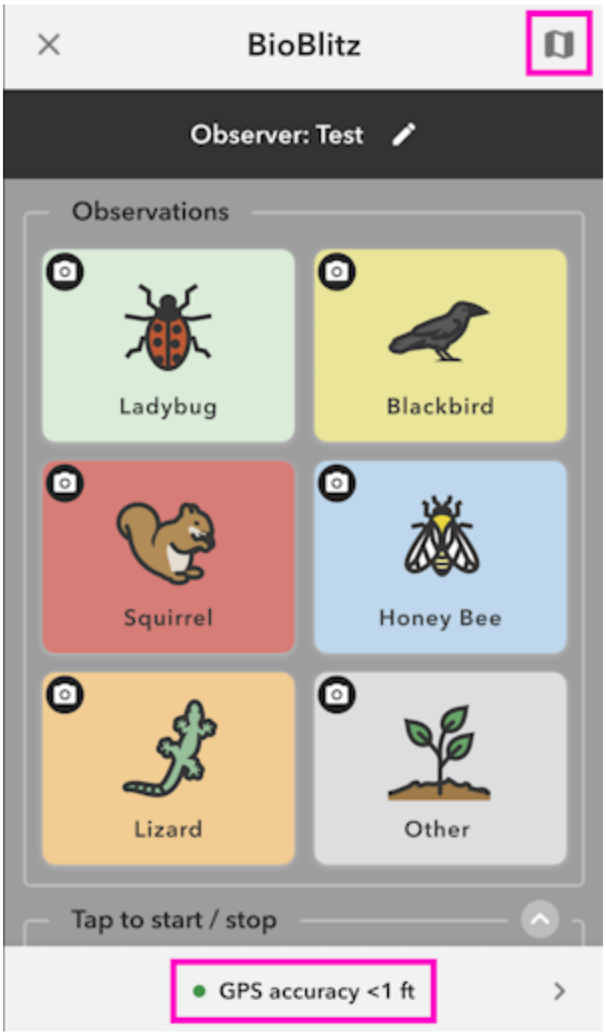
That provider should now be shown connected.



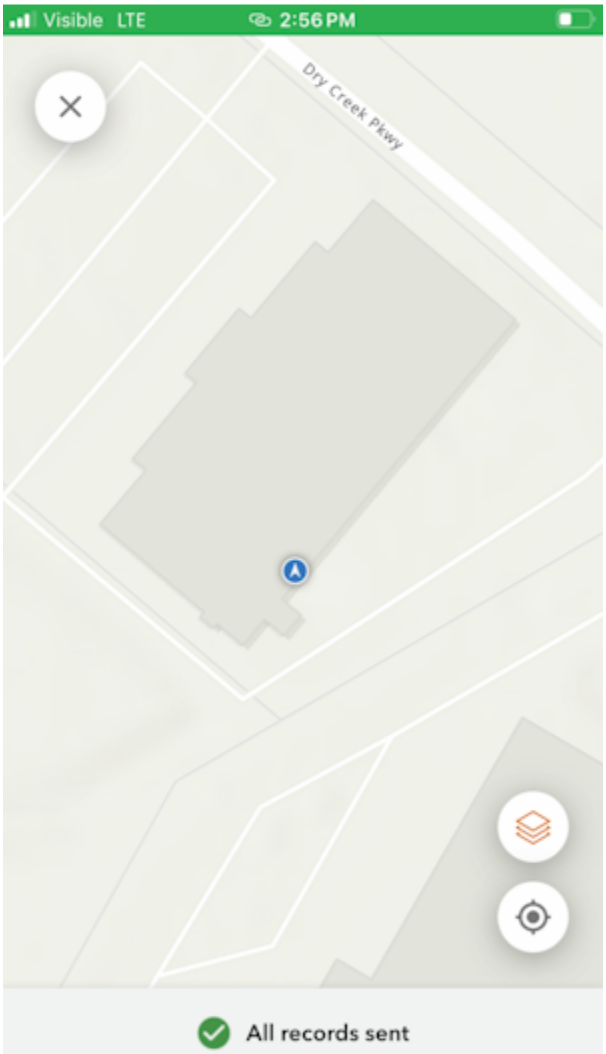
From the main screen, click on the plus in the lower left corner and then **BROWSE PROJECTS**.



For this example, add the BioBlitz project.



Above, we can see the GPS accuracy is better than 1ft. Click on the map icon in the top right corner.



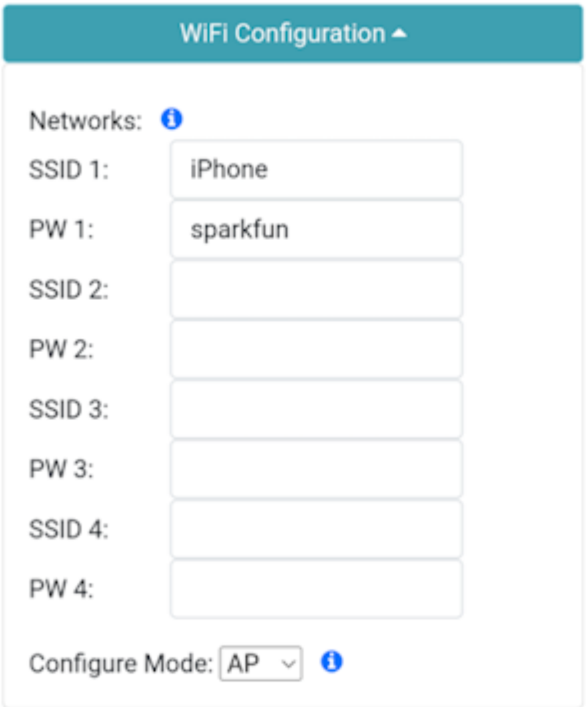
From the map view, we can see our location with very high accuracy. We can now begin gathering point information with millimeter accuracy.

4.2.3 ArcGIS Survey123

ArcGIS Survey123 Home Screen

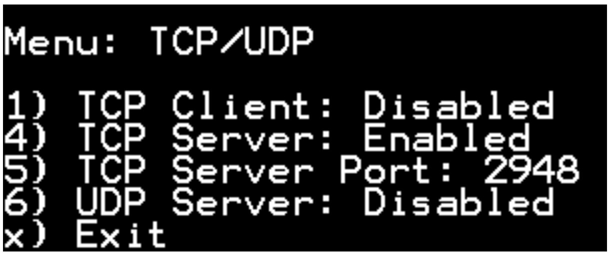
ArcGIS Survey123 is a popular offering from Esri that works well with SparkFun RTK products.

ArcGIS Survey123 connects to the RTK device over TCP. In other words, the RTK device needs to be connected to the same WiFi network as the device running ArcGIS. Generally, this is an iPhone or iPad.



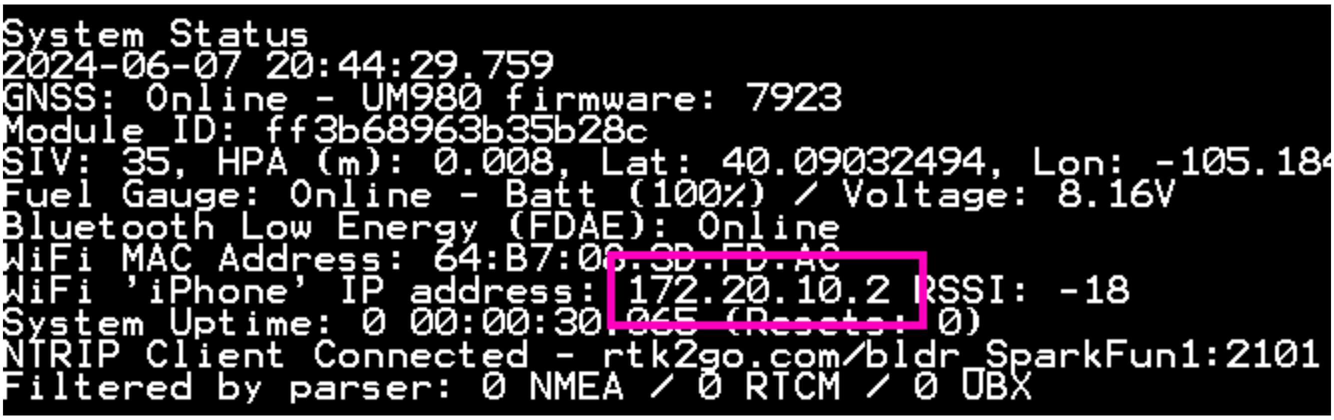
Adding WiFi network to settings

The RTK device must use WiFi to connect to the data collector. Using a cellular hotspot or cellphone is recommended. In the above image, the device will attempt to connect to *iPhone* (a cell phone hotspot) when WiFi is needed.



TCP Server Enabled on port 2948

Next, the RTK device must be configured as a *TCP Server*. The default port of 2948 works well. See TCP/UDP Menu for more information.

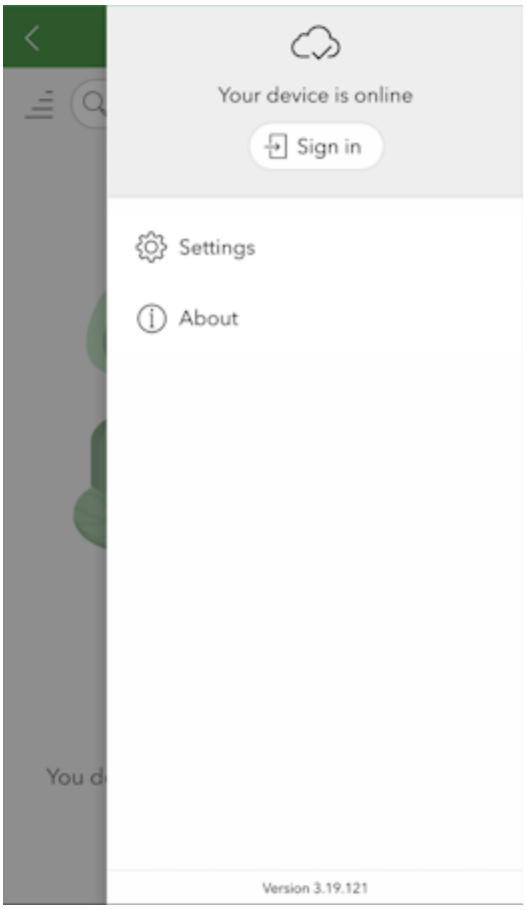


RTK device showing IP address

Once the RTK device connects to the WiFi hotspot, its IP address can be found in the System Menu. This is the number that needs to be entered into ArcGIS Survey123. You can now proceed to the ArcGIS Survey123 app to set up the software connection.

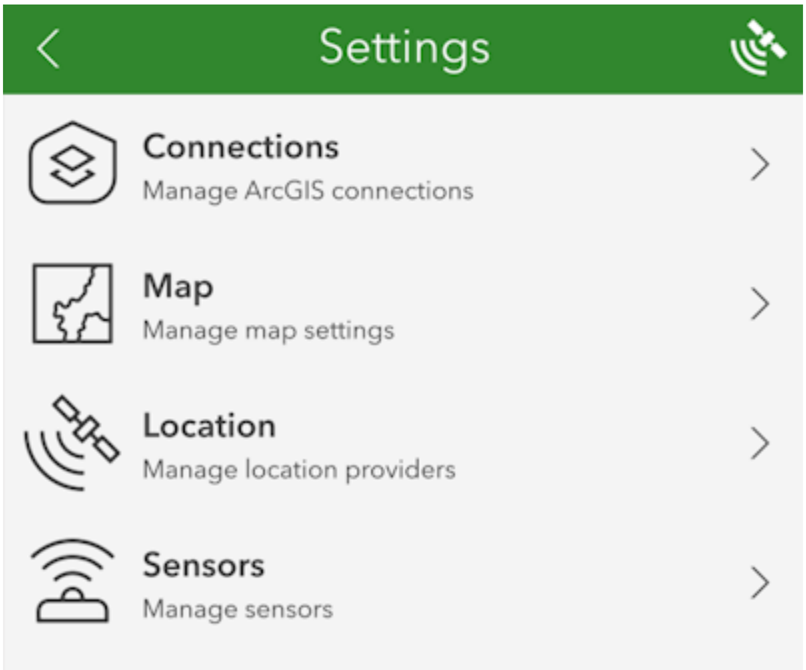
ArcGIS Survey123 Home Screen

From the home screen, click on the 'hamburger' icon in the upper right corner.



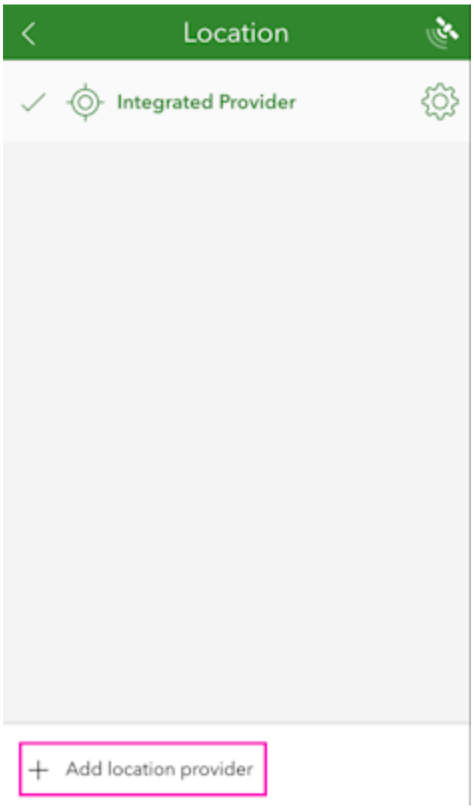
ArcGIS Survey123 Settings Menu

From the settings menu, click on the *Settings* gear.



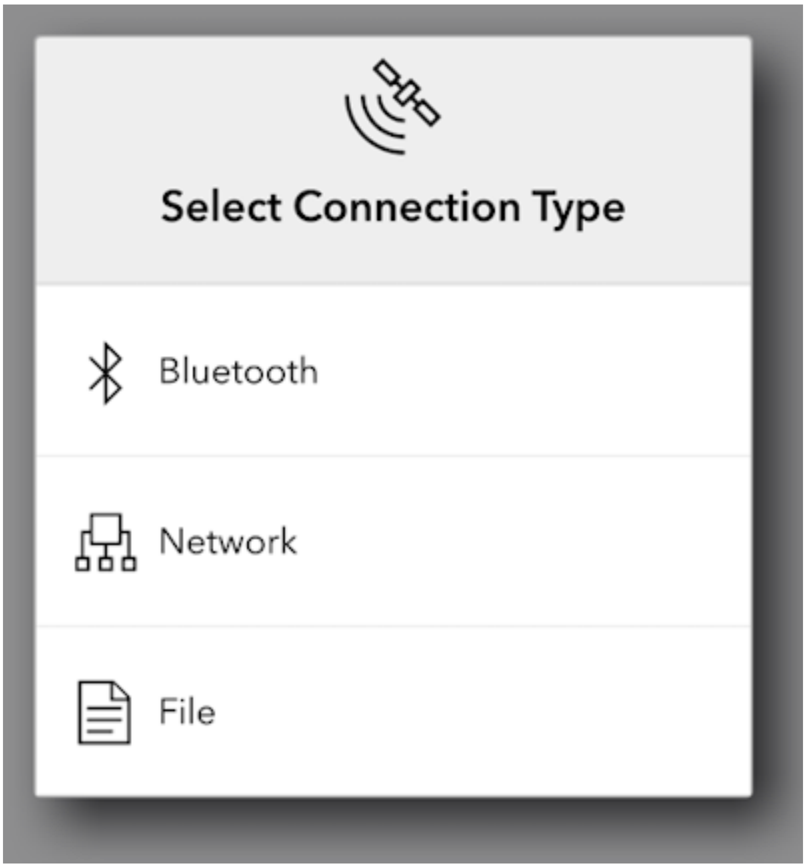
ArcGIS Survey123 Settings List

From the settings list, click on *Location*.



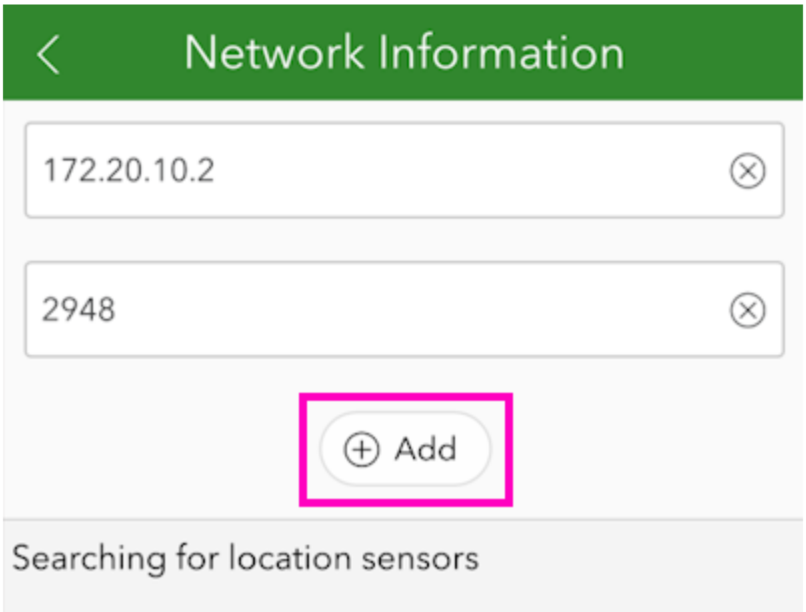
ArcGIS Survey123 List of Location Providers

Click on the *Add location provider*.



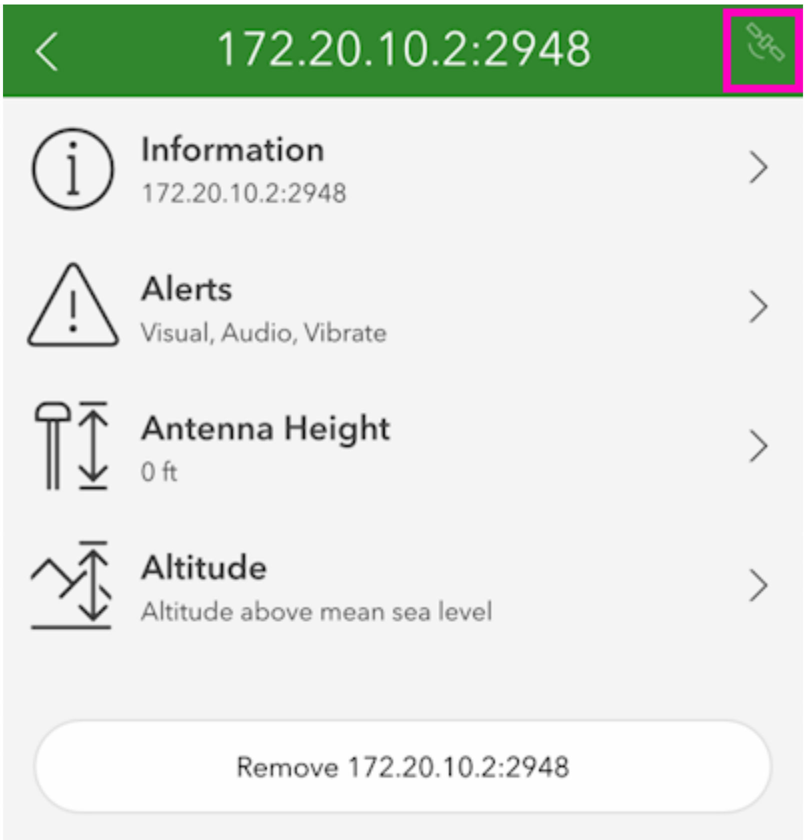
ArcGIS Survey123 Network Connection Type

Select *Network*.



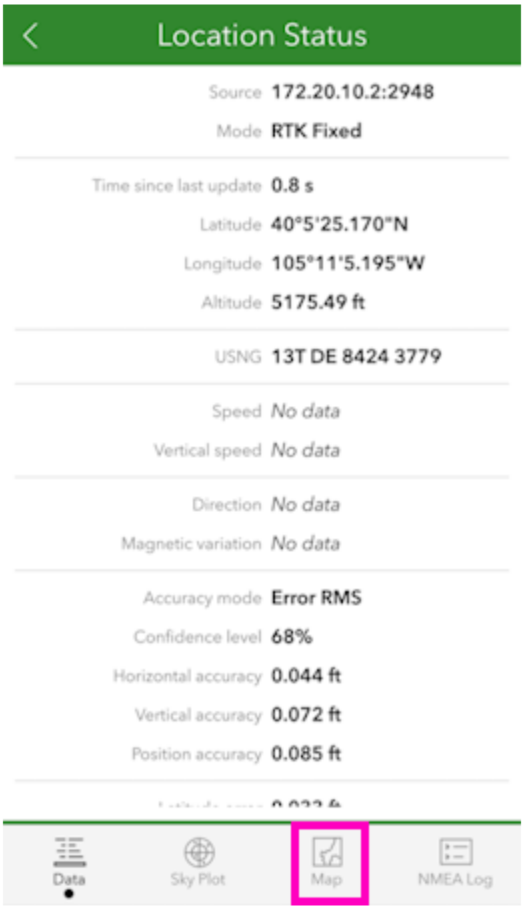
ArcGIS Survey123 TCP Connection Information

Enter the IP address previously found along with the TCP port. Once complete, click *Add*.



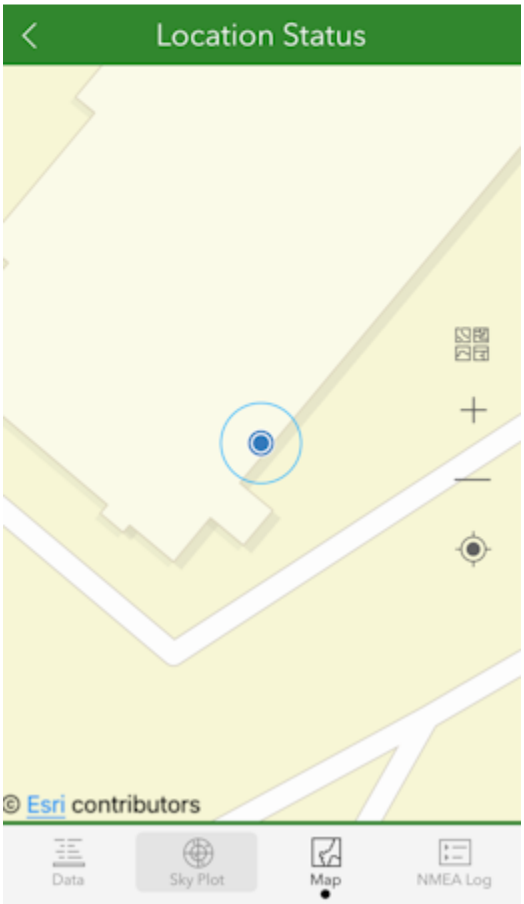
ArcGIS Survey123 Sensor Settings

You may enter various sensor-specific settings including antenna height, if desired. To view real-time sensor information, click on the satellite icon in the upper right corner.



ArcGIS Survey123 Sensor Data

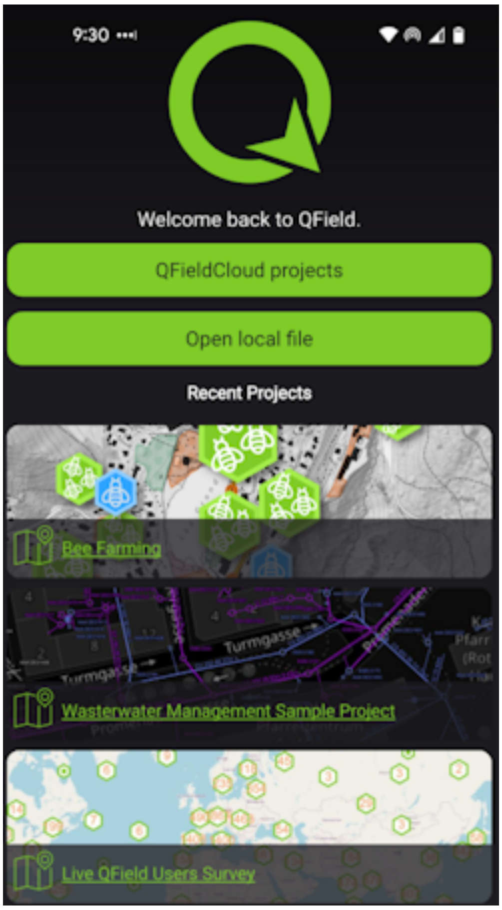
The SparkFun RTK device's data should now be seen. Click on the *Map* icon to return to the mapping interface.



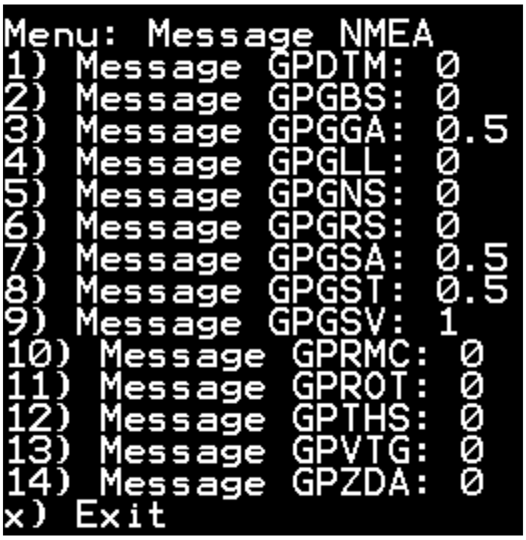
ArcGIS Survey123 Map Interface

Returning to the map view, we can now begin gathering point information with millimeter accuracy.

4.2.4 QField



QField is a free iOS app that runs QGIS.



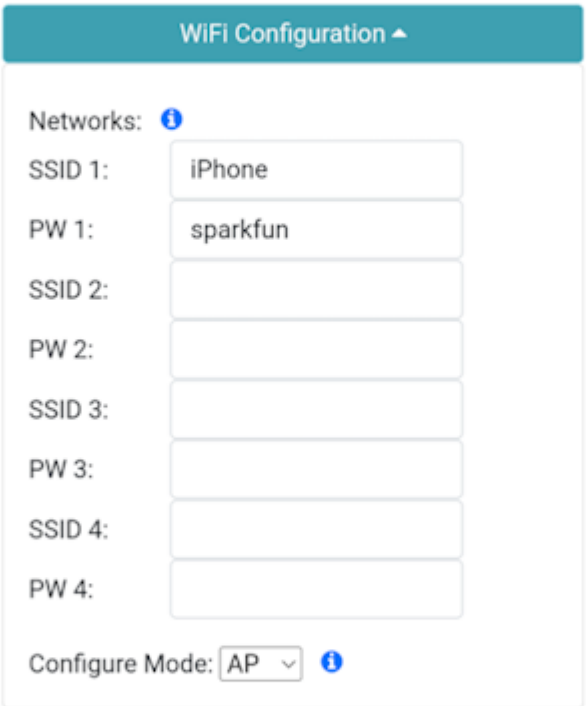
Modified NMEA messages on RTK Torch

First, configure the RTK device to output *only* the following NMEA messages:

- GPGGA
- GPGSA
- GPGST
- GPGSV

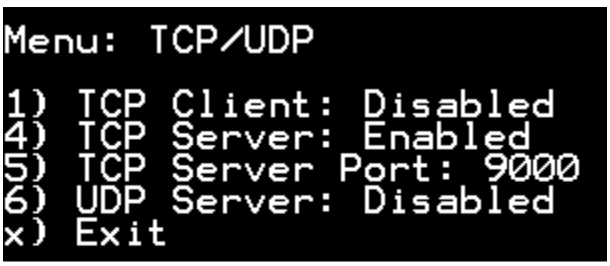
QField currently does not correctly parse other messages such as **GPRMC**, or **RTCM**. These messages will prevent communication if they are enabled.

These NMEA message settings can be found under the Messages menu, using the web config page or the serial config interface.



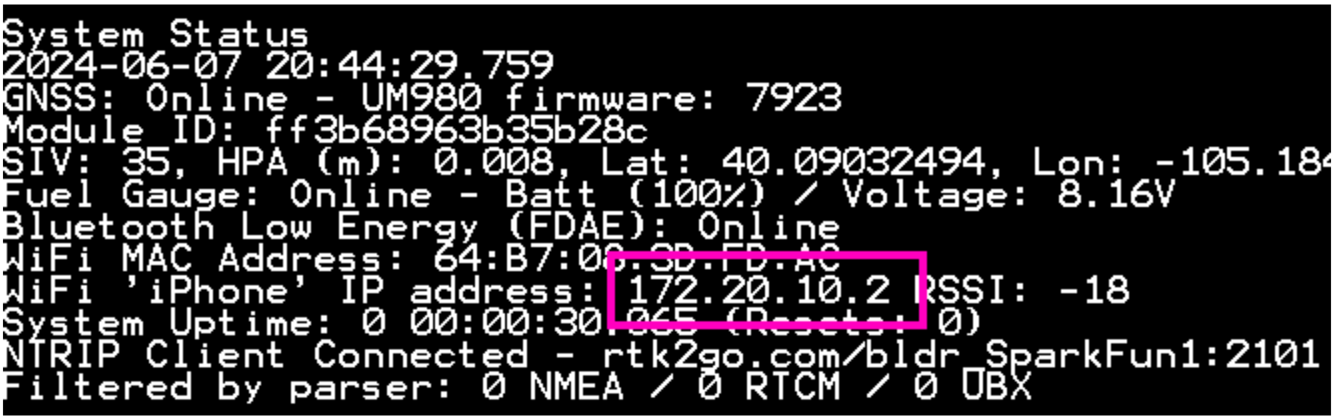
Adding WiFi network to settings

Next, the RTK device must use WiFi to connect to the data collector. Using a cellular hotspot or cellphone is recommended. In the above image, the device will attempt to connect to *iPhone* (a cell phone hotspot) when WiFi is needed.



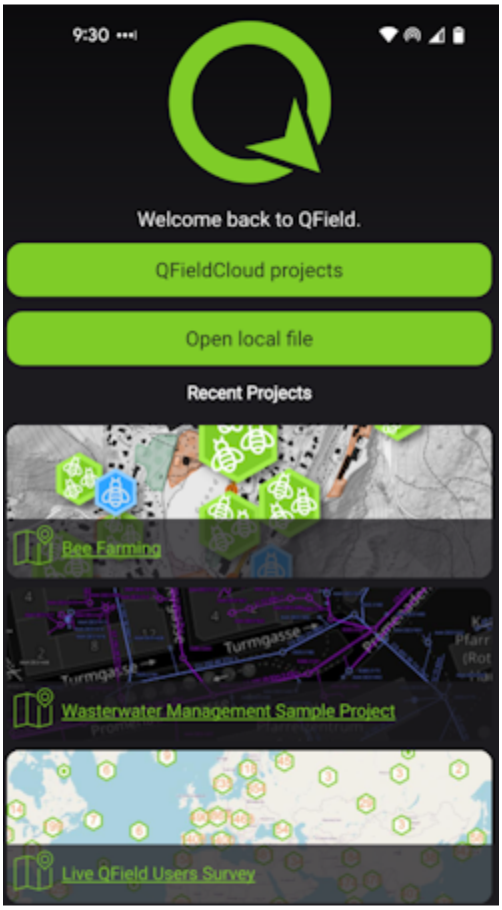
TCP Server Enabled on port 9000

Next, the RTK device must be configured as a *TCP Server*. QField uses a default port of 9000 so that is what we recommend using. See TCP/UDP Menu for more information.



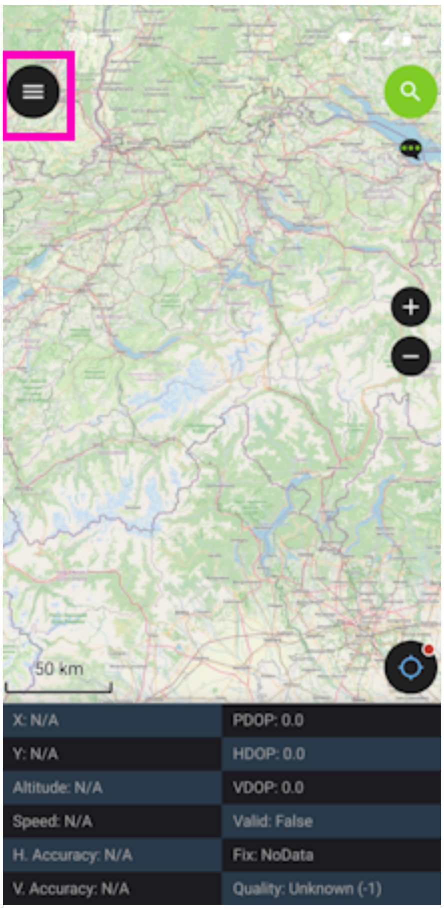
RTK device showing IP address

Once the RTK device connects to the WiFi hotspot, its IP address can be found in the System Menu. This is the number that needs to be entered into QField. You can now proceed to the QField app to set up the software connection.



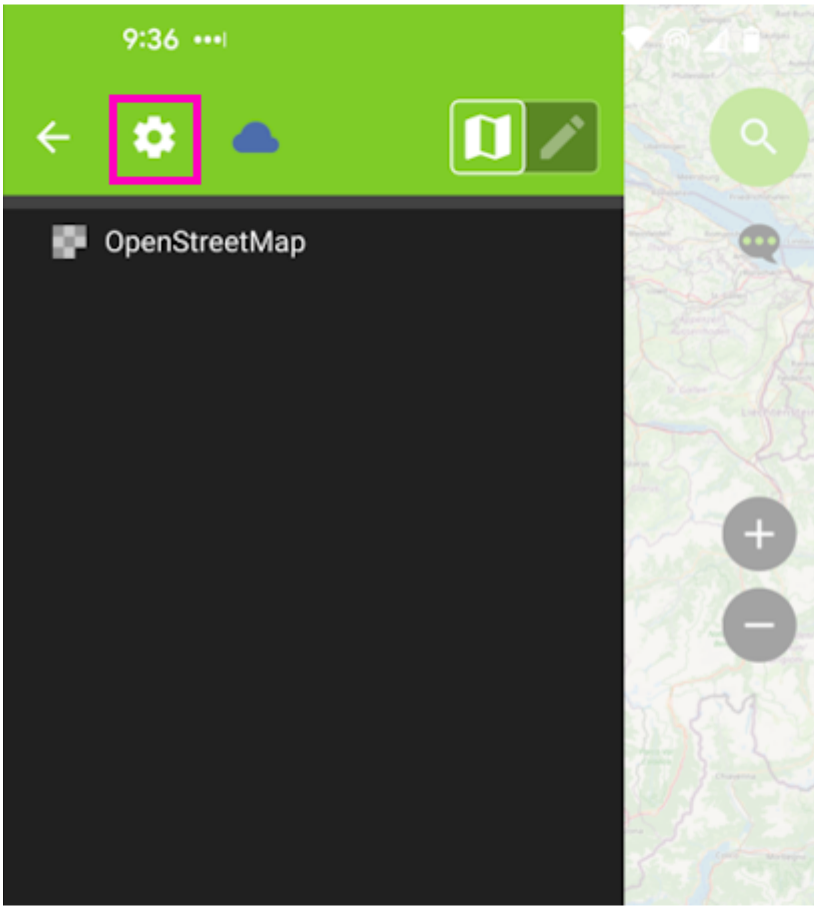
QField Opening Screen

Click on *QFieldCloud projects* to open your project that was previously created on the QField Cloud or skip this step by using one of the default projects (*Bee Farming*, *Wastewater*, etc).



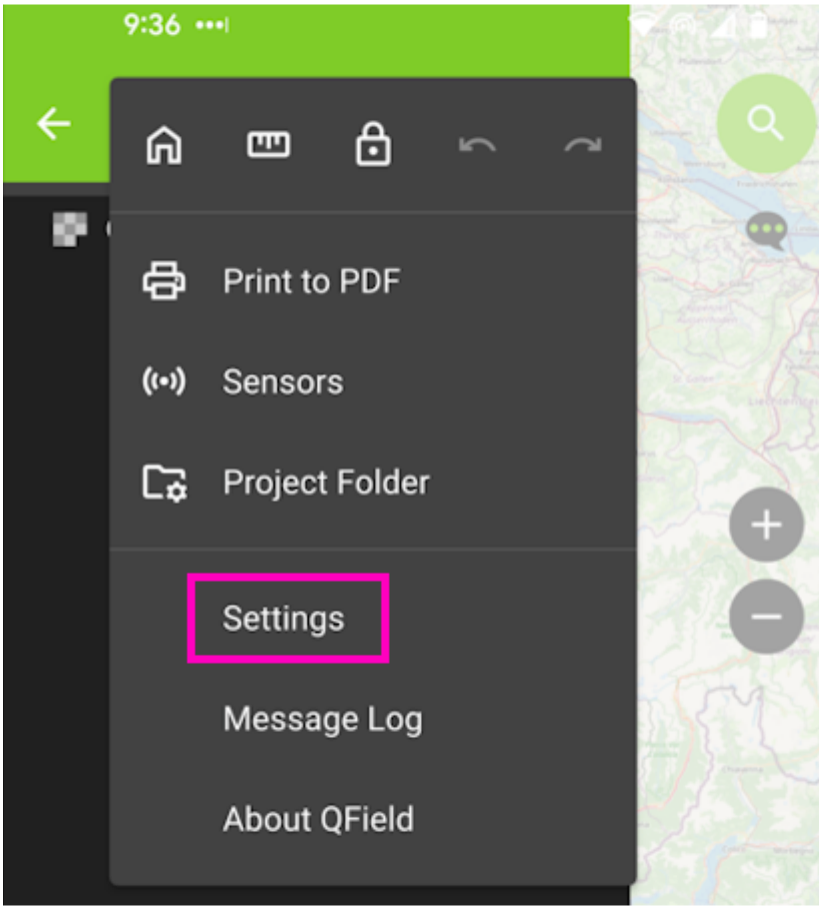
QField Main Map

From the main map, click on the 'hamburger' icon in the upper left corner.

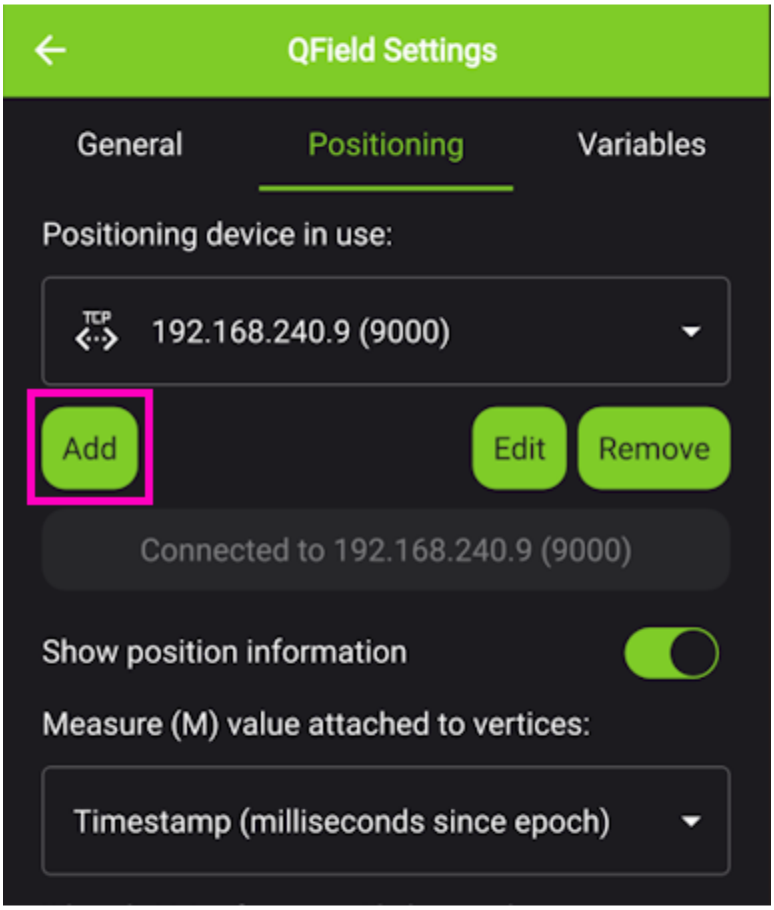


QField Settings Gear

Click on the gear to open settings.

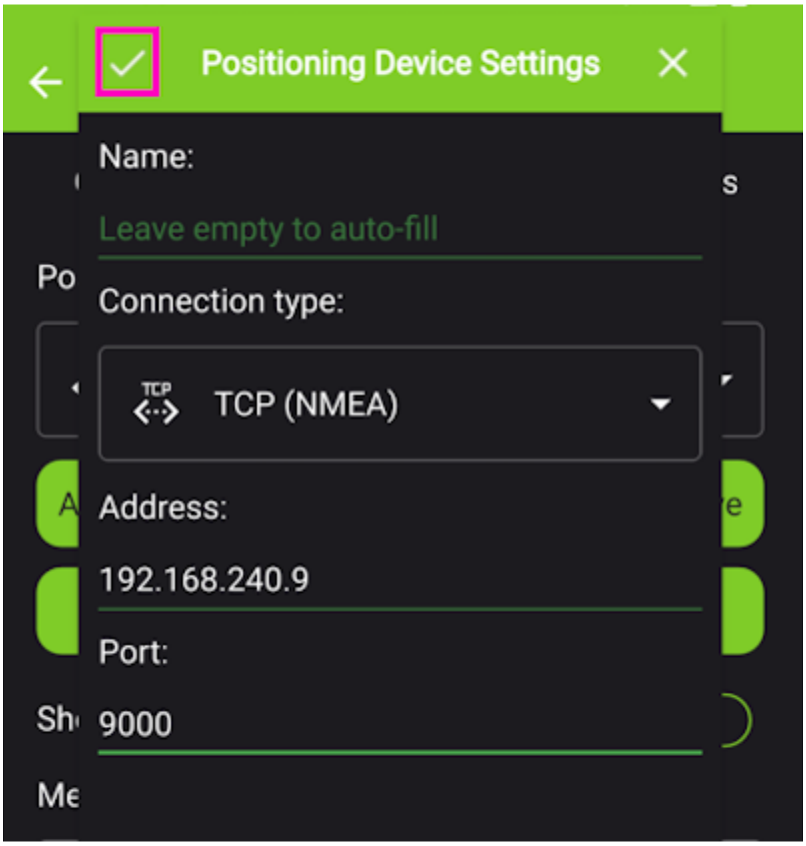


Click on the *Settings* menu.



QField Positioning Menu

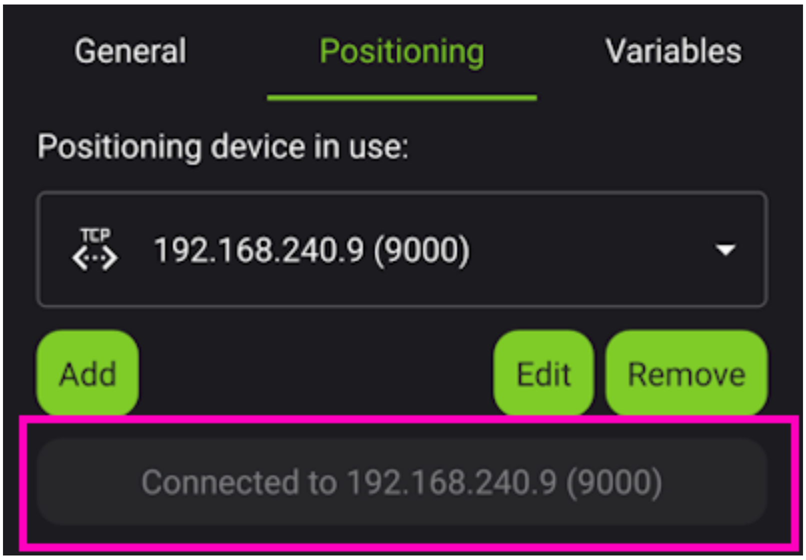
From the *Positioning* menu, click Add.



QField Entering TCP Information

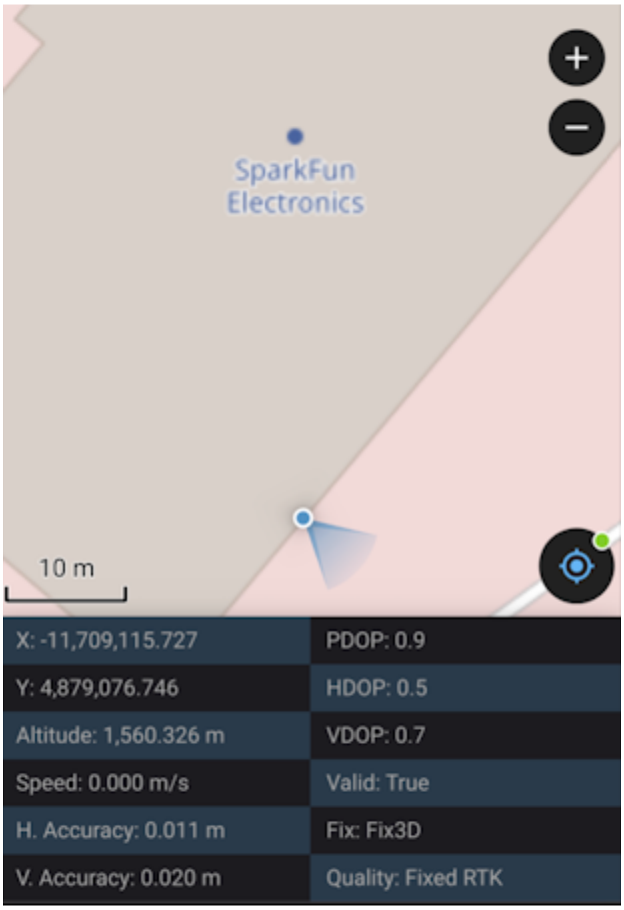
Select TCP as the connection type. Enter the IP address of the RTK device and the port number. Finally, hit the small check box in the upper left corner (shown in pink above) to close the window.

Once this information is entered, QField will automatically attempt to connect to that IP and port.



QField TCP Connected

Above, we see the port is successfully connected. Exit out of all menus.



QField Connected via TCP with RTK Fix

Returning to the map view, we see an RTK Fix with 11mm positional accuracy.

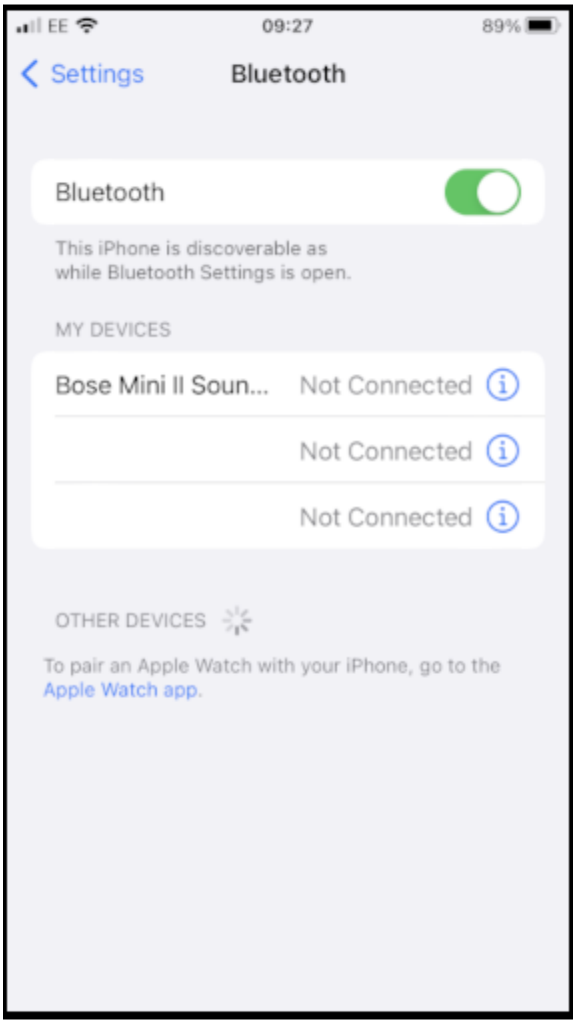
4.2.5 SW Maps

SWMaps is available for iOS here.

Make sure your RTK device is switched on and operating in Rover mode.

Make sure Bluetooth is enabled on your iOS device Settings.

The RTK device will not appear in the *OTHER DEVICES* list. That is OK.

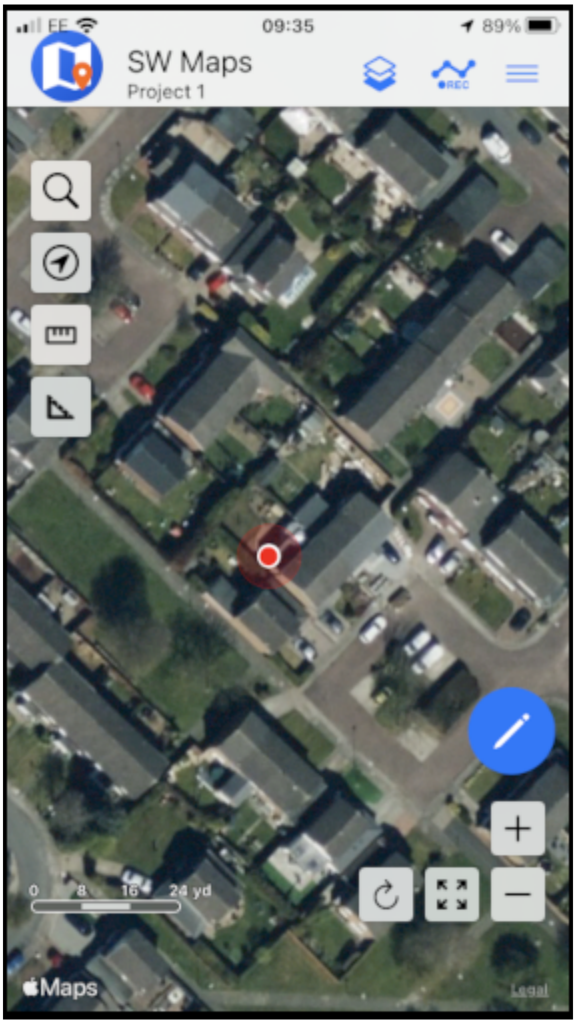


iOS Settings Bluetooth

Open SWMaps.

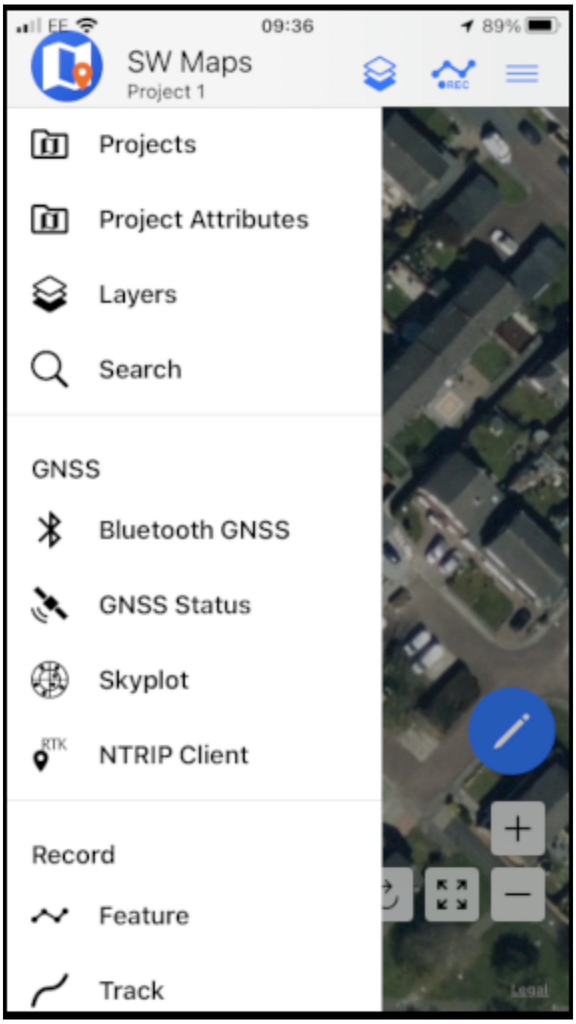
Open or continue a Project if desired.

SWMaps will show your approximate location based on your iOS device's location.



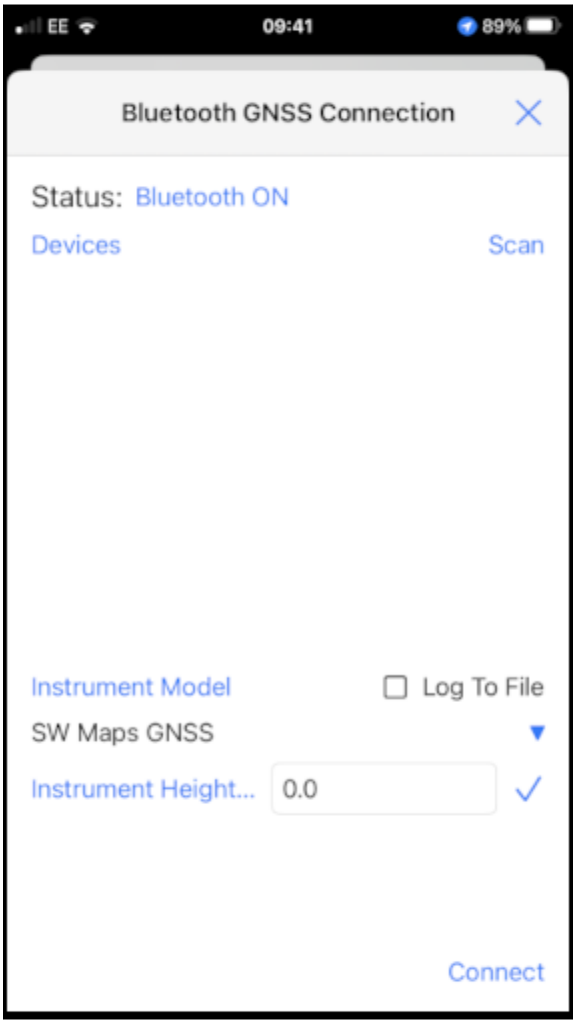
iOS SWMaps Initial Location

Press the 'SWMaps' icon at the top left of the screen to open the menu.



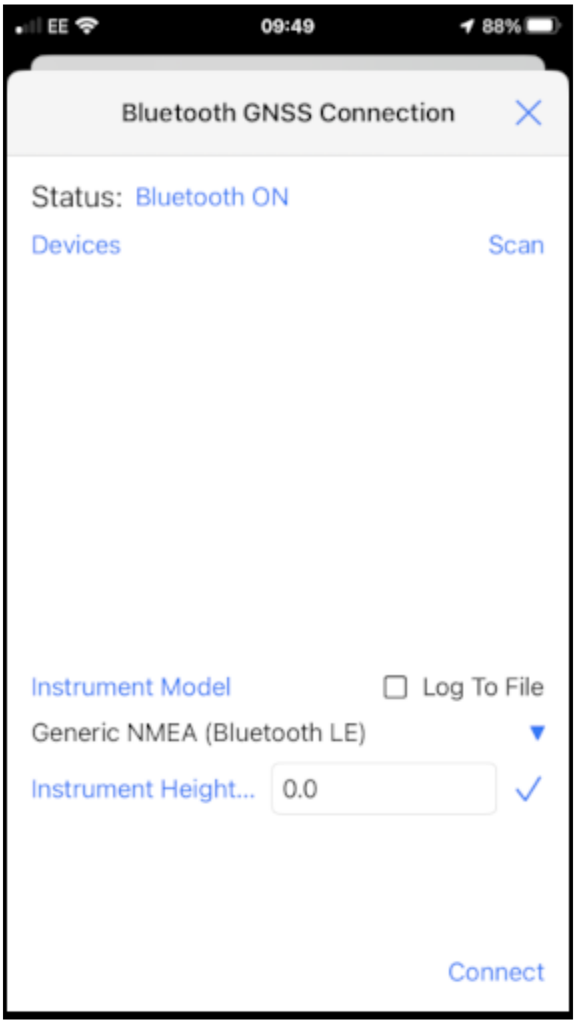
iOS SWMaps Menu

Select Bluetooth GNSS.



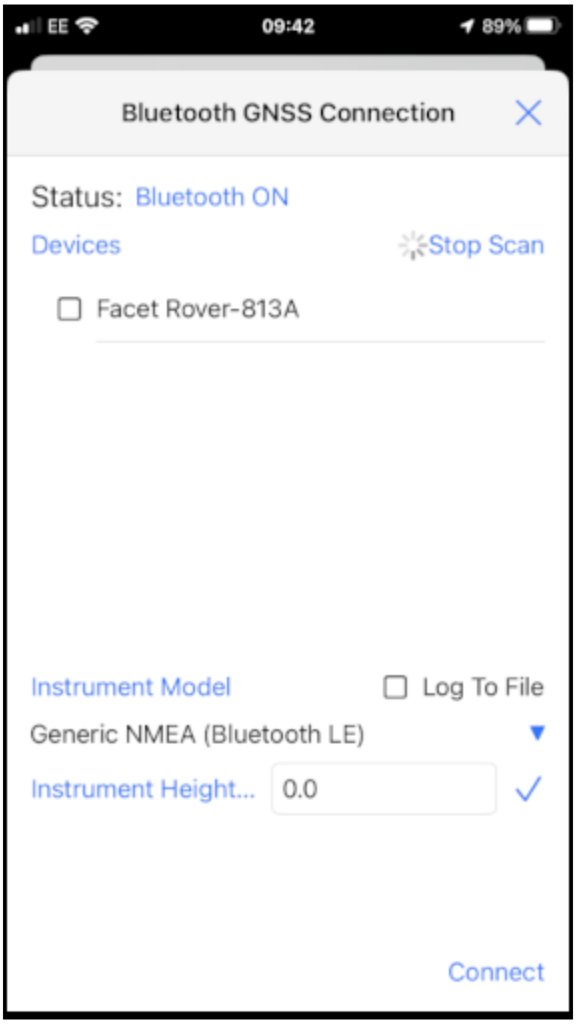
iOS SWMaps Bluetooth Connection

Set the **Instrument Model** to **Generic NMEA (Bluetooth LE)**.



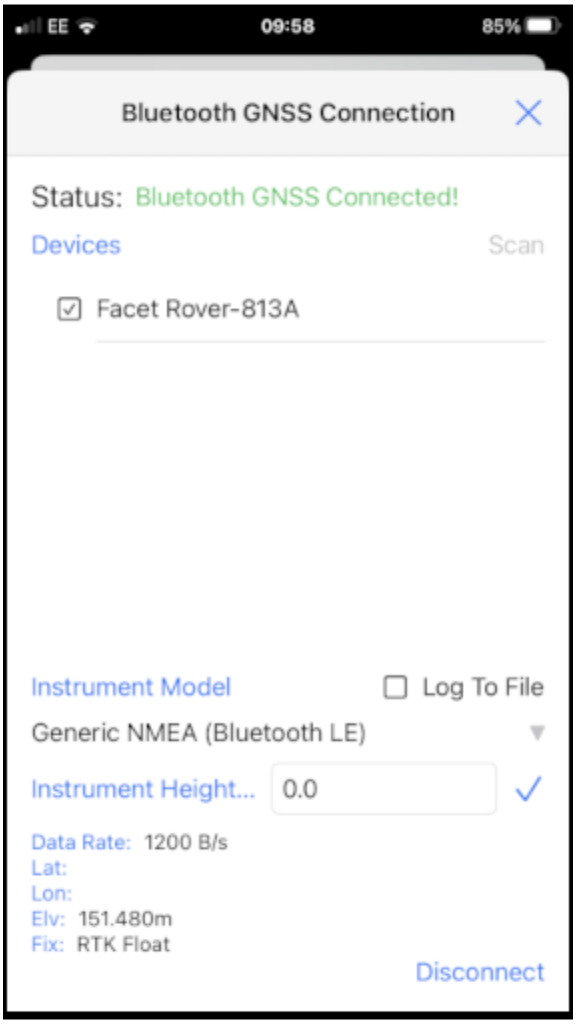
iOS SWMaps Instrument Model

Press 'Scan' and your RTK device should appear.



iOS SWMaps Bluetooth Scan

Select (tick) the RTK device and press 'Connect'.



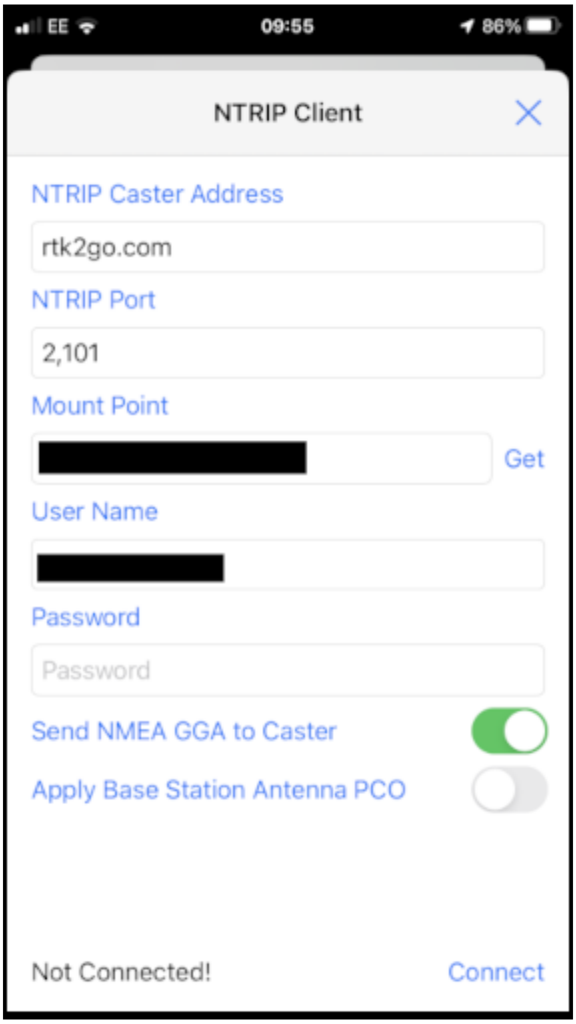
iOS SWMaps Bluetooth Connected

Close the menu and your RTK location will be displayed on the map.

You can now use the other features of SWMaps, including the built-in NTRIP Client.

Re-open the menu and select 'NTRIP Client'.

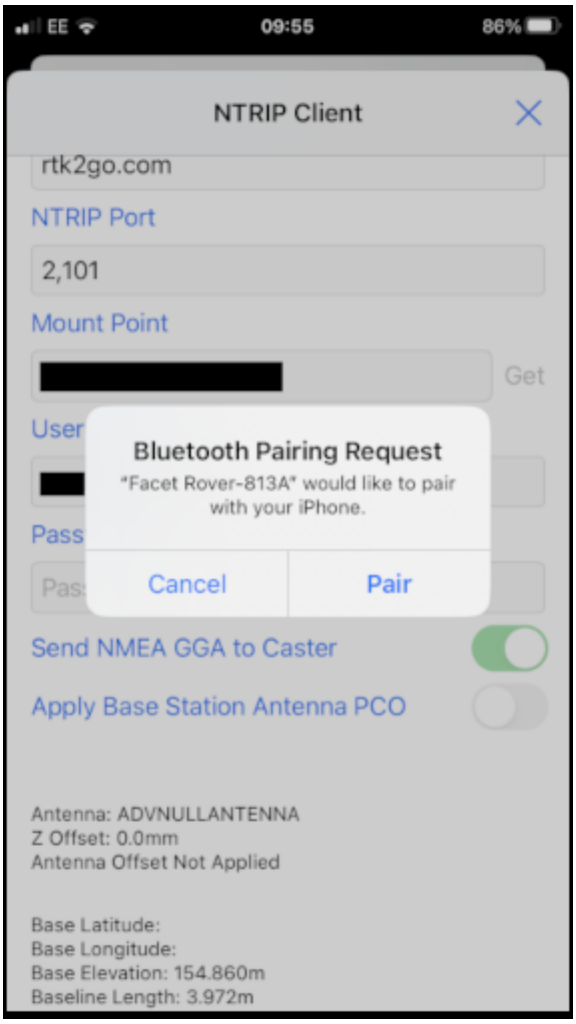
Enter the details for your NTRIP Caster - as shown in the SWMaps section above.



iOS SWMaps NTRIP Client

Click 'Connect'

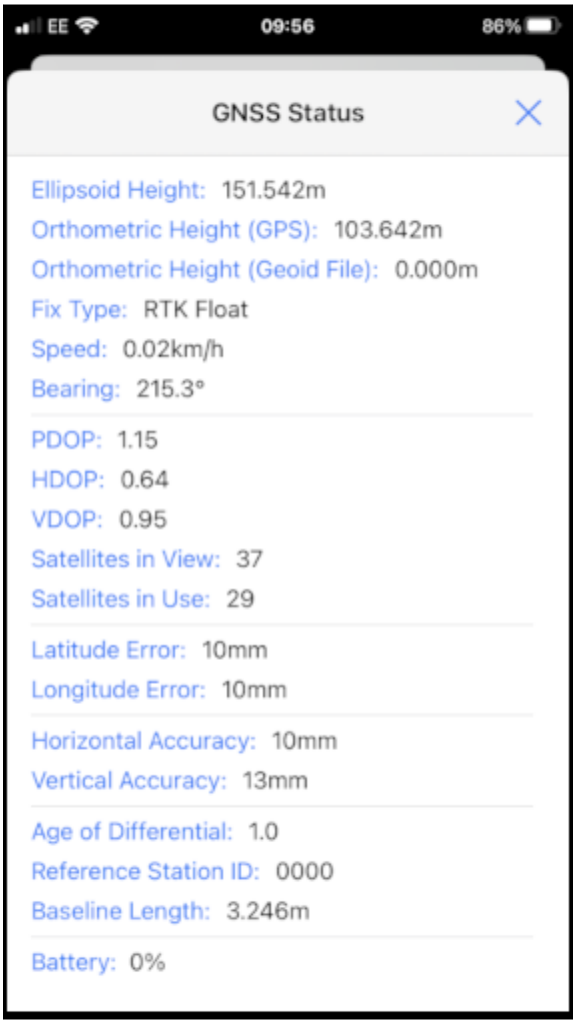
At this point, you should see a Bluetooth Pairing Request. Select 'Pair' to pair your RTK with your iOS device.



iOS Bluetooth Pairing

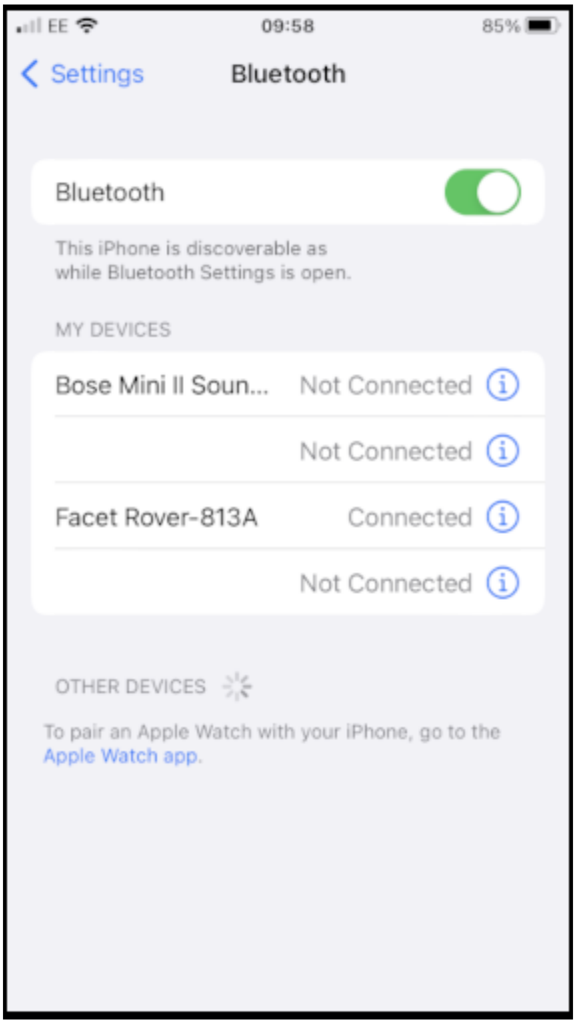
SWMaps will now receive NTRIP correction data from the caster and push it to your RTK over Bluetooth BLE.

From the SWMaps menu, open 'GNSS Status' to see your position, fix type and accuracy.



iOS SWMaps GNSS Status

If you return to the iOS Bluetooth Settings, you will see that your iOS and RTK devices are now paired.



iOS Settings Bluetooth - Paired

4.2.6 Other GIS Packages

Hopefully, these examples give you an idea of how to connect the RTK product line to most any GIS software. If there is other GIS software that you'd like to see configuration information about, please open an issue on the RTK Firmware repo and we'll add it.

4.2.7 What's an NTRIP Caster?

In a nutshell, it's a server that is sending out correction data every second. There are thousands of sites around the globe that calculate the perturbations in the ionosphere and troposphere that decrease the accuracy of GNSS accuracy. Once the inaccuracies are known, correction values are encoded into data packets in the RTCM format. You, the user, don't need to know how to decode or deal with RTCM, you simply need to get RTCM from a source within 10km of your location into the RTK device. The NTRIP client logs into the server (also known as the NTRIP caster) and grabs that data, every second, and sends it over Bluetooth to the RTK device.

4.2.8 Where do I get RTK Corrections?

Be sure to see Correction Sources.

Don't have access to an NTRIP Caster or other RTCM correction source? There are a few options.

The SparkFun RTK Facet L-Band gets corrections via an encrypted signal from geosynchronous satellites. This device gets RTK Fix without the need for a WiFi or cellular connection.

Also, you can use a 2nd RTK product operating in Base mode to provide the correction data. Check out Creating a Permanent Base.

If you're the DIY sort, you can create your own low-cost base station using an ESP32 and a ZED-F9P breakout board. Check out How to Build a DIY GNSS Reference Station.

There are services available as well. Syklark provides RTCM coverage for $49 a month (as of writing) and is extremely easy to set up and use. Point One also offers RTK NTRIP service with a free 14 day trial and easy to use front end.

4.3 Windows

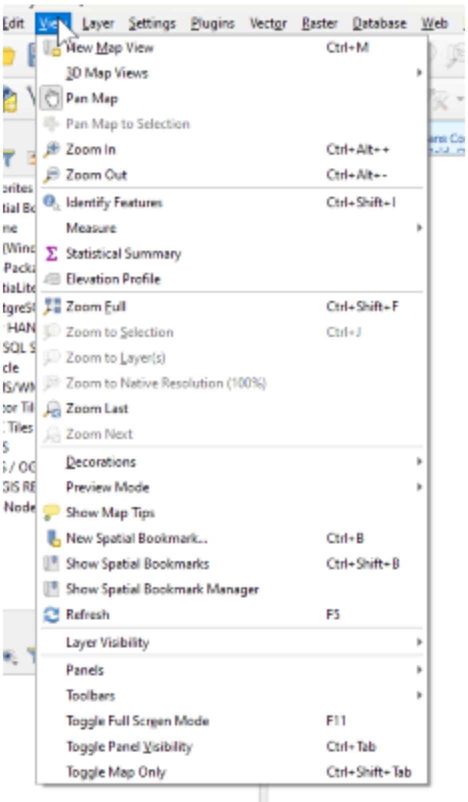
Torch:

There are a variety of 3rd party apps available for GIS and surveying. We will cover a few examples below that should give you an idea of how to get the incoming NMEA data into the software of your choice.

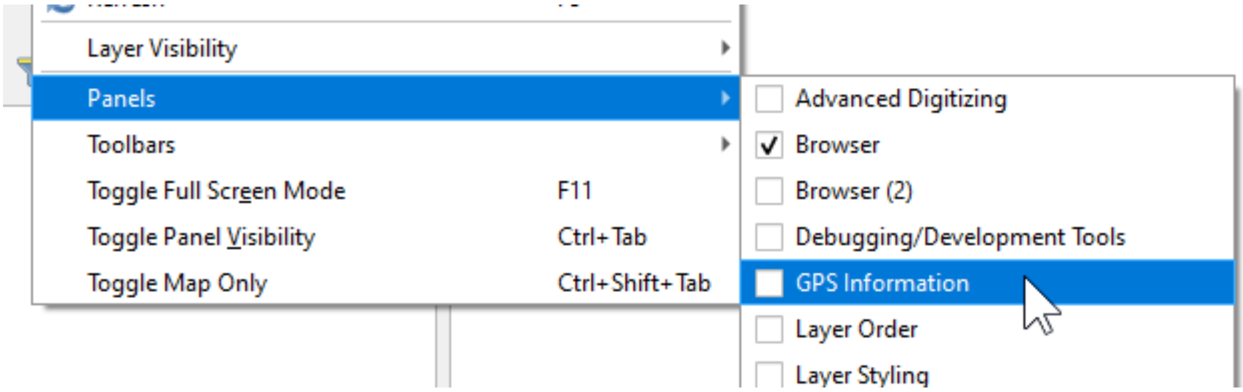
4.3.1 QGIS

QGIS is a free and open-source geographic information system software for desktops. It's available here.

Once the software is installed open QGIS Desktop.

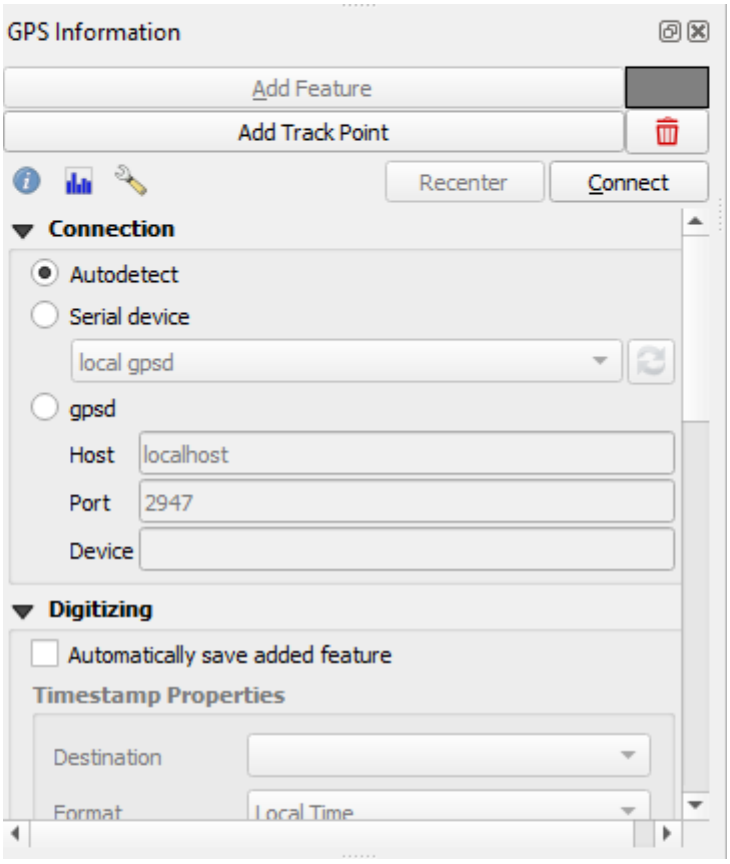


Open the View Menu, then look for the 'Panels' submenu.

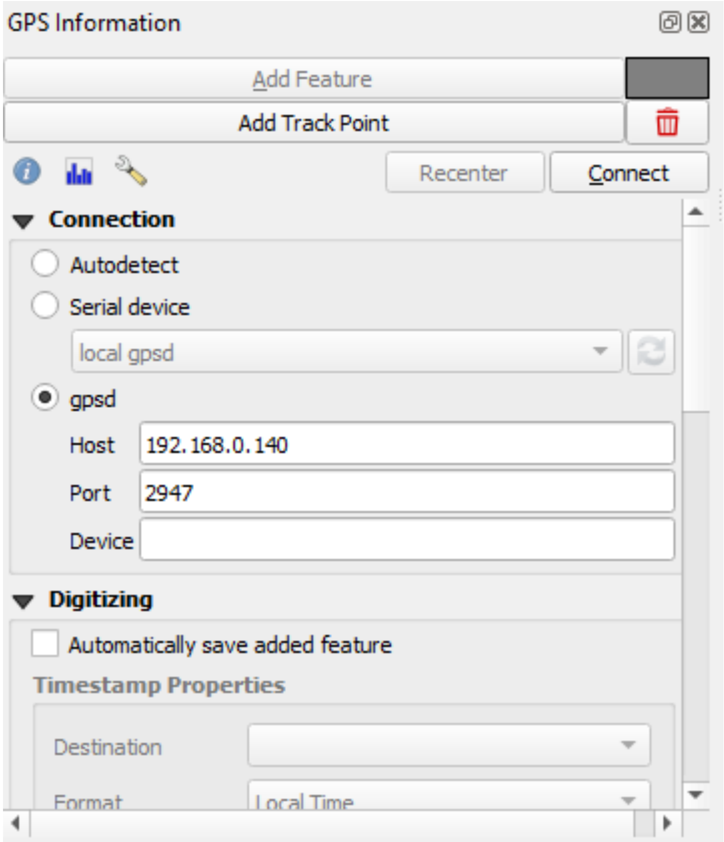


From the Panels submenu, enable 'GPS Information'. This will show a new panel on the left side.

At this point, you will need to enable *TCP Server* mode on your RTK device from the WiFi Config menu. Once the RTK device is connected to local WiFi QGIS will be able to connect to the given IP address and TCP port.

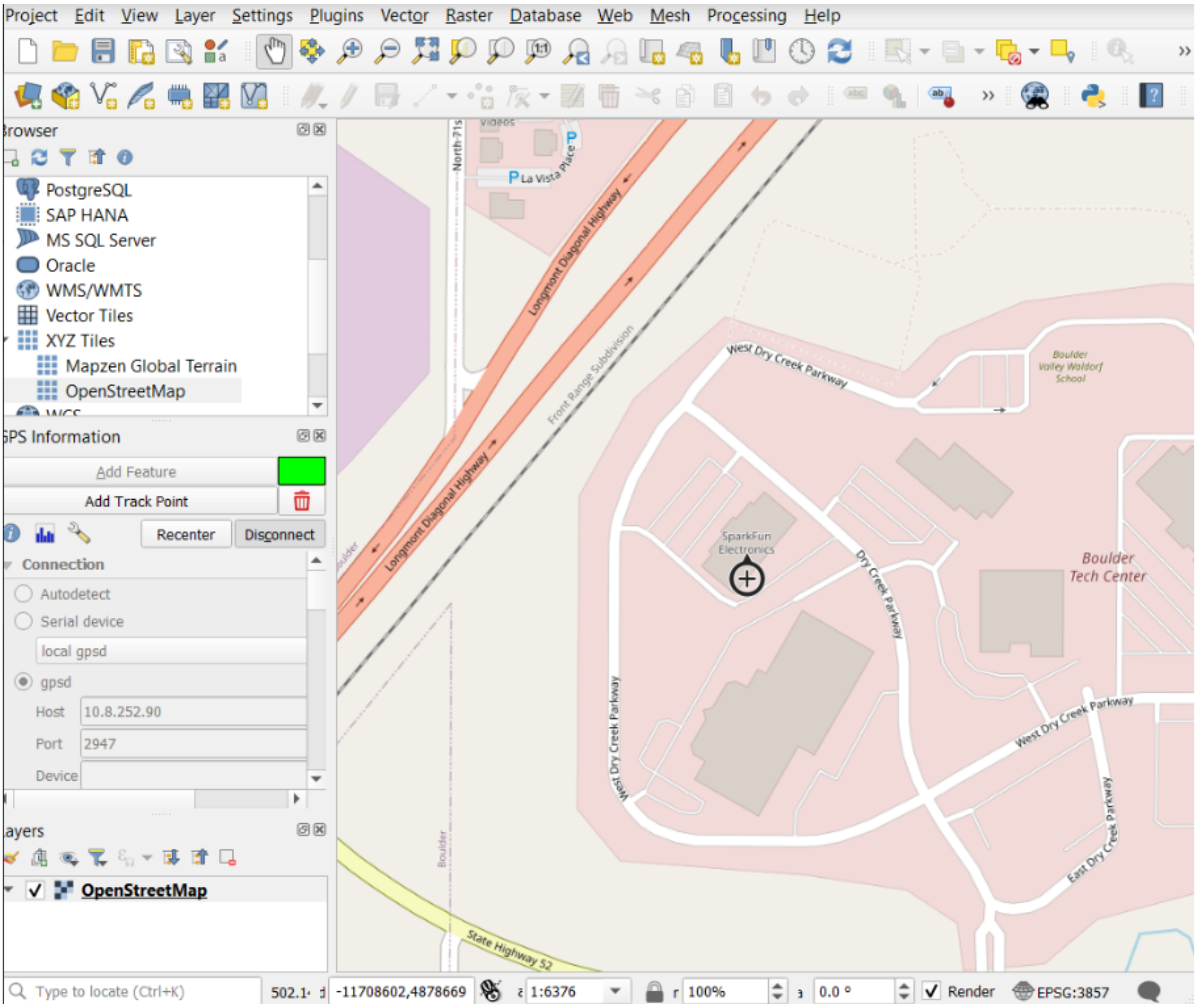


Above: From the subpanel, select 'gpsd'.

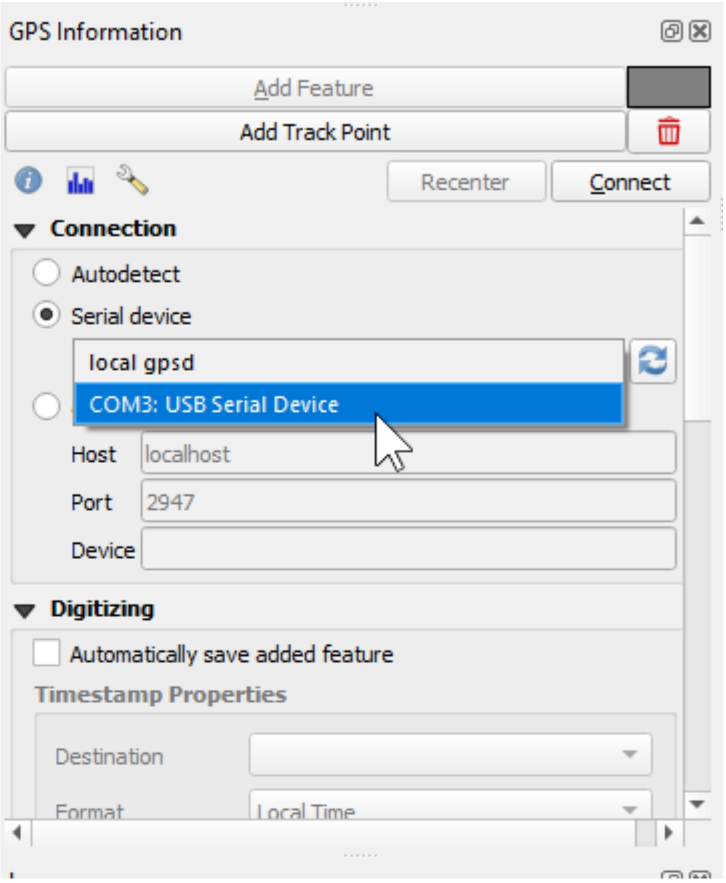


Enter the IP address of your RTK device. This can be found by opening a serial connection to the device. The IP address will be displayed every few seconds. Enter the TCP port to use. By default an RTK device uses 2948.

Press 'Connect'.



The device location will be shown on the map. To see a map, be sure to enable OpenStreetMap under the XYZ Tiles on the Browser.



Alternatively, a direct serial connection to the RTK device can be obtained. Use a USB cable to connect to the 'CONFIG UBLOX' port on RTK Surveyor/Express/Plus and the single USB C port on the RTK Facet/L-Band. Be sure you have the u-blox driver installed. Then select the appropriate COM port for the u-blox module. See Configure with Serial for more information.

4.3.2 Other GIS Packages

Hopefully, these examples give you an idea of how to connect the RTK product line to most any GIS software. If there is other GIS software that you'd like to see configuration information about, please open an issue on the RTK Firmware repo and we'll add it.

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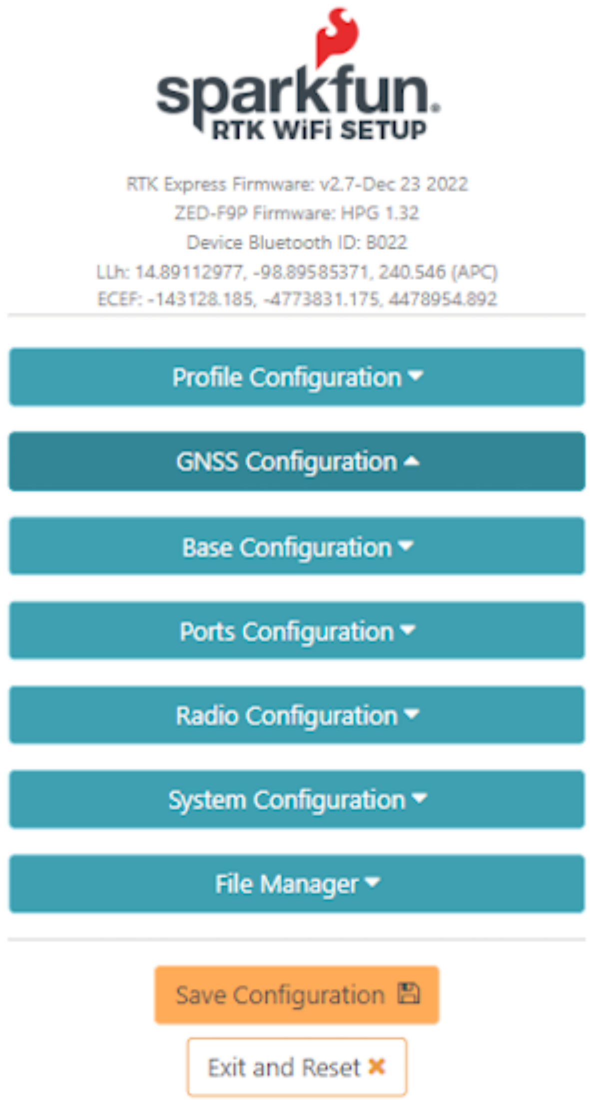
There are services available as well. Syklark provides RTCM coverage for $49 a month (as of writing) and is extremely easy to set up and use. Point One also offers RTK NTRIP service with a free 14 day trial and easy to use front end.

5. Configuration Methods

5.1 Configure with WiFi

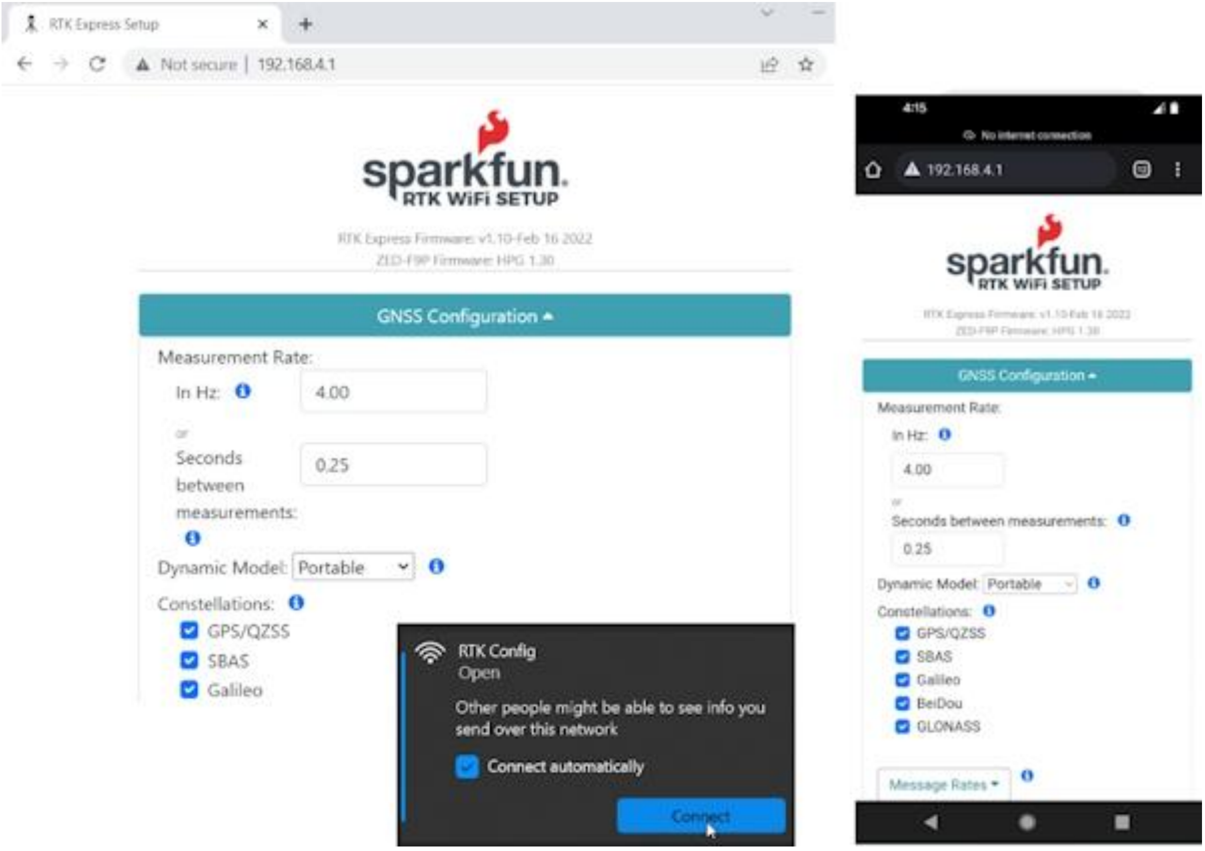
Surveyor: / Express: / Express Plus: / Facet: / Facet L-Band: / Reference Station:



Configuration page via WiFi

Starting with firmware v1.7, WiFi-based configuration is supported. For more information about updating the firmware on your device, please see Updating RTK Firmware.

The RTK device will present a webpage that is viewable from either a desktop/laptop with WiFi or a cell phone. For advanced configurations, a desktop is recommended. For quick in-field changes, a cell phone works great.



Desktop vs Phone display size configuration

5.1.1 RTK Express / Express Plus / Facet

To get into WiFi configuration follow these steps:

1. Power on the RTK Express, Express Plus, or Facet.
2. Once the device has started press the Setup button repeatedly until the *Config* menu is highlighted.
3. The display will blink a WiFi icon indicating it is waiting for incoming connections.
4. Connect to WiFi network named ‘RTK Config’.
5. Open a browser (Chrome is preferred) and type **192.168.4.1** into the address bar.

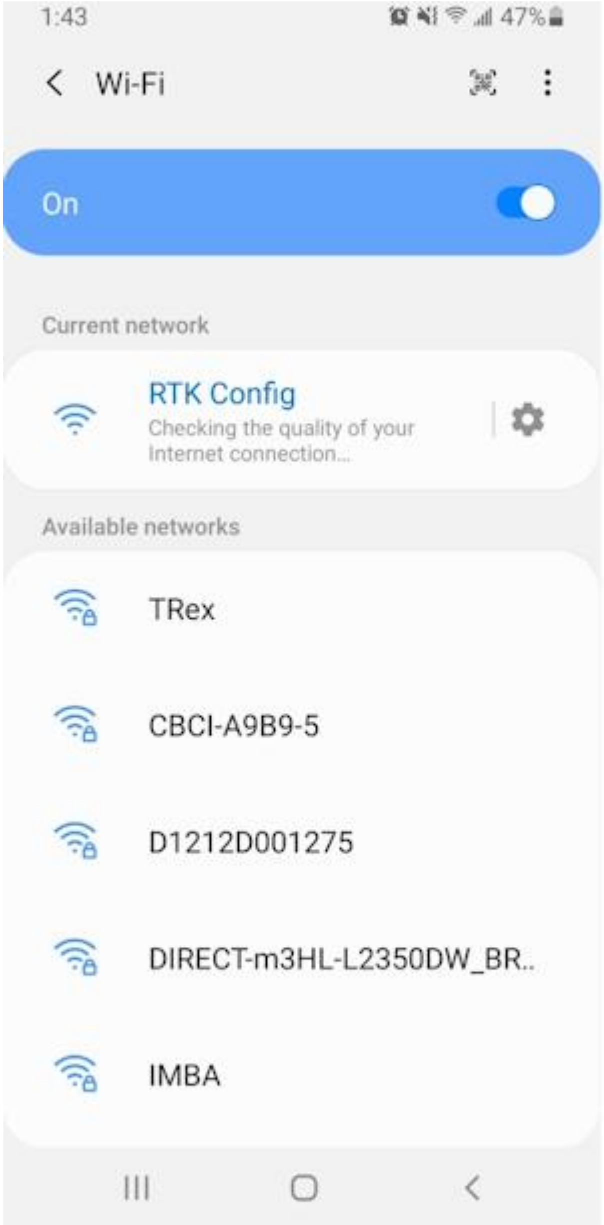
Device ready for cellphone configuration

5.1.2 RTK Surveyor

To get into WiFi configuration follow these steps:

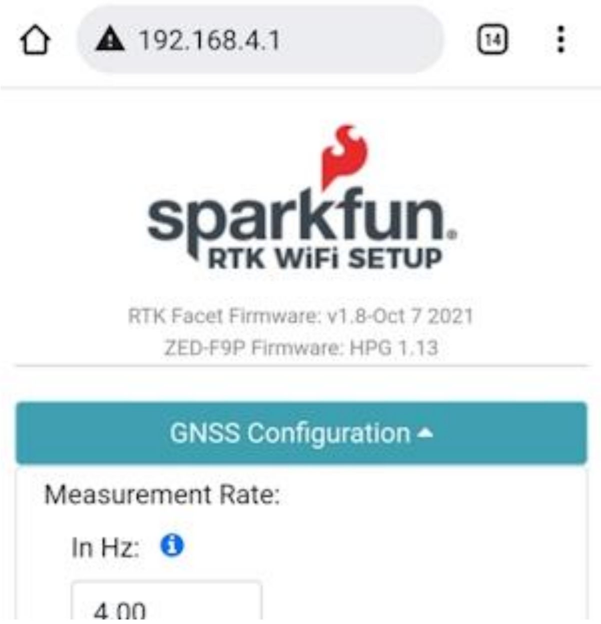
1. Power the RTK Surveyor on in Rover mode.
2. Once the device has started the Bluetooth status LED should be blinking at 1Hz. Now toggle the **SETUP** switch from Base and back to Rover within 1 second. If successful, the Bluetooth status LED will begin fading in/out. The device is now broadcasting as a WiFi access point.
3. Connect to WiFi network named 'RTK Config'.
4. Open a browser (Chrome is preferred) and type **192.168.4.1** into the address bar.

5.1.3 Connecting to WiFi Network



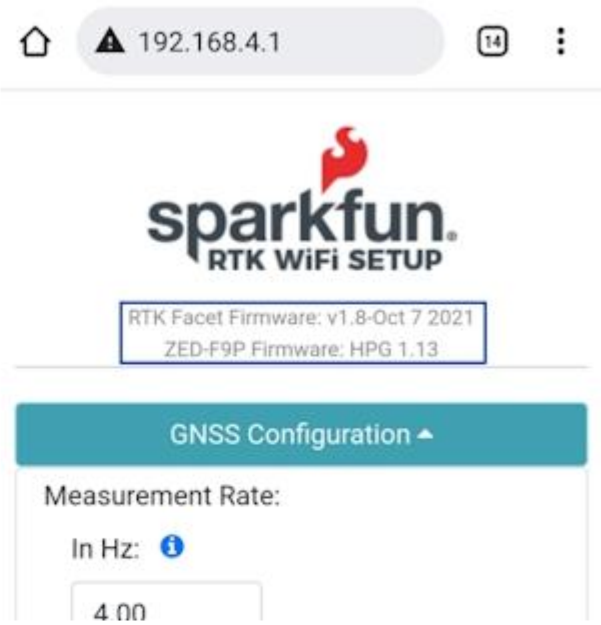
The WiFi network RTK Config as seen from a cellphone

Note: Upon connecting, your phone may warn you that this WiFi network has no internet. That's ok. Stay connected to the network and open a browser. If you still have problems turn off Mobile Data so that the phone does not default to cellular for internet connectivity and instead connects to the RTK Device.



Connected to the RTK WiFi Setup Page

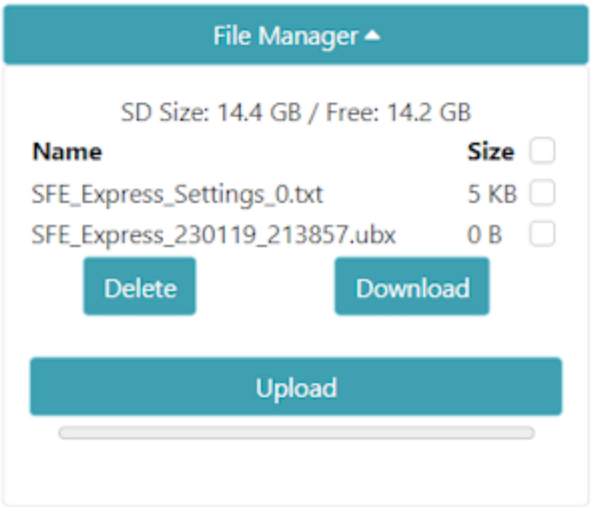
Clicking on the category 'carrot' will open or close that section. Clicking on an ‘i’ will give you a brief description of the options within that section.



This unit has firmware version 1.8 and a ZED-F9P receiver

Please note that the firmware for the RTK device and the firmware for the ZED receiver is shown at the top of the page. This can be helpful when troubleshooting or requesting new features.

5.1.4 File Manager



Added in v3.0 firmware, a file manager is shown if an SD card is detected. This is a handy way to download files to a local device (cell phone or laptop) as well as delete any unneeded files. The SD size and free space are shown. And files may be uploaded to the SD card if needed.

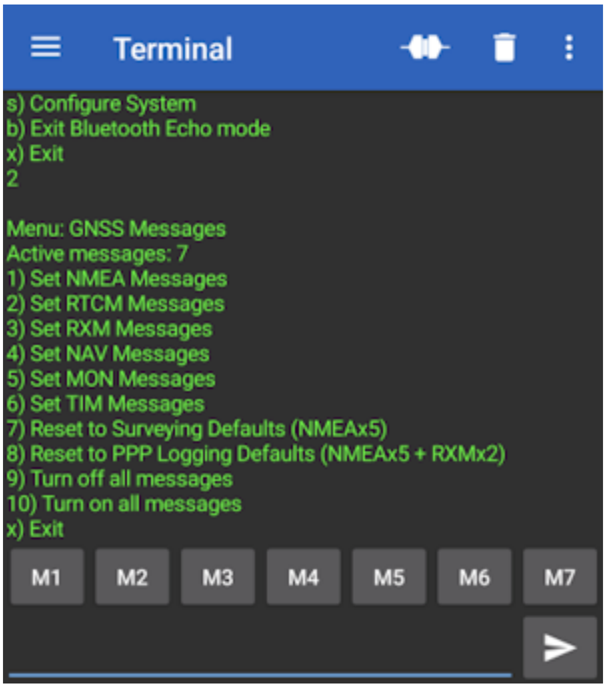
Additionally, clicking on the top checkbox will select all files for easy removal of a large number of files.

5.1.5 Saving and Exit

Once settings are input, please press 'Save Configuration'. This will validate any settings, show any errors that need adjustment, and send the settings to the unit. The page will remain active until the user presses 'Exit to Rover Mode' at which point the unit will exit WiFi configuration and return to standard Rover mode.

5.2 Configure with Bluetooth

Surveyor: / Express: / Express Plus: / Facet: / Facet L-Band: / Reference Station:

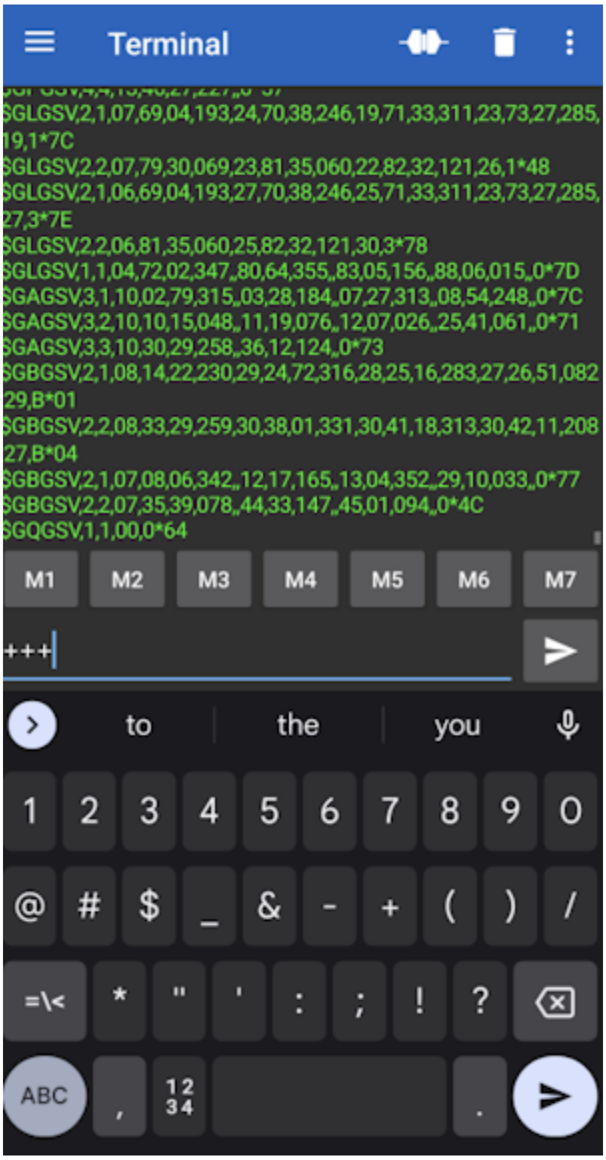


Configuration menu via Bluetooth

Starting with firmware v3.0, Bluetooth-based configuration is supported. For more information about updating the firmware on your device, please see Updating RTK Firmware.

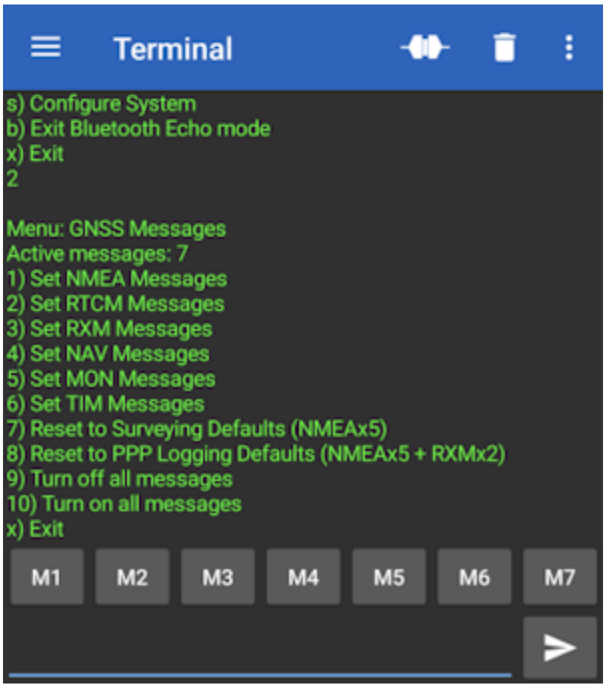
The RTK device will be a discoverable Bluetooth device (both BT SPP and BLE are supported). For information about Bluetooth pairing, please see Connecting Bluetooth.

5.2.1 Entering Bluetooth Echo Mode



Once connected, the RTK device will report a large amount of NMEA data over the link. To enter Bluetooth Echo Mode send the characters +++.

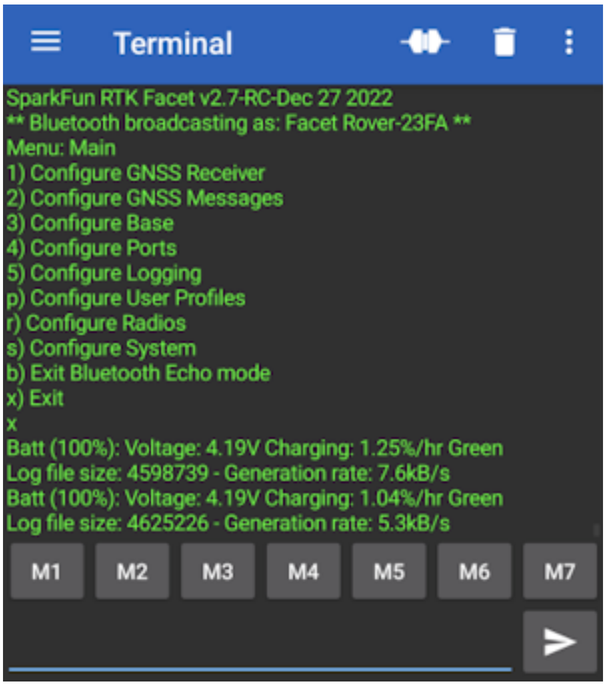
Note: There must be a 2 second gap between any serial sent from a phone to the RTK device, and any escape characters. In almost all cases this is not a problem. Just be sure it's been 2 seconds since an NTRIP source has been turned off and attempting to enter Bluetooth Echo Mode.



The GNSS Messages menu shown over Bluetooth Echo Mode

Once in Bluetooth Echo Mode, any character sent from the RTK unit will be shown in the Bluetooth app, and any character sent from the connected device (cell phone, laptop, etc) will be received by the RTK device. This allows the opening of the config menu as well as the viewing of all regular system output.

For more information about the Serial Config menu please see Configure with Serial.

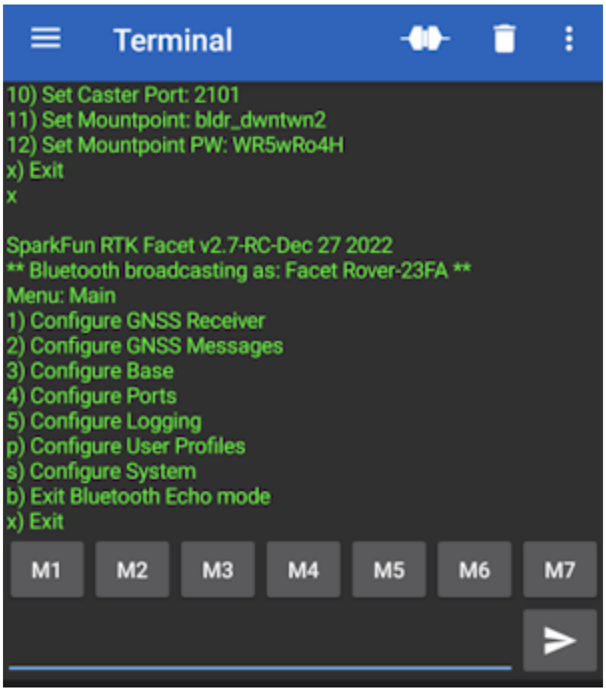


Exit from the Serial Config Menu

Bluetooth can also be used to view system status and output. Simply exit the config menu using option 'x' and the system output can be seen.

5.2.2 Exit Bluetooth Echo Mode

To exit Bluetooth Echo Mode simply disconnect Bluetooth. In the Bluetooth Serial Terminal app, this is done by pressing the 'two plugs' icon in the upper right corner.

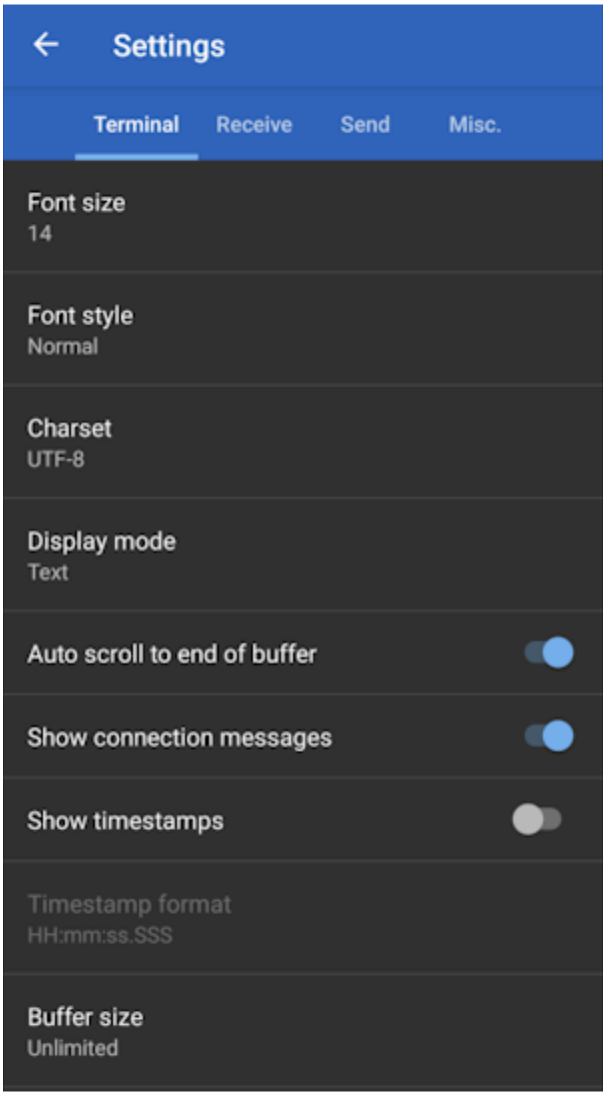


Menu option 'b' for exiting Bluetooth Echo Mode

Alternatively, if you wish to stay connected over Bluetooth but need to exit Bluetooth Echo Mode, use the 'b' menu option from the main menu.

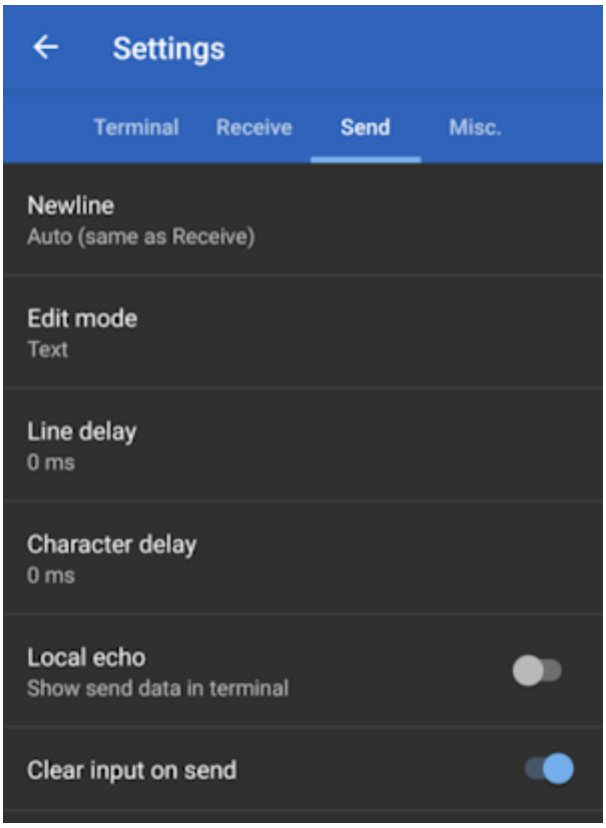
5.2.3 Serial Bluetooth Terminal Settings

Here we provide some settings recommendations to make the terminal navigation of the RTK device a bit easier.



Terminal Settings with Timestamps disabled

Disable timestamps to make the window a bit wider, allowing the display of longer menu items without wrapping.



Clear on send and echo off

Clearing the input box when sending is very handy as well as turning off local echo.

5.3 Configure with Serial

Surveyor: / Express: / Express Plus: / Facet: / Facet L-Band: / Reference Station:

Note: Starting with v3.0 of the firmware any serial menu that is shown can also be accessed over Bluetooth. This makes any configuration of a device much easier in the field. Please see Configure With Bluetooth for more information.

To configure an RTK device using serial attach a USB C cable to the device. The device can be on or off.

5.3.1 RTK Surveyor / Express / Express+

The SparkFun RTK Surveyor has a variety of connectors

Connect the USB cable to the connector labeled **Config ESP32**.

Once connected a COM port will enumerate. Open the `Device Manager` in Windows and look under the Ports branch to see what COM port the device is assigned to.

5.3.2 RTK Facet

Connect the USB cable to the USB connector.

There is a USB hub built into the RTK Facet. When you attach the device to your computer it will enumerate two COM ports.

In the image above, the `USB Serial Device` is the ZED-F9P and the `USB-SERIAL CH340` is the ESP32.

Don't See 'USB-Serial CH340'? If you've never connected a CH340 device to your computer before, you may need to install drivers for the USB-to-serial converter. Check out our section on "How to Install CH340 Drivers" for help with the installation.

Don't See 'USB Serial Device'? The first time a u-blox module is connected to a computer you may need to adjust the COM driver. Check out our section on "How to Install u-blox Drivers" for help with the installation.

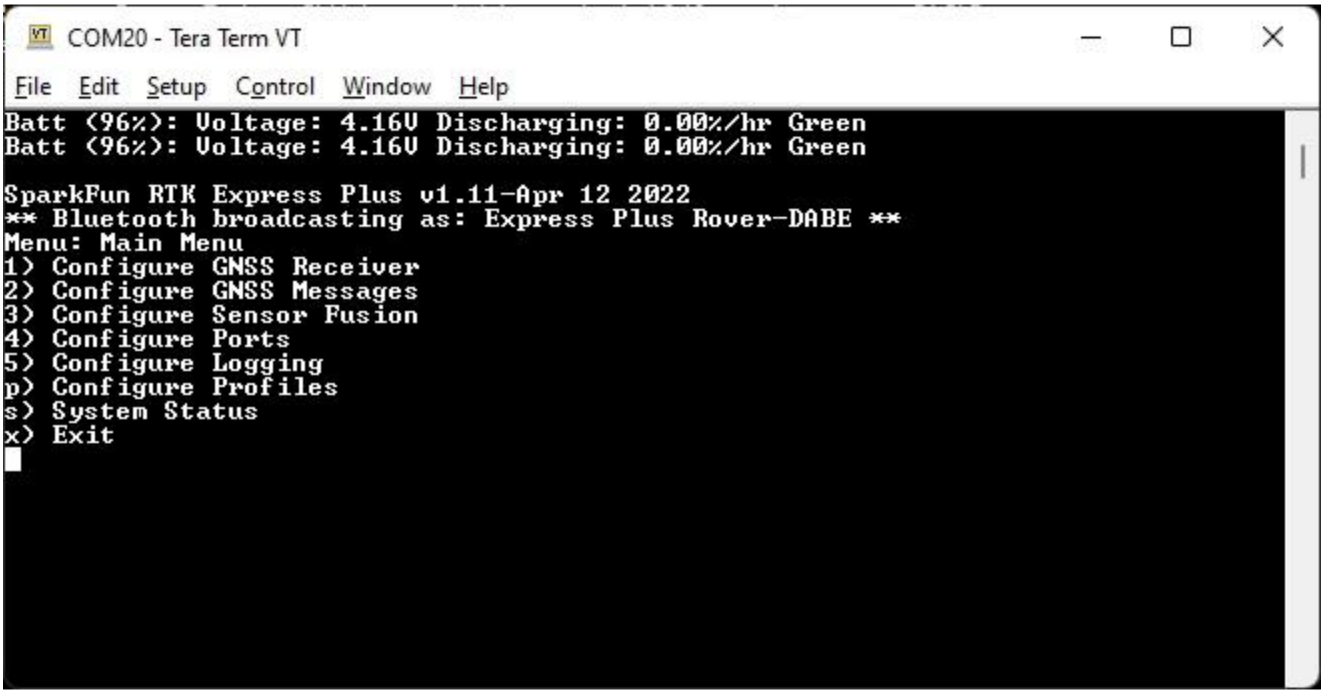
Configuring the RTK device is done over the *USB-Serial CH340* COM port via the serial text menu. Various debug messages are printed to this port at 115200bps and a serial menu can be opened to configure advanced settings.

Configuring the ZED-F9P is done over the *USB Serial Device* port using u-center. It's not necessary for normal operation but is handy for tailoring the receiver to specific applications. As an added perk, the ZED-F9P can be detected automatically by some mobile phones and tablets. If desired, the receiver can be directly connected to a compatible phone or tablet removing the need for a Bluetooth connection.

5.3.3 Terminal Window

Open a terminal window at 115200bps; you should see various status messages every second. Press any key to open the configuration menu. Not sure how to use a terminal? Check out our Serial Terminal Basics tutorial.

Note that some Windows terminal programs (e.g. Tera Term) may reboot the Facet when the terminal connection is closed. You can disconnect the USB cable first to prevent this from happening.

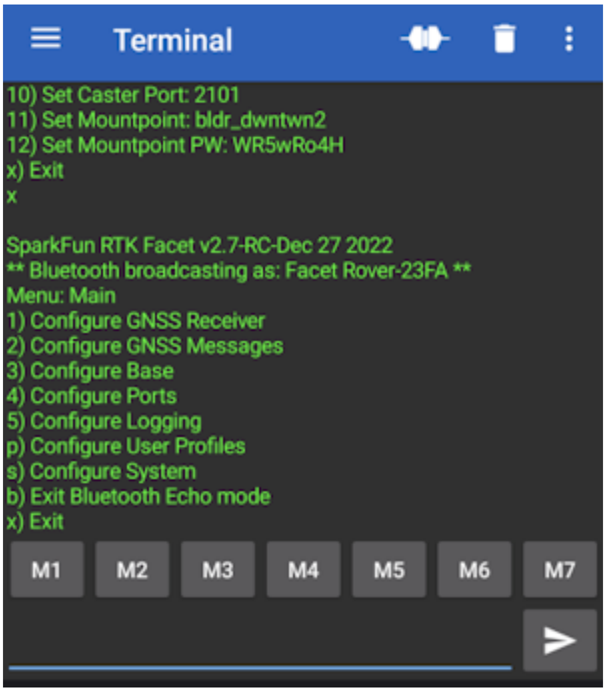


Main Menu

Pressing any button will display the Main menu. The Main menu will display the current firmware version and the Bluetooth broadcast name. Note: When powered on, the RTK device will broadcast itself as either *[Platform] Rover-XXXX* or *[Platform] Base-XXXX* depending on which state it is in. The Platform is 'Facet', 'Express', 'Surveyor', etc.

Pressing '1' or 's' for example, will open those submenus.

The menus will timeout after 10 minutes of inactivity, so if you do not press a key the device will exit the menu and return to reporting status messages.

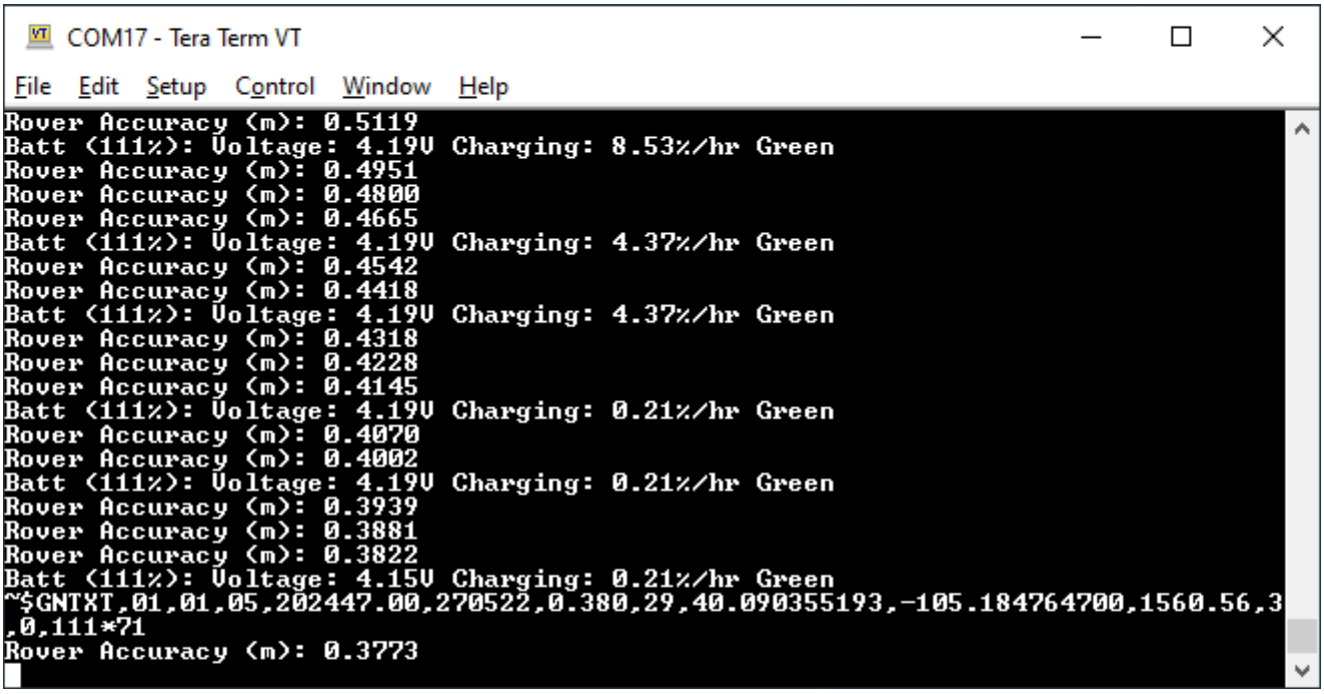


Configuration menu via Bluetooth

Note: Starting with firmware v3.0, Bluetooth-based configuration is supported. Please see Configure With Bluetooth for more information.

5.3.4 System Report

Sending the `~` character to the device over the serial port will trigger a system status report. This is a custom NMEA-style sentence, complete with CRC.



Terminal showing System Status

Below is an example system status report sentence:

$GNTXT,01,01,05,202447.00,270522,0.380,29,40.090355193,-105.184764700,1560.56,3,0,86*71

- $GNTXT : Start of custom NMEA sentence
- 01 : Number of sentences
- 01 : Sentence number
- 05 : Sentence type ID (5 is for System Status messages)
- 202447.00 : Current hour, minute, second, milliseconds
- 270522 : Current day, month, year
- 0.380 : Current horizontal positional accuracy (m)
- 29 : Satellites in view
- 40.090355193 : Latitude
- -105.184764700 : Longitude
- 1560.56 : Altitude
- 3 : Fix type (0 = no fix, 2 = 2D fix, 3 = 3D fix, 4 = 3D + Dead Reackoning, 5 = Time)
- 0 : Carrier solution (0 = No RTK, 1 = RTK Float, 2 = RTK Fix)
- 86 : Battery level (% remaining)
- *71 : The completion of the sentence and a CRC

Note: This is a custom NMEA sentence, can vary in length, and may exceed the maximum permitted sentence length of 61 characters.

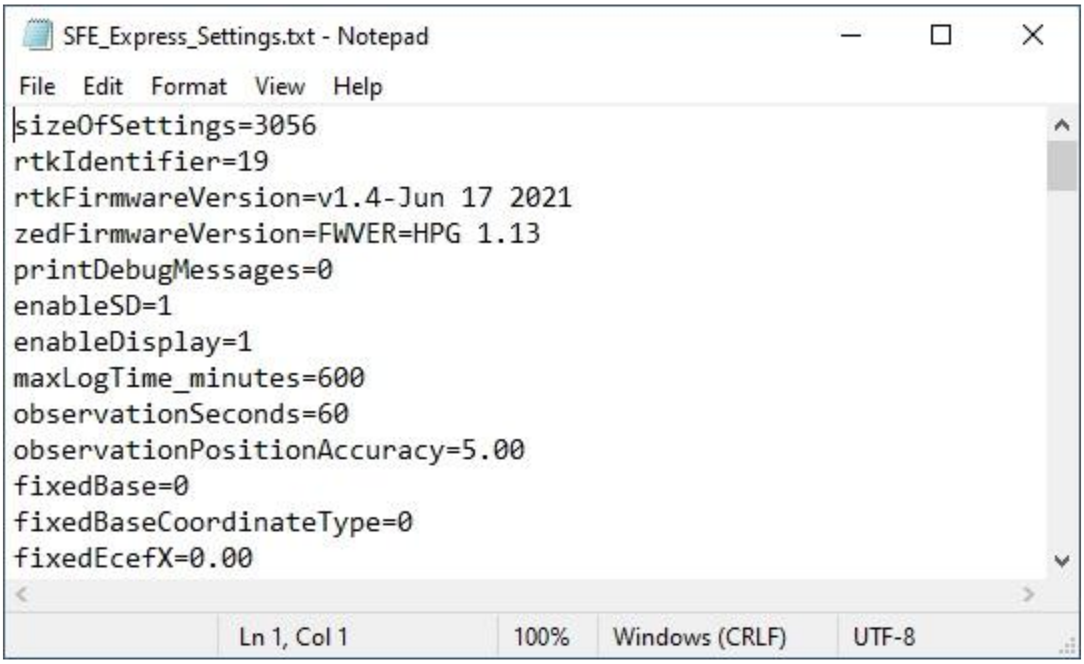
5.3.5 L-Band Performance During Serial Configuration

Because of the way corrections are provided between the sub modules (NEO-D9S and ZED-F9P), the corrections will be interrupted while the configuration menu is open. RTK Fix may be lost if the menu is open for more than ~30s. RTK Fix will return once the configuration is complete and the menu is closed.

Note: This only affects the RTK Facet L-Band model.

5.4 Configure with Settings File

Surveyor: / Express: / Express Plus: / Facet: / Facet L-Band: / Reference Station:



SparkFun RTK Settings File

All device settings are stored both in internal memory and an SD card if one is detected. The device will load the latest settings at each power on. If there is a discrepancy between the internal settings and a settings file then the settings file will be used. This allows a collection of RTK products to be identically configured using one 'golden' settings file loaded onto an SD card.

All system configuration can be done by editing the *SFE_[Platform]_Settings_0.txt* file (example shown above) where [Platform] is Facet, Express, Surveyor, etc and 0 is the profile number (0, 1, 2, 3). This file is created when a microSD card is installed. The settings are clear text but there are no safety guards against setting illegal states. It is not recommended to use this method unless You Know What You're Doing®.

Keep in mind:

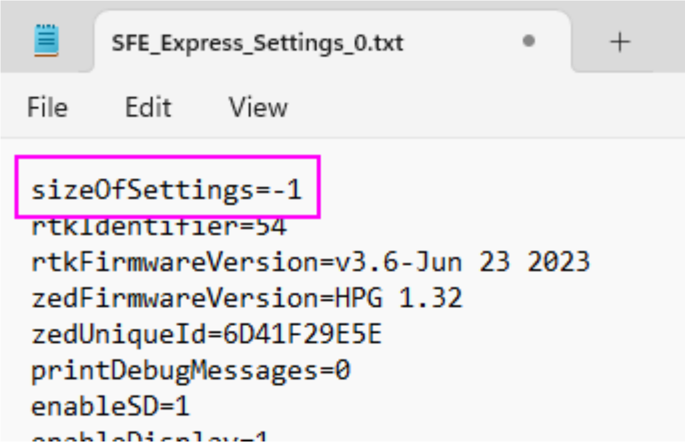
- The settings file contains hundreds of settings.
- The SD card file "SFE_Express_Settings_0.txt" is used for Profile 1, SD card file "SFE_Express_Settings_1.txt" is used for Profile 2, etc. (note that setting 0 is for profile 1, ...)
- When switching to a new profile, the settings file on the SD card with all settings will be created or updated. The internal settings will not be updated until you switch to the profile. Additionally, the file for a particular profile will not be created on the SD card until you switch to that profile.
- It is not necessary that the settings file on the SD card have all of the settings.

For example, if you only wanted to set up two wireless networks for profile 2, you could create a file named "SFE_Express_Settings_1.txt" that only contained the following settings:

```
profileName=a name you choose
enablePvtServer=1
wifiNetwork0SSID=your SSID name 1
wifiNetwork0Password=your SSID password 1
wifiNetwork1SSID=your SSID name 2
wifiNetwork1Password=your SSID password 2
wifiConfigOverAP=0
```

These settings on the SD card will overwrite the settings in the RTK Express internal memory. Once you select this profile on your RTK Express, the SD card file will be overwritten with all of the merged settings.

5.4.1 Forcing a Factory Reset



If the device has been configured into an unknown state the device can be reset to factory defaults. Power down the RTK device and remove the SD card. Using a computer and an SD card reader, open the SFE_[Platform]_Settings_0.txt file where [Platform] is Facet, Express, Surveyor, etc and 0 is the profile number (0, 1, 2, 3). Modify the **sizeOfSettings** value to -1 and save the file to the SD card. Reinsert the SD card into the RTK unit and power up the device. Upon power up, the device will display 'Factory Reset' while it clears the settings.

Note: A device may have multiple profiles, ie multiple settings files (SFE_Express_Settings_**0**.txt, SFE_Express_Settings_**1**.txt, etc). All settings files found on the SD card must be modified to guarantee the factory reset.

5.5 Configure with u-center

Surveyor: / Express: / Express Plus: / Facet: / Facet L-Band: / Reference Station:

The ZED-F9P module can be configured independently using the u-center software from u-blox by connecting a USB cable to the *Config u-blox* USB connector. Settings can be saved to the module between power cycles. For more information please see SparkFun's Getting Started with u-center by u-blox.

However, because the ESP32 does considerable configuration of the ZED-F9P at power on it is not recommended to modify the settings of the ZED-F9P using u-center. Nothing will break but your changes will likely be overwritten at the next power cycle.